

ASASIPINTAR PROGRAM
MATHEMATICS
PNAP0143-LOGICAL REASONING

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COURSE MODULE: LOGICAL REASONING.

TOPIC 1: LOGIC AND PROOF

1.1 Logic

Objectives:

Students should be able to

- Define statements (simple and compound statements).
- Construct truth tables of statements
- Determine whether two compound statements are logically equivalent
- Negate a statement

What is a statement?

A statement is a sentence that is either definitely true (T) or false (F).

Example 1: Decide whether or not the following are statements. In the case of a statement, say if it is true or false.

a) Kuala Lumpur is the capital of Malaysia

b) $5 > 6$

c) $x + 1 = 2$

d) Hello !

Solution: a) Statement. True b) Statement. False c) Not a statement d) Not a statement

A statement is divided into two types: 1) Simple statement 2) Compound statement

Statements such as a) Kuala Lumpur is the capital of Malaysia

b) $5 > 6$

are simple both simple statements.

A compound statement is a combination of two or more simple statements.

Connectives

There are five common connectives:

1. Negation $\sim$ (this symbol is called tilde)
2. Conjunction $\wedge$

3. Disjunction $\vee$
4. Implication $\rightarrow$
5. Equivalence $\leftrightarrow$

Example 2: Let p be the statement “I am 8 feet tall”. What is the negation of p ?

Solution: The negation of p denoted by $\sim p$ is “I am **not** 8 feet tall”.

Example 3: Let p be the statement “They are talking” and q be the statement “The door is closed”. Write the conjunction statement.

Solution: The conjunction statement (denoted by $p \wedge q$) is “They are talking **and** the door is closed”.

Example 4: Let p be the statement “A policeman has to cross a river using a small boat” and q be the statement “A thief has to cross a river using a small boat”. Write the disjunction statement.

Solution: The disjunction statement (denoted by $p \vee q$) is “**Either** a policeman **or** a thief has to cross a river using a small boat”.

Example 5: Let p be the statement “you participate in class” and q be the statement “you will get extra points”. Write the implication statement $p \rightarrow q$.

Solution: The implication statement $p \rightarrow q$ is “**If** you participate in class **then** you will get extra points”.

The implication statement is also called the conditional statement. In the “If...then” statements, the “if-part” is called the antecedent while the “then-part” is called the consequent. Other ways to say “If you participate in class then you will get extra points” are

“You participate in class **only if** you will get extra points”

“You will get extra points **if** you participate in class”

“You participate in class **implies** you will get extra points”

Example 6: Let p be the statement “He will push button A” and q be the statement “The red light flashes”. Write the statement $p \leftrightarrow q$.

Solution: The statement $p \leftrightarrow q$ is “He will push button A **if and only if** the red light flashes”.

The statement $p \leftrightarrow q$ is called equivalence or bi-conditional. Some other books use “ $\equiv$ ” instead.

The statement $p \leftrightarrow q$ is a combination of two conditionals: $p \rightarrow q$ and $q \rightarrow p$. For example, “He will push button A **if and only if** the red light flashes” is actually the same as saying “He will push button A **if** the red light flashes **and** He will push button A **only if** the red light flashes”.

Note that, here the usage of “only if” is not the same as “if”. Do not confuse with English sentences.

Quantifiers

Consider this sentence:

x plays the piano

This sentence is not a statement because we cannot determine whether it is True or False. This sentence has a variable and therefore it is called a Predicate. Let us denote the sentence “ x plays the piano” as $U(x)$. Compare $U(x)$ with this sentence:

Ahmad plays the piano

This sentence is definitely a statement and it actually replaces x in “ x plays the piano” with the word “Ahmad”. Hence we denote this statement as $U(\text{Ahmad})$. Note that $U(x)$ is not a statement but $U(\text{Ahmad})$ is a statement.

A predicate is not a statement. If the variables in the predicate is specified (being given a quantity or a name), then the predicate becomes a statement. We can use quantifiers to specify the variables in a predicate.

There are two types of quantifiers: Universal $\forall$ and Existential $\exists$

Existential $\exists$: “There exists a” or “There is a” or “There is at least one”

Universal $\forall$: “For all” or “For every”

Some symbols:

N : Set of positive integers = $\{1, 2, 3, 4, 5, 6, \dots, 100, 101, 102, \dots\}$

R : Set of real numbers = $(-\infty, \infty)$

Z : Set of integers = $\{\dots -2000, -1999, -1998, -1997, -1996, \dots 0, 1, 2, 3, 4, \dots, 1996, 1997, \dots\}$

Q : Set of rational numbers = any numbers that can be written as $\frac{a}{b}$ where a and b are integers with $b \neq 0$.

Example 7

Let $P(x)$ be “ x is even”. Express the following statement in symbolic form.

“For all integer x , x is even”.

Solution: $\forall x \in \mathbf{Z}, P(x)$.

Example 8

Let $Q(x)$ be “ $x < 4$ ”. Express the following statement in symbolic form.

“There is a real number x such that $x < 4$ ”

Solution: $\exists x \in \mathbf{R}, Q(x)$.

Negation of a statement involving a quantifier

Let say we have a statement:

All **actress** are **singers**

The negation of the above statement is:

There is at least one **actress** who is not a **singer**

What is the negation of $\forall x P(x)$?

The negation of $\forall x P(x)$ is $\exists x \sim P(x)$.

Example 9

Let $P(x)$ be “ x is even”. Express negation of $\forall x \in \mathbf{Z}, P(x)$ in sentence.

Solution: Negation of $\forall x \in \mathbf{Z}, P(x)$ is $\exists x \in \mathbf{Z}, \sim P(x)$. In sentence, the negation is written as

“There is an integer x such that x is not even (or x is odd)”.

Let say we have a statement:

There is at least one **international undergraduate student** in **UKM Bangi campus**

The negation of this statement is:

All **international undergraduate students** are not in **UKM Bangi campus**

What is the negation of $\exists x Q(x)$?

The negation of $\exists x Q(x)$ is $\forall x \sim Q(x)$.

Example 10

Let $Q(x)$ be " $x \leq 0$ ". Express the negation of $\exists x \in \mathbf{N}, Q(x)$ in sentence.

Solution: Negation of $\exists x \in \mathbf{N}, Q(x)$ is $\forall x \in \mathbf{N}, \sim Q(x)$. In sentence, the negation is written as

"For all integer positive x , $x > 0$ ".

Exercises

1. Decide whether or not the following are statements. In the case of a statement, say if it is true or false.
 - a) Today is raining
 - b) For every real number x , $x > 2$.
 - c) $x \neq 3$
 - d) Please open that door
 - e) I am a good student
 - f) There is a positive integer x such that $x \leq 2$
2. Express each statement in symbolic form.
 - a) Either my parents are out of town or visiting my sick neighbor
 - b) If it rains, then he will not come
 - c) I will score A if and only if I study hard
 - d) For all integer a , $2a - 1$ is odd.
 - e) There is a real number y such that $2099 < y < 2100$.

Solution:

1. a) Statement. The truth value depends on whether or not it is raining today
 b) Statement. False. There exist a real number x such that $x \leq 2$.
 c) Not a statement

- d) Not a statement
- e) Statement. The truth value depends on whether you are a good student or not
- f) Statement. True.

2. a) $p \vee q$ (p =my parents are out of town, q =my parents are visiting my sick neighbor)
 b) $p \rightarrow \sim q$ (p = it rains, q =he will come)
 c) $p \leftrightarrow q$ (p = I will score A, q = I study hard)
 d) $\forall a \in \mathbf{Z}, P(a)$ ($P(a)$ = $2a - 1$ is odd)
 e) $\exists y \in \mathbf{R}, Q(y)$ ($Q(y)$ = $2099 < y < 2100$)

Truth tables

Truth tables are tables giving all possible truth values of statements.

Negation

p	$\sim p$
T	F
F	T

A statement and its negation have opposite truth values.

Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

A conjunction $p \wedge q$ is True only when p and q are both True.

Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

A disjunction $p \vee q$ is False only when p and q are both False.

Implication

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication $p \rightarrow q$ is only False when q is False given that p is True.

Bi-conditional

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Bi-conditional $p \leftrightarrow q$ is True when p and q are both True or are both False.

Logically equivalent

Two compound statements are called logically equivalent if they have the same truth value for every possible truth value assignment to the component statements.

Example 11

Determine whether $p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$.

Solution:

Step 1: Construct truth tables for each statement

p	q	$\sim q$	$\sim p$	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

Step 2: Compare the truth values

The statements $p \rightarrow q$ and $\sim q \rightarrow \sim p$ have the same truth values.

Therefore, $p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$.

Exercises 1.1

1. Determine whether $p \rightarrow q$ is logically equivalent to $\sim p \vee q$.
2. Determine whether $p \rightarrow q$ is logically equivalent to $p \wedge \sim q$.
3. Negate the following statements
 - a) For every positive integer r , $r > 1$
 - b) There is a real number y such that $0 < y < 1$
 - c) $5 > 1$ and $1 > 0$

Solution

1. $p \rightarrow q$ is logically equivalent to $\sim p \vee q$ (Follow Steps as in Example 11)
2. $p \rightarrow q$ is not logically equivalent to $p \wedge \sim q$ (Follow Steps as in Example 11)
3. a) There exist a positive integer r such that $r \leq 1$
 b) y is not in the interval $(0, 1)$ for all real numbers y .
 c) $5 \leq 1$ or $1 \leq 0$

Negation of $p \wedge q$ is logically equivalent to $\sim p \vee \sim q$.

Similarly, negation of $p \vee q$ is logically equivalent to $\sim p \wedge \sim q$.

Show that $\sim(p \wedge q)$ is logically equivalent to $\sim p \vee \sim q$ using truth tables as in Example 11.

1.2 Proof

Objectives:

Students should be able to

- State the differences between a theorem, a proof and a definition
- Construct simple proofs using direct proof method
- Construct simple proofs using contrapositive proof method
- Construct simple proofs using proof by contradiction method
- Construct simple proofs using proof by mathematical induction method

Mathematical terms that you should know

Theorem: A mathematical statement that is true and has been proved to be true

Proof of a theorem: A written verification that shows that the theorem is definitely true.

Definition: An exact, unambiguous (no double interpretations) explanation of the meaning of a mathematical word or phrase

Methods of proving

Many mathematical theorems are either conditional or bi-conditional statements.

This notes will cover only theorems in conditional (i.e. $p \rightarrow q$) form. Bi-conditionals can be proved by first prove $p \rightarrow q$ and then prove $q \rightarrow p$.

There are four methods of proving:

- Direct Proof
- Contrapositive Proof
- Proof by Contradiction
- Proof by Mathematical Induction

Direct proof

Let say we have a theorem which is in conditional form $p \rightarrow q$. Outline of direct proof is given by

Theorem: Statement “ $p \rightarrow q$ ”

Proof: Suppose “ p ”

:

Therefore “ q ”.

From the outline, firstly we assume that p is True. By using this assumption and other mathematical knowledge, we prove that q is also True.

Example 1

Prove that if x is odd then x^2 is odd.

Solution:

Step 1: Write the statement in symbol.

p : x is odd

q : x^2 is odd

Statement: " $p \rightarrow q$ "

Step 2: Use Direct Proof outline to prove the statement.

Proof: Suppose " x is odd".

Since x is odd, x can be written as

$$x = 2k + 1 \text{ where } k \text{ is an integer}$$

$$x^2 = (2k + 1)(2k + 1)$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

$2k^2 + 2k$ is an integer, say $2k^2 + 2k = m$.

x^2 can also be written as $x^2 = 2m + 1$.

Therefore, " x^2 is odd".

Contrapositive proof

$p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$. You may verify this by using a truth table. Outline of contrapositive proof is given by

Theorem: Statement " $p \rightarrow q$ "

Proof: Suppose " $\sim q$ "

:

Therefore " $\sim p$ ".

From the outline, firstly we assume that $\sim q$ is True. By using this assumption and other mathematical knowledge, we prove that $\sim p$ is also True.

Example 2

Prove that if $x^2 - 6x + 5$ is even then x is odd.

Solution:

Step 1: Write the statement in symbol.

p : $x^2 - 6x + 5$ is even

q : x is odd

Statement: " $p \rightarrow q$ "

Step 2: Use Contrapositive Proof outline to prove the statement.

Proof: Suppose " x is even".

Since x is even, x can be written as

$$x = 2k, k \text{ is an integer}$$

$$x^2 - 6x + 5 = (2k)^2 - 6(2k) + 5$$

$$= 4k^2 - 12k + 4 + 1$$

$$= 2(2k^2 - 6k + 2) + 1$$

$2k^2 - 6k + 2$ is an integer, let say $2k^2 - 6k + 2 = n$.

$x^2 - 6x + 5$ can be written as $x^2 - 6x + 5 = 2n + 1$.

Therefore, " $x^2 - 6x + 5$ is odd".

Proof by contradiction

For proof by contradiction, the theorem is assumed to False which means that the statement $p \rightarrow q$ is assumed to be False.

$p \rightarrow q$	$\sim(p \rightarrow q)$
T	F
F	T

From the above table (last row), $p \rightarrow q$ False is the same as $\sim(p \rightarrow q)$ True.

$\sim(p \rightarrow q)$ is logically equivalent to $p \wedge \sim q$. You may verify this using a truth table.

Outline of proof by contradiction is given by

Theorem: Statement “ $p \rightarrow q$ ”

Proof: Suppose “ $\sim(p \rightarrow q)$ ” (which is logically equivalent to “ $p \wedge \sim q$ ”)

:

Contradiction.

From the outline, firstly we assume that $p \rightarrow q$ is False which is the same as saying that $\sim(p \rightarrow q)$ is True. By using this assumption and other mathematical knowledge, we must get a contradiction. The contradiction shows that the statement $p \rightarrow q$ is indeed True.

Example 3

Prove that if $x^2 - 6x + 5$ is even then x is odd by using the method of proof by contradiction.

Solution:

Step 1: Write the statement in symbol

p : $x^2 - 6x + 5$ is even

q : x is odd

Statement: “ $p \rightarrow q$ ”

Step 2: Use Proof by Contradiction outline to prove the statement.

Proof: Suppose the statement “if $x^2 - 6x + 5$ is even then x is odd” is false.

This is logically equivalent to $x^2 - 6x + 5$ is even and x is even.

Since x is even, x can be written as

$$x = 2k, k \text{ is an integer}$$

$$x^2 - 6x + 5 = (2k)^2 - 6(2k) + 5$$

$$= 4k^2 - 12k + 4 + 1$$

$$= 2(2k^2 - 6k + 2) + 1$$

$$2k^2 - 6k + 2 \text{ is an integer, let say } 2k^2 - 6k + 2 = n.$$

$$x^2 - 6x + 5 \text{ can be written as } x^2 - 6x + 5 = 2n + 1.$$

This means that $x^2 - 6x + 5$ is odd which contradicts the assumption that $x^2 - 6x + 5$ is even.

This contradiction shows that the statement “if $x^2 - 6x + 5$ is even then x is odd” is indeed True.

Remark:

The method of proof by contradiction is not just limited to prove conditional statement. It can be used to prove any kind of statement. General outline for proof by contradiction is given by

Statement: “ p ”

Proof: Suppose “ $\sim p$ ”.

:

Contradiction.

Example 4

Prove that the number $\sqrt{2}$ is irrational.

Solution:

Step 1: Let p be the statement “ $\sqrt{2}$ is irrational”.

Step 2: Use the General outline for Proof by Contradiction to prove the statement.

Proof: Suppose “ $\sqrt{2}$ is rational”.

Since $\sqrt{2}$ is rational, we can write

$$\sqrt{2} = \frac{a}{b} \text{ where } a \text{ and } b \text{ are integers with } b \neq 0.$$

Let us assume that a and b have no common factors other than 1, for if they have, the factors will be cancelled out.

$$2 = \frac{a^2}{b^2}$$

$2b^2 = a^2$ implies that a^2 is even. Thus, a is even.

Since a is even, a can be written as

$$a = 2k \text{ where } k \text{ is an integer.}$$

So,

$$2b^2 = (2k)^2$$

$$2b^2 = 4k^2$$

$b^2 = 2k^2$ implies that b^2 is even. Thus, b is even.

Since a and b are even, they have a common factor that is 2. This is a contradiction to the fact that a and b have no common factors other than 1. Hence, $\sqrt{2}$ is indeed an irrational number.

Proof by mathematical induction

Principle of mathematical induction is used to prove the statement “ $\forall x \in \mathbf{N}, P(x)$ ”. The outline of the principle of mathematical induction is given by

Statement: “ $\forall x \in \mathbf{N}, P(x)$ ”

Proof:

1. Basis step
Prove “ $P(1)$ ” is True.
2. Inductive step
Prove “ $P(k) \rightarrow P(k+1)$ ” is True using Direct Proof method. So, the outline is

Suppose “ $P(k)$ ”
 :
 Therefore “ $P(k+1)$ ”

$P(1)$ is True.

$P(k) \rightarrow P(k+1)$ is True.

Therefore $P(x)$ is True for all $x \in \mathbf{N}$.

Example 5

Prove that $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$ for all $n \in \mathbf{N}$.

Solution:

Step 1: Write the statement in symbol.

$$P(n): 1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$$

Statement: “ $\forall n \in \mathbf{N}, P(n)$ ”

Note that the predicate $P(n)$ is an equation where the left side of the equation is a series, while the right side of the equation is the sum of the series (i.e. n^2).

$$P(n): \underbrace{1 + 3 + 5 + 7 + \dots + (2n - 1)}_{\text{Left side(series)}} = \underbrace{n^2}_{\text{Right side(sum of series)}}$$

Left side(series)

Right side(sum of series)

“ $1 + 3 + 5 + 7 + \dots + (2n - 1)$ ” is an arithmetic series where the first term T_1 is “1”, the second term T_2 is “3”, the third term T_3 is “5” ... and the n th term T_n is “ $2n - 1$ ”.

Step 2: Use the outline of the principle of mathematical induction to prove the statement.

Proof:

1. $P(1)$: $1 = 1^2$ is a True statement.
2. $P(k)$: “ $1 + 3 + 5 + 7 + \dots + (2k - 1) = k^2$ ”
 $P(k + 1)$: $1 + 3 + 5 + 7 + \dots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$
 Suppose “ $1 + 3 + 5 + 7 + \dots + (2k - 1) = k^2$ ”

$$\begin{aligned} 1 + 3 + 5 + 7 + \dots + (2k - 1) + (2(k + 1) - 1) &= k^2 + (2(k + 1) - 1) \\ &= k^2 + 2k + 2 - 1 \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

 Therefore “ $1 + 3 + 5 + 7 + \dots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$ ”.

$P(1)$ is True.

$P(k) \rightarrow P(k + 1)$ is True.

Therefore $P(n)$ is True for all $n \in \mathbf{N}$.

Example 6

Concerning the Fibonacci sequence, prove that $F_{n+1}^2 - F_{n+1}F_n - F_n^2 = (-1)^n$ for all positive integers n .

Solution:

The Fibonacci Sequence is the sequence of numbers:

1, 1, 2, 3, 5, 8, 13, 21, 34, ...

First term F_1 is fixed to be 1.

Second term F_2 is also fixed to be 1.

Third term $F_3 = F_1 + F_2$

$$= 1 + 1 = 2$$

Fourth term $F_4 = F_2 + F_3$

$$= 1 + 2 = 3$$

Fifth term $F_5 = F_3 + F_4$

$$= 2 + 3 = 5$$

So, n th term $F_n = F_{n-2} + F_{n-1}$.

Step 1: Write the statement in symbol.

$$P(n): F_{n+1}^2 - F_{n+1}F_n - F_n^2 = (-1)^n$$

Statement: " $\forall n \in \mathbf{N}, P(n)$ ".

Note that the predicate $P(n)$ is an equation where the left side of the equation is $F_{n+1}^2 - F_{n+1}F_n - F_n^2$ while the right side of the equation is $(-1)^n$.

$$P(n): F_{n+1}^2 - F_{n+1}F_n - F_n^2 = (-1)^n$$



Step 2: Use the outline of the principle of mathematical induction to prove the statement.

Proof:

$$1. P(1): F_2^2 - F_2F_1 - F_1^2 = -1.$$

To prove $P(1)$ is True, we must show that the left side of $P(1)$ is equal to the right side of $P(1)$.

$$\text{Left side: } F_2^2 - F_2F_1 - F_1^2 = 1 - 1(1) - 1 = -1.$$

$$\text{Right side: } -1.$$

Therefore, Left side = Right side.

Hence, $P(1)$ is True.

$$2. P(k): F_{k+1}^2 - F_{k+1}F_k - F_k^2 = (-1)^k$$

$$P(k+1): F_{k+2}^2 - F_{k+2}F_{k+1} - F_{k+1}^2 = (-1)^{k+1}$$

$$\text{Suppose } "F_{k+1}^2 - F_{k+1}F_k - F_k^2 = (-1)^k"$$

To prove $P(k+1)$, we must show that the left side of $P(k+1)$ is equal to the right side of $P(k+1)$.

$$\text{Left side of } P(k+1): F_{k+2}^2 - F_{k+2}F_{k+1} - F_{k+1}^2$$

$$\text{Right side of } P(k+1): (-1)^{k+1}$$

$$\begin{aligned} \text{Left side: } F_{k+2}^2 - F_{k+2}F_{k+1} - F_{k+1}^2 &= (F_k + F_{k+1})^2 - (F_k + F_{k+1})F_{k+1} - F_{k+1}^2 \\ &= F_k^2 + 2F_kF_{k+1} + F_{k+1}^2 - F_kF_{k+1} - F_{k+1}^2 - F_{k+1}^2 \end{aligned}$$

$$\begin{aligned}
&= F_k^2 + F_k F_{k+1} - F_{k+1}^2 \\
&= (-1)(-F_k^2 - F_k F_{k+1} + F_{k+1}^2) \\
&= (-1)(-1)^k \\
&= (-1)^{k+1} : \text{Right side}
\end{aligned}$$

Therefore, “ $F_{k+2}^2 - F_{k+2} F_{k+1} - F_{k+1}^2 = (-1)^{k+1}$ ”.

$P(1)$ is True.

$P(k) \rightarrow P(k+1)$ is True.

Therefore $P(n)$ is True for all $n \in \mathbf{N}$.

Example 7

Prove that $1 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbf{N}$.

Solution:

Similar to Example 5, we use principle of mathematical induction to prove the statement.

Example 8

Prove that the integer n is odd if and only if n^2 is odd.

Solution:

This statement is a bi-conditional.

Let p : n is odd

q : n^2 is odd

The statement can be written as $p \leftrightarrow q$. Statement $p \leftrightarrow q$ is a combination of two conditionals: $p \rightarrow q$ and $q \rightarrow p$. Thus, we must prove

1. If n is odd then n^2 is odd - using Direct Proof method
2. If n^2 is odd then n is odd - using Contrapositive Proof method

Exercises 1.2

1. Prove that $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for all positive integers n .
2. Prove by contradiction that, if $a < b$ then $a < b + 1$ for any integers a and b .

Solution

1. Similar to Example 5, use principle of mathematical induction to prove the statement.
2. Proof: Suppose $a < b$ and $a \geq b + 1$.

$$a < b \rightarrow a - b < 0$$

$$a \geq b + 1 \rightarrow a - b \geq 1 \text{ contradicts with } a - b < 0.$$

The contradiction shows that “if $a < b$ then $a < b + 1$ ” is True.

TUTORIAL 1

1. Write the negation for each of the following statement:
 - a) $\exists t \in \mathbf{Z}, 0 \leq t \leq 1$
 - b) I feel better today
 - c) The restaurant has neither the takeaway nor the dine-in menu
 - d) $\forall x \in \mathbf{Q}, \frac{1}{x-1} > 0$
2. Express each statement in symbolic form. Explain the symbols that you use.
 - a) If you are not willing to risk the usual, then you will have to settle for the ordinary
 - b) There is a prime number that is even
 - c) $\frac{1}{2y+1} < 0$ if and only if $y < -\frac{1}{2}$ for every real number y .
3. Determine whether the statement $[\sim p \wedge (p \rightarrow q)] \rightarrow \sim q$ is logically equivalent to $[[p \vee (q \rightarrow p)] \vee q$
4. Determine whether the statement $(p \wedge q) \vee (p \wedge r)$ is logically equivalent to $p \wedge (q \vee r)$
5. Use truth table(s) to verify that $p \leftrightarrow q$ is logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$
6. Prove the following theorems using direct proof method.
 - a) If n is an odd integer, $4 \leq n \leq 9$ then n is the sum of two prime numbers.
 - b) If x and y are both even integers, then the product xy is an even integer.
 - c) The sum of two odd integers is even.
7. Prove the following theorems using contrapositive proof method.
 - a) If $m + n$ is even, then m and n are either both odd or both even.
 - b) If $x > 0$ then $x^2 + x > 0$ for every real numbers x .
8. Prove the following theorems using proof by contradiction.

- a) If a number added to itself gives itself then the number is zero.
- b) The product of odd integers is never even.

9. Prove that $1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + 4 \cdot 6 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$ for all positive integers n .

10. Prove that $3^1 + 3^2 + 3^3 + \dots + 3^n = \frac{3^{n+1} - 3}{2}$ for all positive integers n .

TOPIC 2: FUNCTION AND GRAPH

2.1 Real numbers

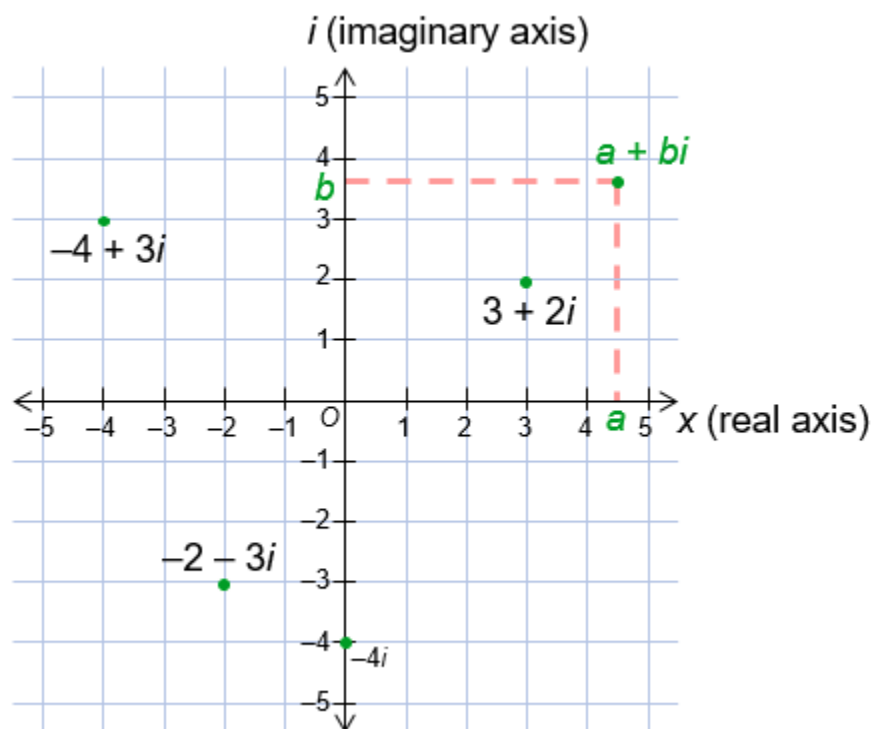
Objectives:

Students should be able to

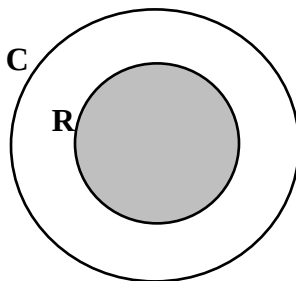
- identify types of intervals
- solve some linear inequalities
- solve some polynomial inequalities
- solve some inequalities involving absolute values

Real numbers are numbers that can be expressed as decimals. Set of real numbers is denoted as **R**. Complex numbers are numbers that can be expressed as $a + bi$ where a and b are real numbers while $i = \sqrt{-1}$. Complex numbers are points on an Argand Diagram:

Argand Diagram



Set of complex numbers is denoted as **C**. You may find more on set of complex numbers **C** from the internet. For example: <https://www.onlinemathlearning.com/complex-numbers-a5.html>. Set of real numbers **R** is a subset of **C** because, for any real number a , we can write $a = a + 0i$.

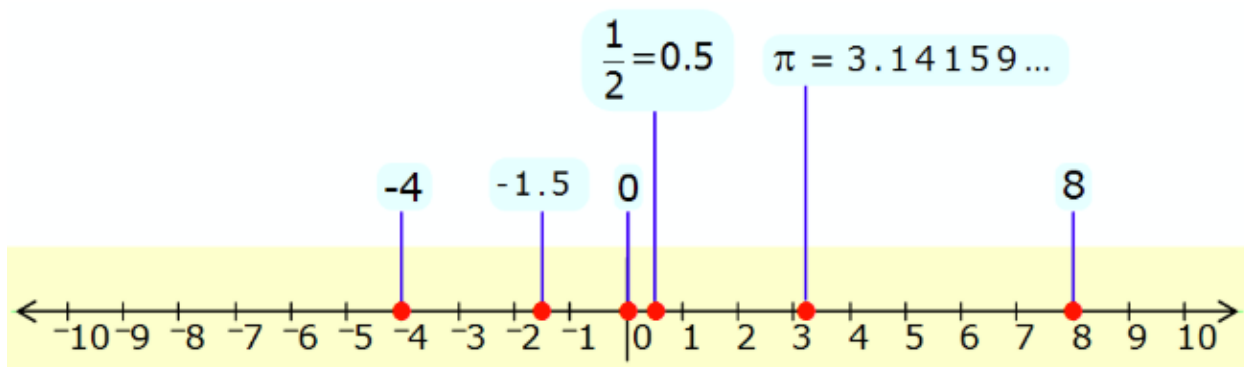


Properties of real numbers

- If a , b , and c are real numbers, then
- $a < b \rightarrow a + c < b + c$
- $a < b \rightarrow a - c < b - c$
- $a < b$ and $c > 0 \rightarrow ac < bc$
- $a < b$ and $c < 0 \rightarrow ac > bc$. Special case: $a < b \rightarrow -a > -b$.
- $a > 0 \rightarrow \frac{1}{a} > 0$
- If a and b are both positive or both negative, then $a < b \rightarrow \frac{1}{b} < \frac{1}{a}$.

Real number line

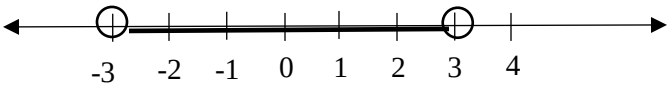
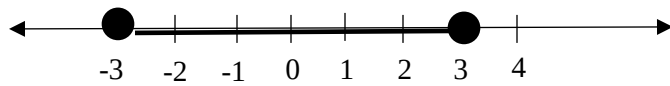
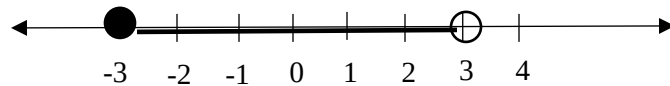
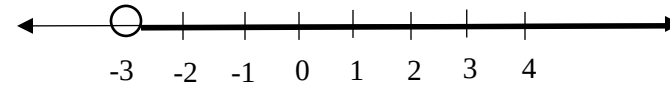
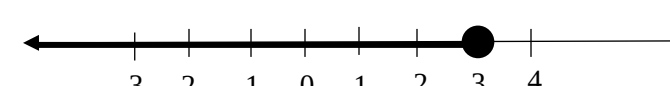
Any real number can be represented as a point on a real number line (or real line).



Interval

A subset of the real line is called an interval if it contains two real numbers or one real number and an infinity symbol, and contains all the real numbers lying between any two of its elements.

Types of interval:

Notation	Type	Illustration
(a, b)	Open interval	$(-3, 3)$ 
$[a, b]$	Closed interval	$[-3, 3]$ 
$[a, b)$	Half-open (or half-closed)	$[-3, 3)$ 
(a, ∞)	Open interval	$(-3, \infty)$ 
$(-\infty, b]$	Closed interval	$(-\infty, 3]$ 

Inequalities

Inequalities are mathematical expressions involving the symbols $>$, $<$, $\geq$ and $\leq$. To ‘solve’ an equality means to find values of an unknown x can take and still satisfy the inequality. We will solve inequalities using algebra.

Example 1 (linear inequalities)

Solve for x :

- a) $-2x \geq 4$
- b) $\frac{x+3}{4} < \frac{2x-4}{3}$
- c) $4 < 2x - 8 \leq 10$
- d) $3 + x > 7x - 2 \geq 5x - 10$

Solution:

a) $-2x \geq 4 \rightarrow x \leq -2$

b) $\frac{x+3}{4} < \frac{2x-4}{3}$
 $\frac{x+3}{4} - \frac{2x-4}{3} < 0$
 $\frac{3x+9-8x+16}{12} < 0$

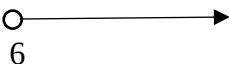
Since $12 > 0$, therefore $3x+9-8x+16 < 0$

$$-5x + 25 < 0$$

$$5 < x$$

c) $4 < 2x - 8 \leq 10$ means that $2x - 8 \leq 10$ and $2x - 8 > 4$.

AND

$2x - 8 \leq 10$	$2x - 8 > 4$
$2x \leq 18$	$2x > 12$
$x \leq 9$	$x > 6$
	
$6 < x \leq 9$	

d) $3 + x > 7x - 2 \geq 5x - 10$ means that $7x - 2 \geq 5x - 10$ and $3 + x > 7x - 2$.

Exercise: Fill in the following table.

AND

$7x - 2 \geq 5x - 10$	$3 + x > 7x - 2$
$\frac{5}{6} > x \geq -4$	

Example 2 (polynomial inequalities)

Solve for x . Write your answers in interval notation.

- a) $x^2 < -12 + 13x$
- b) $2x^2 \geq 3 - x$
- c) $9x^2 \geq 24x - 16$
- d) $2x^3 + 3x^2 - 3x < 2$

Solution:

- a) $x^2 < -12 + 13x$
 $x^2 - 13x + 12 < 0$
 $(x - 1)(x - 12) < 0$

	1	12	
Factor	$x < 1$	$1 < x < 12$	$x > 12$
$(x - 1)$	–	+	+
$(x - 12)$	–	–	+
	+	–	+

Answer: $x \in (1, 12)$.

- b) $2x^2 \geq 3 - x$
 $2x^2 + x - 3 \geq 0$
 $(2x + 3)(x - 1) \geq 0 \rightarrow (2x + 3)(x - 1) > 0$
 $\rightarrow (2x + 3)(x - 1) = 0$

Exercise: Fill in the following table.

	$-\frac{3}{2}$	1	
Factor	$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 1$	$x > 1$
$(2x + 3)$			
$(x - 1)$			

Answer for $(2x + 3)(x - 1) > 0$ is $x \in (-\infty, -\frac{3}{2}) \cup (1, \infty)$.

Answer for $(2x + 3)(x - 1) = 0$ is $x = -\frac{3}{2}$ or $x = 1$.

Answer: $x \in (-\infty, -\frac{3}{2}] \cup [1, \infty)$.

c) $9x^2 \geq 24x - 16$

$$9x^2 - 24x + 16 \geq 0$$

Exercise: Fill in the following space and table.

$$(\quad)(\quad) \geq 0 \rightarrow (\quad)(\quad) > 0$$

$$\rightarrow (\quad)(\quad) = 0$$

$$\frac{4}{3}$$

Factor		
	+	+

Answer for $(\quad)(\quad) > 0$ is $x \in (-\infty, \frac{4}{3}) \cup (\frac{4}{3}, \infty)$.

Answer for $(\quad)(\quad) = 0$ is $x = \frac{4}{3}$.

Answer: $x \in (-\infty, \infty)$.

d) $2x^3 + 3x^2 - 3x < 2$

$$2x^3 + 3x^2 - 3x - 2 < 0$$

To know how to factor any cubic equation, you may visit:

<https://youtu.be/vUNgcN6MbjA>

Cubic: $2x^3 + 3x^2 - 3x - 2$

a_n	a_0
1, 2	1, 2

		Factor of the last term: a_0	
		1	2
Factor of coefficient of $2x^3$: a_n	1	$\frac{a_0}{a_n} = \frac{1}{1} = 1$	$\frac{a_0}{a_n} = \frac{2}{1} = 2$
	2	$\frac{a_0}{a_n} = \frac{1}{2} = 0.5$	$\frac{a_0}{a_n} = \frac{2}{2} = 1$

All possible roots of the cubic are: ± 1 , ± 0.5 , and ± 2 .

Try $x = 1$.

Substitute $x = 1$ into the cubic to get

$$2(1^3) + 3(1^2) - 3(1) - 2 = 0$$

Thus, $x = 1$ is a root of the cubic.

Since $x = 1$ is a root, therefore, $x - 1$ is a factor of the cubic. We write:

$$2x^3 + 3x^2 - 3x - 2 = (x-1) \underbrace{(\quad)(\quad)}_{P(x)}$$

Long division:

$$\begin{array}{r}
 2x^2 + 5x + 2 \\
 x-1 \overline{) 2x^3 + 3x^2 - 3x - 2} \\
 \underline{-2x^3} \\
 0 3x^2 \\
 \underline{- -2x^2} \\
 5x^2 \\
 \underline{- 5x^2} \\
 0 -3x \\
 \underline{- -5x} \\
 2x \\
 \underline{- 2x} \\
 0 -2 \\
 \underline{- -2} \\
 0
 \end{array}$$

$$\begin{aligned}
 \text{So, } 2x^3 + 3x^2 - 3x - 2 &= (x-1)(2x^2 + 5x + 2) < 0 \\
 &= (x-1)(2x+1)(x+2) < 0
 \end{aligned}$$

$$\begin{array}{ccccc}
 & -2 & & -\frac{1}{2} & & 1
 \end{array}$$

Factor	$x < -2$	$-2 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < 1$	$x > 1$
$(x-1)$	-	-	-	+
$(2x+1)$	-	-	+	+
$(x+2)$	-	+	+	+
	$\ominus$	+	$\ominus$	+

Answer: $x \in (-\infty, -2) \cup (-\frac{1}{2}, 1)$.

Example 3 (*rational inequalities*)

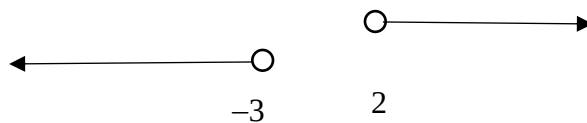
Solve for x . Write your answers in interval notation.

- a) $\frac{x+3}{x-2} < 0$
- b) $\frac{x^2+10x+25}{x^2-x-2} \leq 0$
- c) $\frac{x^2-3x-10}{1-x} \geq 2$
- d) $\frac{2}{x-1} < \frac{3}{x+7}$

Solution:

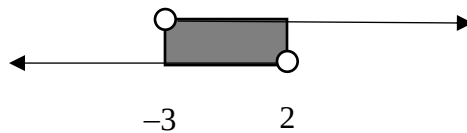
a) $\frac{x+3}{x-2} < 0 \Leftrightarrow x+3 < 0 \text{ and } x-2 > 0, \text{ or, } x+3 > 0 \text{ and } x-2 < 0.$

Case 1: $x+3 < 0$ and $x-2 > 0$
 $x < -3$ and $x > 2$.



No intersection.

Case 2: $x+3 > 0$ and $x-2 < 0$.
 $x > -3$ and $x < 2$.



Intersection: $x \in (-3, 2)$.

Answer: $x \in (-3, 2)$.

$$\begin{aligned} \text{b) } \frac{x^2 + 10x + 25}{x^2 - x - 2} \leq 0 &\rightarrow \frac{x^2 + 10x + 25}{x^2 - x - 2} < 0 \\ &\rightarrow \frac{x^2 + 10x + 25}{x^2 - x - 2} = 0 \end{aligned}$$

$$\frac{x^2 + 10x + 25}{x^2 - x - 2} < 0$$

$$x^2 + 10x + 25 = (x + 5)(x + 5) = (x + 5)^2$$

$$x^2 - x - 2 = (x - 2)(x + 1)$$

	-5		-1	2
Factor	$x < -5$	$-5 < x < -1$	$-1 < x < 2$	$x > 2$
$(x + 5)^2$	+	+	+	+
$(x - 2)$	-	-	-	+
$(x + 1)$	-	-	+	+
	+	+	-	+

$$\frac{x^2 + 10x + 25}{x^2 - x - 2} = 0 \text{ implies } (x + 5)^2 = 0 \text{ and hence, } x = -5.$$

Answer: $x \in (-1, 2) \cup \{-5\}$.

$$\begin{aligned} \text{c) } \frac{x^2 - 3x - 10}{1 - x} &\geq 2 \\ \frac{x^2 - 3x - 10}{1 - x} - 2 &\geq 0 \\ \frac{x^2 - 3x - 10 - 2(1 - x)}{1 - x} &\geq 0 \\ \frac{x^2 - 3x + 2x - 10 - 2}{1 - x} &\geq 0 \\ \frac{x^2 - x - 12}{1 - x} &\geq 0 \end{aligned}$$

$$\begin{aligned} \frac{x^2 - x - 12}{1 - x} \geq 0 &\rightarrow \frac{x^2 - x - 12}{1 - x} > 0 \\ &\rightarrow \frac{x^2 - x - 12}{1 - x} = 0 \end{aligned}$$

$$\frac{x^2 - x - 12}{1 - x} > 0$$

$$x^2 - x - 12 = (x + 3)(x - 4)$$

	-3		1	4
Factor	$x < -3$	$-3 < x < 1$	$1 < x < 4$	$x > 4$
$(x + 3)$	-	+	+	+
$(x - 4)$	-	-	-	+
$(1 - x)$	+	+	-	-
	$\oplus$	-	$\oplus$	-

$$\frac{x^2 - x - 12}{1 - x} = 0 \text{ implies } (x + 3)(x - 4) = 0 \text{ and hence, } x = 4 \text{ or } -3.$$

Answer: $(-\infty, -3] \cup (1, 4]$.

$$\begin{aligned}
 \text{d) } \frac{2}{x-1} &< \frac{3}{x+7} \\
 \frac{2}{x-1} - \frac{3}{x+7} &< \frac{3}{x+7} - \frac{3}{x+7} \\
 \frac{2}{x-1} - \frac{3}{x+7} &< 0 \\
 \frac{2(x+7) - 3(x-1)}{(x-1)(x+7)} &< 0 \\
 \frac{2x+14-3x+3}{(x-1)(x+7)} &< 0 \\
 \frac{-x+17}{(x-1)(x+7)} &< 0
 \end{aligned}$$

Exercise: Fill in the following table and write the answer.

	-7	1	17
Factor			
$(-x + 17)$			
$(x - 1)$			
$(x + 7)$			

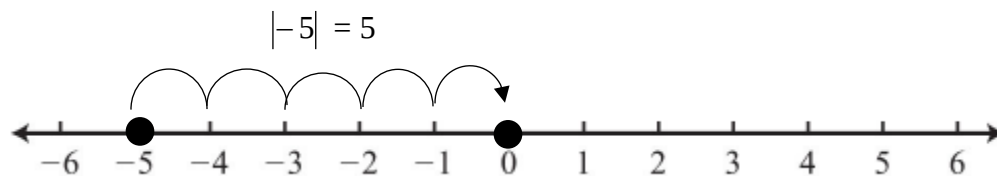
Answer: _____

Absolute value

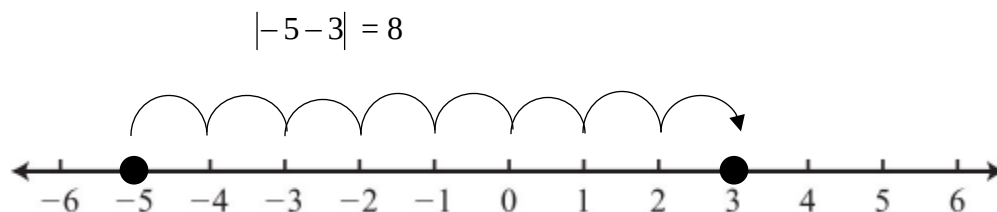
The absolute value of a number x , denoted by $|x|$, is defined by the formula

$$\sqrt{x^2} = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Remark: For geometric interpretation, $|x|$ is the distance between point x and point 0 on the real line. For example, $|-5|$ is the distance between point -5 and point 0.



In general, $|x - y|$ is the distance between point x and point y on the real line. For example, $|-5 - 3|$ is the distance between point -5 and point 3.



Triangle inequality on the real line:

$$|a + b| \leq |a| + |b|$$

$$|5 + (-6)| \leq |5| + |-6|$$

Properties of absolute values

If a is any positive number, then

- $|x| = a$ if and only if $x = \pm a$
- $|x| < a$ if and only if $-a < x < a$
- $|x| > a$ if and only if $x > a$ or $x < -a$
- $|x| \leq a$ if and only if $-a \leq x \leq a$
- $|x| \geq a$ if and only if $x \geq a$ or $x \leq -a$

Example 4 (inequalities involving absolute value)

Solve for x :

- a) $|2x - 3| < 5$
- b) $|5 - 6x| > 7$
- c) $x^2 < 2$
- d) $|x + 1| > 2|x - 3|$
- e) $|2x + 1| < 4x - 2$

$$f) \left| \frac{x-2}{x+1} \right| < 3$$

$$g) \frac{|x|+3}{|x|-3} > 0$$

Solution:

$$a) -5 < 2x - 3 < 5$$

$$\text{Answer: } -1 < x < 4.$$

$$b) 5 - 6x > 7 \text{ or } 5 - 6x < -7$$

$$\text{Answer: } x < -\frac{1}{3} \text{ or } x > 2.$$

$$c) x^2 < 2$$

$$x^2 - 2 < 0$$

Let $x^2 = 2$. We have

$$|x| = \sqrt{2}$$

$$x = \pm \sqrt{2}$$

	$-\sqrt{2}$		$\sqrt{2}$
Factor	$x < -\sqrt{2}$	$-\sqrt{2} < x < \sqrt{2}$	$x > \sqrt{2}$
$(x^2 - 2)$	+	⊖	+

$$\text{Answer: } -\sqrt{2} < x < \sqrt{2}$$

$$d) |x+1| > 2|x-3|$$

$$|x+1| = \begin{cases} x+1, & x+1 \geq 0 \\ -(x+1), & x+1 < 0 \end{cases}$$

$$|x-3| = \begin{cases} x-3, & x-3 \geq 0 \\ -(x-3), & x-3 < 0 \end{cases}$$

	-1		3
Absolute value	$x < -1$	$-1 < x < 3$	$x > 3$
$ x+1 $	$-(x+1) = -x-1$	$x+1$	$x+1$
$ x-3 $	$-(x-3) = 3-x$	$-(x-3) = 3-x$	$x-3$
	Case 1	Case 2	Case 3

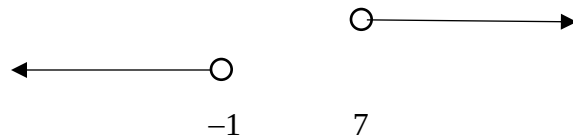
Case 1 ($x < -1$)

$$|x+1| > 2|x-3|$$

$$-(x+1) > 2(3-x)$$

$$-x - 1 > 6 - 2x$$

$$x > 7$$



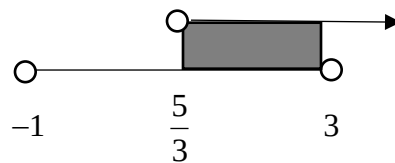
No intersection.

Case 2 ($-1 < x < 3$)

$$|x + 1| > 2|x - 3|$$

$$x + 1 > 2(3 - x)$$

$$x > \frac{5}{3}$$



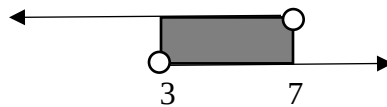
Intersection: $\frac{5}{3} < x < 3$.

Case 3 ($x > 3$)

$$|x + 1| > 2|x - 3|$$

$$x + 1 > 2(x - 3)$$

$$x < 7$$



Intersection: $3 < x < 7$.

Case $x = -1$.

$$|-1 + 1| > 2|-1 - 3|$$

$$0 > 8 \quad ? \quad 0 \not> 8.$$

Thus, $x = -1$ is not included in the answer.

Case $x = 3$.

$$|3 + 1| > 2|3 - 3|$$

$$4 > 0$$

Thus, $x = 3$ is included in the answer.

From Case 2, we have $\frac{5}{3} < x < 3$. In the interval notation: $x \in (\frac{5}{3}, 3)$. From Case 3, we have $3 < x < 7$ where in the interval notation: $x \in (3, 7)$. In addition, $x = 3$ is also included in the answer.

$$\text{Answer: } x \in (\frac{5}{3}, 3) \cup (3, 7) \cup \{3\} \leftrightarrow x \in (\frac{5}{3}, 7).$$

e) $|2x+1| < 4x-2$

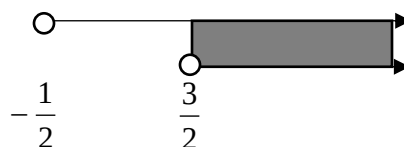
$$|2x+1| = \begin{cases} 2x+1, & 2x+1 \geq 0 \\ -(2x+1), & 2x+1 < 0 \end{cases}$$

Case 1 $\left(x > -\frac{1}{2}\right)$

$$|2x+1| < 4x-2$$

$$2x+1 < 4x-2$$

$$x > \frac{3}{2}.$$



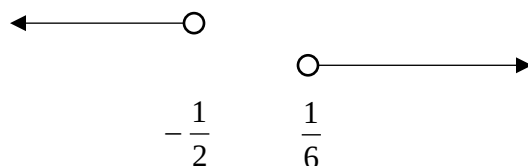
Intersection: $x > \frac{3}{2}$.

Case 2 $\left(x < -\frac{1}{2}\right)$

$$|2x+1| < 4x-2$$

$$-(2x+1) < 4x-2$$

$$x > \frac{1}{6}.$$



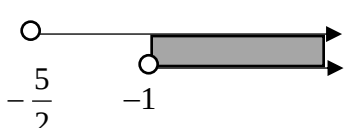
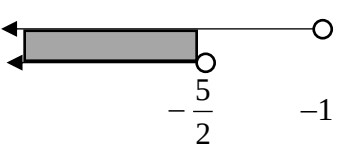
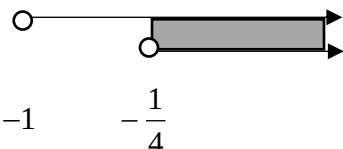
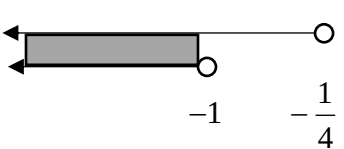
No intersection.

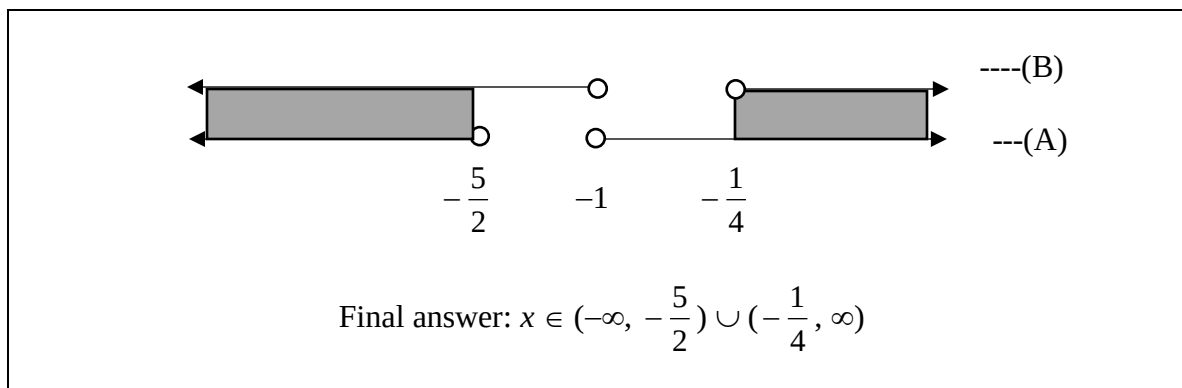
Answer: $x > \frac{3}{2}$

f) $\left| \frac{x-2}{x+1} \right| < 3$

$-3 < \frac{x-2}{x+1} < 3$ means that $\frac{x-2}{x+1} < 3$ and $\frac{x-2}{x+1} > -3$.

AND

$\frac{x-2}{x+1} < 3$	$\frac{x-2}{x+1} > -3$
$\frac{x-2}{x+1} - 3 < 0$ $\frac{x-2-3(x+1)}{x+1} < 0$ $\frac{-2x-5}{x+1} < 0$ Case 1: $-2x-5 < 0$ and $x+1 > 0$ $x > -\frac{5}{2}$ and $x > -1$  Intersection: $x > -1$. Case 2: $-2x-5 > 0$ and $x+1 < 0$ $x < -\frac{5}{2}$ and $x < -1$  Intersection: $x < -\frac{5}{2}$	$\frac{x-2}{x+1} + 3 > 0$ $\frac{x-2+3(x+1)}{x+1} > 0$ $\frac{4x+1}{x+1} > 0$ Case 1: $4x+1 > 0$ and $x+1 > 0$ $x > -\frac{1}{4}$ and $x > -1$  Intersection: $x > -\frac{1}{4}$. Case 2: $4x+1 < 0$ and $x+1 < 0$ $x < -\frac{1}{4}$ and $x < -1$  Intersection: $x < -1$
Answer: $x \in (-1, \infty) \cup (-\infty, -\frac{5}{2})$ ---(A)	Answer: $x \in (-\frac{1}{4}, \infty) \cup (-\infty, -1)$ ---(B)



g) $\frac{|x|+3}{|x|-3} > 0$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Case 1: $x \geq 0$

$$\frac{|x|+3}{|x|-3} > 0$$

$$\frac{x+3}{x-3} > 0$$

Case 2: $x < 0$

$$\frac{|x|+3}{|x|-3} > 0$$

$$\frac{-x+3}{-x-3} > 0$$

		-3		3
Case	Factor	$x < -3$	$-3 < x < 3$	$x > 3$
Case 1	$(x+3)$	-	+	+
	$(x-3)$	-	-	+
		(+)	-	(+)
Case 2	$(-x+3)$	+	+	-
	$(-x-3)$	+	-	-
		(+)	-	(+)

Answer: $x \in (-\infty, -3) \cup (3, \infty)$.

Exercises 2.1

1. Solve for x :

a) $|x - 4| = 5$

b) $|x + 5| = 4$

2. Solve the inequality

a) $3x^2 - 7x + 2 \geq 0$

b) $\frac{x+5}{(x-1)(x+2)} > 0$

c) $x \leq \frac{x+5}{x-3}$

3. Solve for x :

$$\left| \frac{x-1}{x+3} \right| \geq 2$$

Answer:

1. a) $x = 9$, or -1

b) $x = -1$, or -9

2. a) $x \leq \frac{1}{3}$, or $x \geq 2$

b) $-5 < x < -2$, or $x > 1$

c) $x \leq -1$, $3 < x \leq 5$

3. $-7 \leq x < -3$, $-3 < x \leq -\frac{5}{3}$

2.2 Functions and their graphs

Objectives:

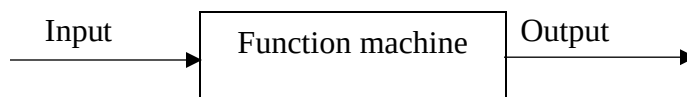
Students should be able to

- distinguish graphs that are functions and that are not functions
- determine whether a function is even or odd
- recognize essential functions
- find the domain and range if given a function
- sketch the graphs obtained by transformations of essential functions
- recognize injective functions

Function

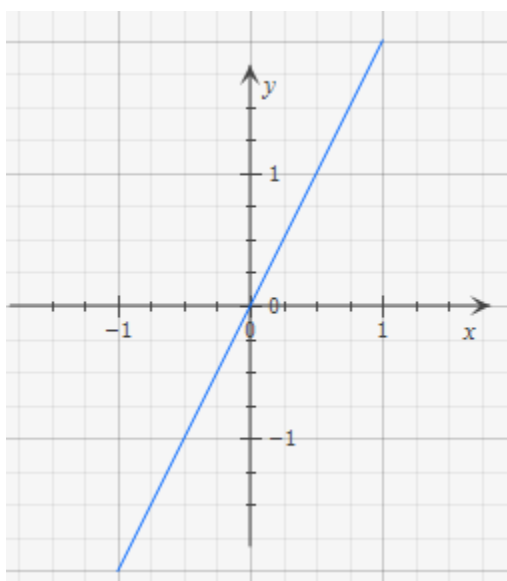
A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$ in a set E . The set D is called the domain of the function. The set E is called the co-domain.

Think of a function as a machine with x as the input while $f(x)$ as the output. Each input has one output.



The range of f is the set of all possible values of $f(x)$ as x varies throughout the domain. A symbol that represents a number in the domain of a function f is called an independent variable. A symbol that represents a number in the range of f is called a dependent variable.

Graph $f(x) = 2x$



Domain: $x \in \mathbf{R}$. Range: $y \in \mathbf{R}$.

Independent variable: x

Dependent variable: y

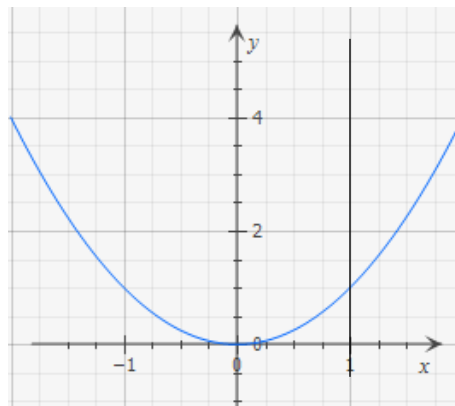
Vertical line test

A graph represents a function if and only if it passes the vertical line test; Every vertical line intersects the graph at most once. A graph that fail this test does not represent a function.

Example of a graph that is a function

Graph $y = x^2$ is a function because every vertical line intersects the graph at most once.

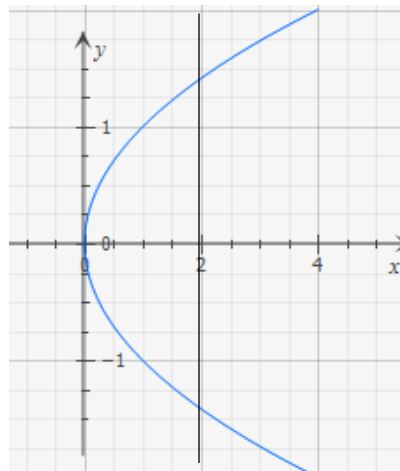
Graph $y = x^2$



Example of a graph that is not a function

Graph $x = y^2$ is not a function because there is a vertical line that intersect the graph more than once.

Graph $x = y^2$



Piecewise defined functions

Functions defined by different formulas in different parts of their domains

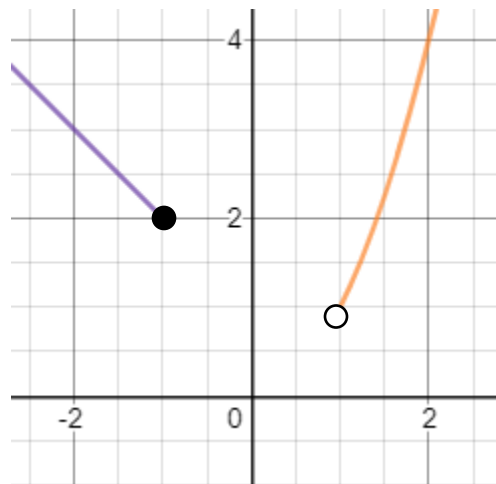
Example (piecewise defined function)

A function f is defined by

$$f(x) = \begin{cases} 1 - x, & \text{if } x \leq -1 \\ x^2, & \text{if } x > 1 \end{cases}$$

Evaluate $f(-2)$, $f(-1)$ and $f(0)$ and sketch the graph.

Solution:



$$f(-2) = 3.$$

$$f(-1) = 2.$$

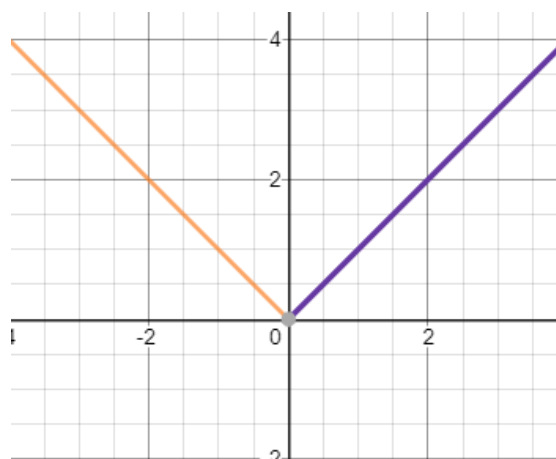
$$f(0) = \text{undefined}.$$

Example 1

Sketch the graph of the absolute value function $f(x) = |x|$.

Solution:

$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Graph $f(x) = |x|$.

Symmetry of a graph

If a graph does not change when reflected over a line or rotated around a point, the graph is symmetric with respect to that line or point. There are three types of symmetry:

1. Symmetric with respect to the y-axis
A graph is symmetric with respect to the y-axis if, whenever (x, y) is a point on the graph, then $(-x, y)$ is also a point on the graph.
2. Symmetric with respect to the x-axis
A graph is symmetric with respect to the x-axis if, whenever (x, y) is a point on the graph, then $(x, -y)$ is also a point on the graph.
3. Symmetric with respect to the origin
A graph is symmetric with respect to the origin if, whenever (x, y) is a point on the graph, then $(-x, -y)$ is also a point on the graph.

A function that is symmetric with respect to the y-axis is called an even function. A function that is symmetric with respect to the origin is called an odd function.

Testing even and odd functions

Even function

If $f(-x) = f(x)$ for every number x in the domain D , then the function f is even.

Example 2

Determine whether $f(x) = x^2$ is an even function.

Solution:

$$\begin{aligned}
 f(-x) &= (-x)^2 \\
 &= -x \cdot -x \\
 &= x^2 \\
 &= f(x)
 \end{aligned}$$

Therefore, $f(x) = x^2$ is an even function.

Odd function

If $f(-x) = -f(x)$ for every number x in the domain D , then the function f is odd.

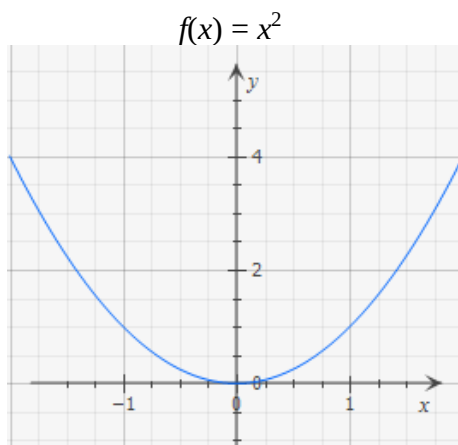
Example 3

Determine whether $f(x) = x^3$ is an odd function.

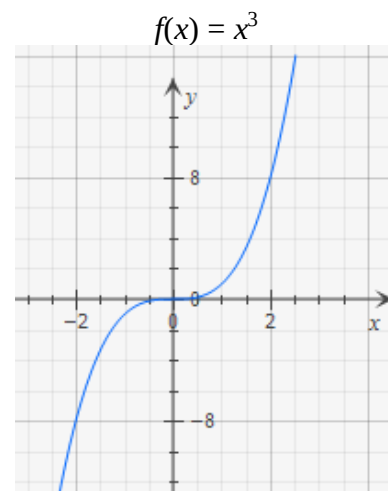
Solution:

$$\begin{aligned}
 f(-x) &= (-x)^3 \\
 &= -x \cdot -x \cdot -x \\
 &= -x^3 \\
 &= -f(x)
 \end{aligned}$$

Therefore, $f(x) = x^3$ is an odd function.



Graph of an even function is symmetric with respect to the y-axis.



Graph of an odd function is symmetric about the origin.

Example 4

Determine whether each of the following functions is even, odd or neither even nor odd.

- a) $f(x) = x^5 + x$
- b) $g(x) = 1 - x^4$
- c) $h(x) = 2x - x^2$

Answer:

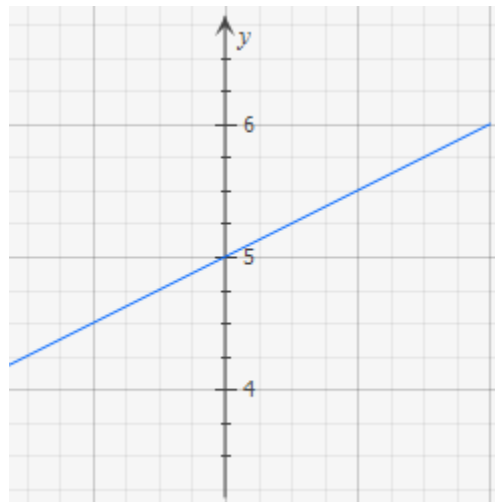
- a) Odd function
- b) Even function
- c) Neither even nor odd

Essential functions

1. Linear functions

Formula: $y = mx + c$ where m is the slope and c is the y-intercept

Graph $y = \frac{1}{2}x + 5$



2. Polynomials

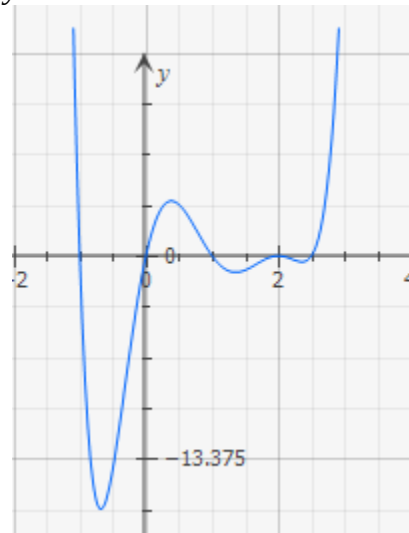
A polynomial function in variable x is a function of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

where n a nonnegative integer, and the numbers $a_0, a_1, \dots, a_n$ are constants called the coefficients of the polynomial. The degree of a polynomial in x is the highest power of x in the polynomial.

$f(x) = 2x^6 - 13x^5 + 26x^4 - 7x^3 - 28x^2 + 20x$ is a polynomial of degree 6.

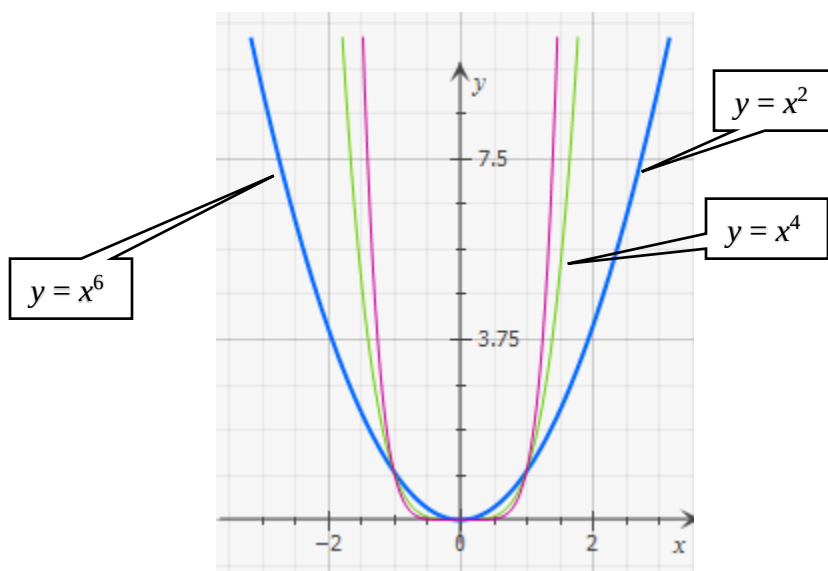
Graph $y = 2x^6 - 13x^5 + 26x^4 - 7x^3 - 28x^2 + 20x$



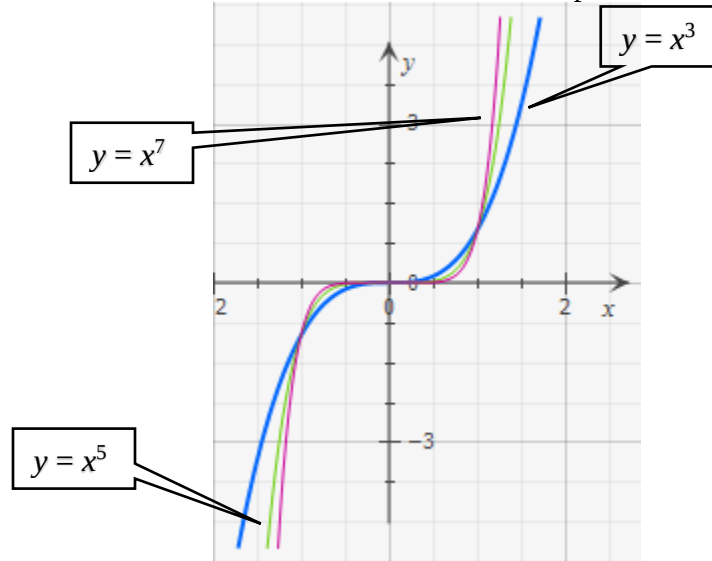
3. Power functions

Power functions are functions of the form $y = x^a$ where a is a constant.

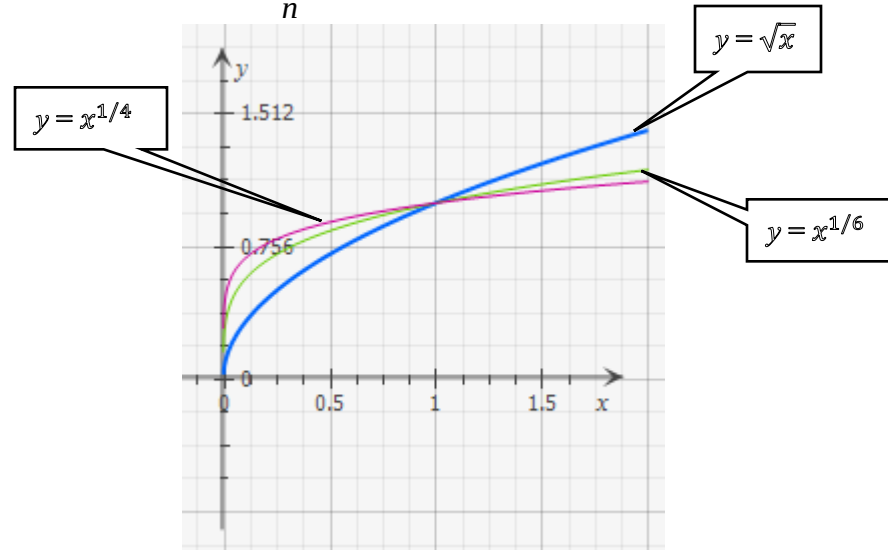
Case $a = n$ where n is even and positive



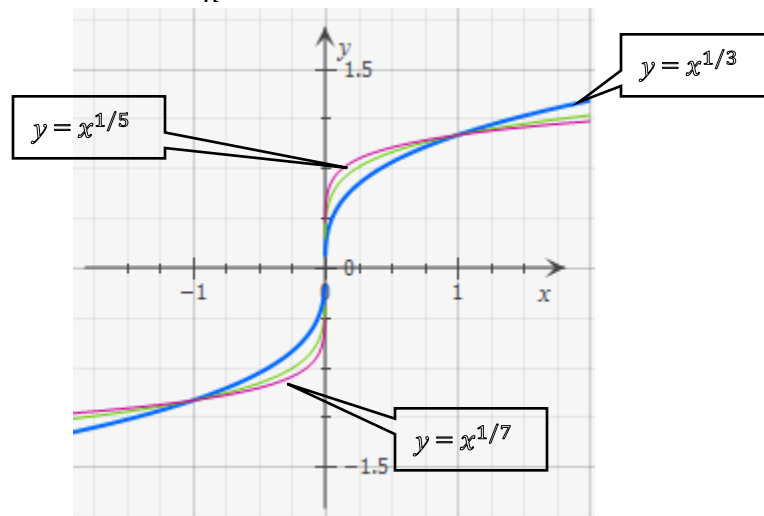
Case $a = n$ where n is odd and positive



Case $a = \frac{1}{n}$ where n is even and positive.

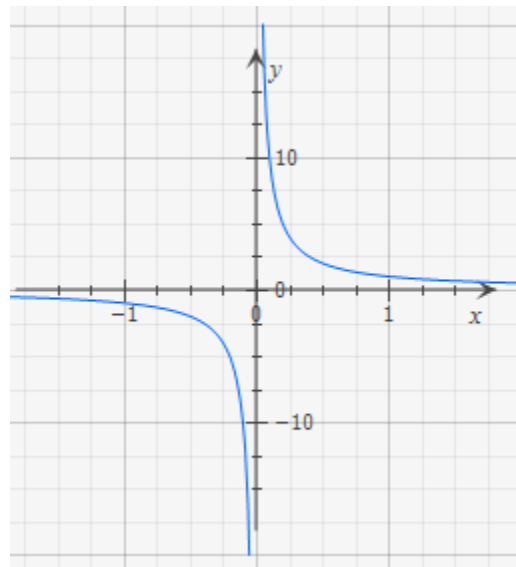


Case $a = \frac{1}{n}$ where n is odd and positive.



Case $a = -1$.

Reciprocal function $y = \frac{1}{x}$.



4. Rational functions

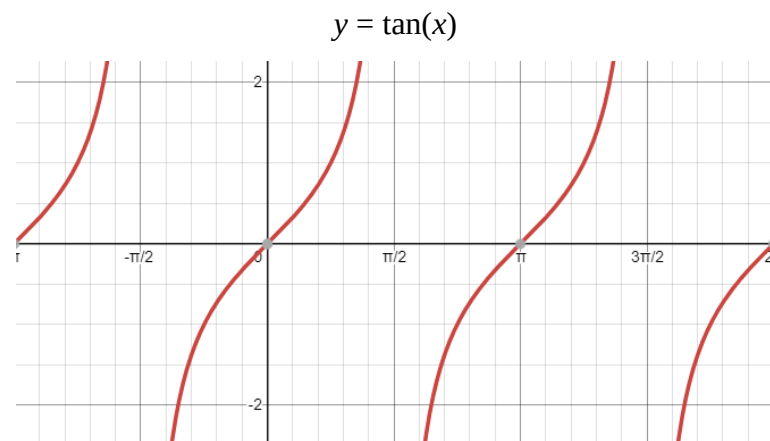
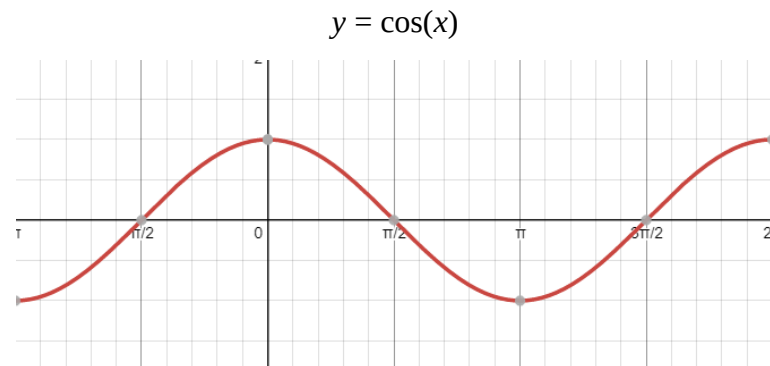
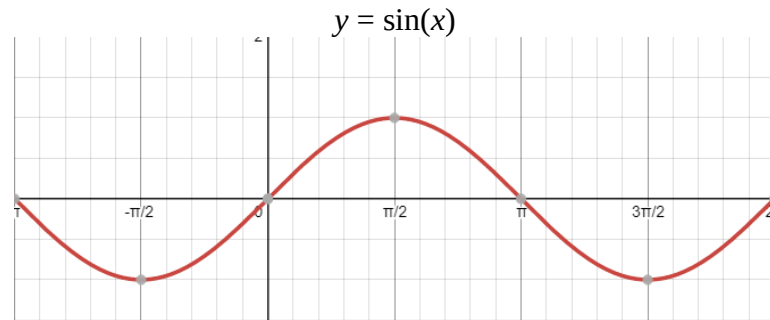
Rational functions are ratio of two polynomials.

$$y = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

5. Algebraic functions

A function f is called an algebraic function if it can be constructed using algebraic operations (addition, subtraction, multiplication, division and taking roots) starting with polynomials. Any rational function is an algebraic function.

6. Trigonometric functions



7. Exponential functions

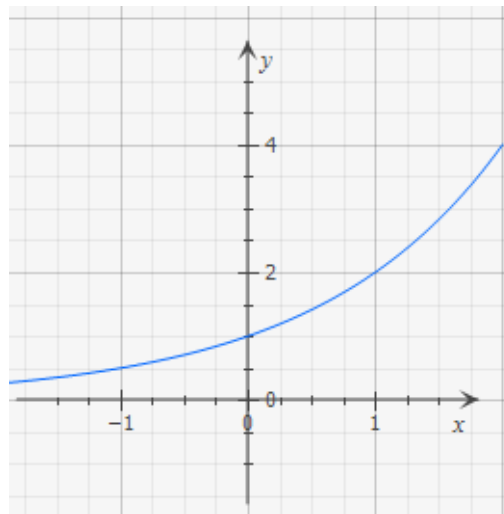
Formula: $y = b^x$.

Case $b > 1$

Properties of the graph:

- The graph passes through the point (0,1)
- The domain is all real numbers
- The range is $y > 0$
- The graph is increasing
- The graph is asymptotic to the x-axis as x approaches negative infinity
- The graph increases without bound as x approaches positive infinity
- The graph is continuous
- The graph is smooth

$$y = 2^x$$

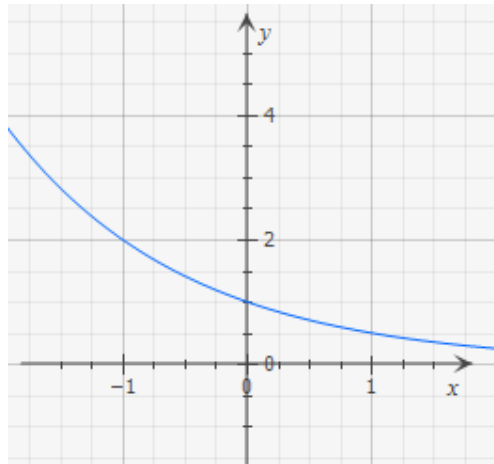


Case $b < 1$

Properties of the graph:

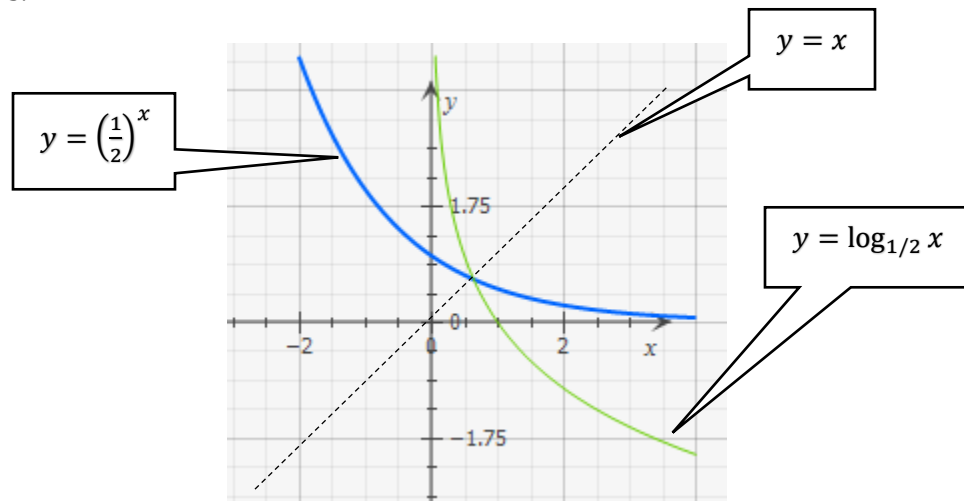
- The graph passes through the point (0,1)
- The domain is all real numbers
- The range is $y > 0$
- The graph is decreasing
- The graph is asymptotic to the x-axis as x approaches positive infinity
- The graph increases without bound as x approaches negative infinity
- The graph is continuous
- The graph is smooth

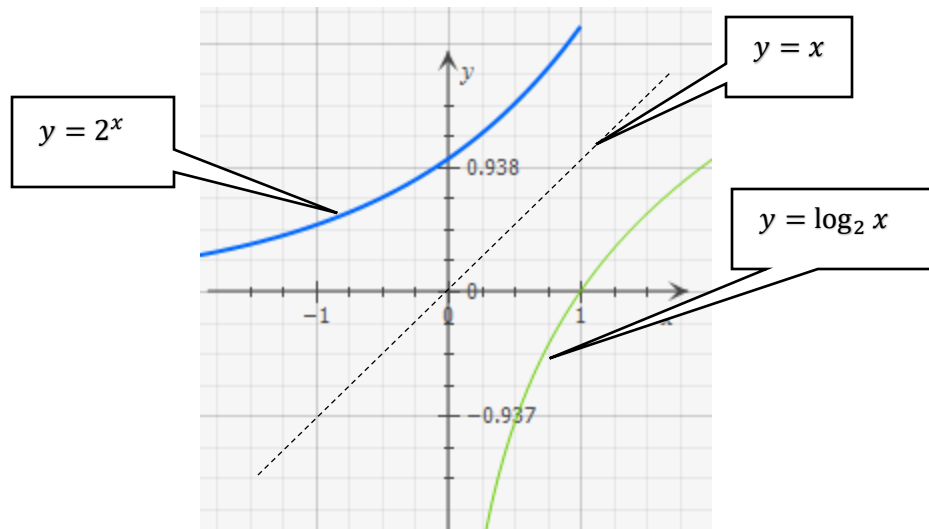
$$y = \left(\frac{1}{2}\right)^x$$



8. Logarithmic functions

Formula: $y = \log_b x$. Logarithmic functions are the inverse functions of the exponential functions.





Remark:

The mathematical constant e is often called Euler's number after Leonhard Euler (pronounced "Oiler"). The number e is approximately equal to 2.718281828459 and it is the base of the natural logarithm.

The natural exponential function $f(x) = e^x$ is the unique function which is equal to its own derivative and has initial value $f(0) = 1$, and one may define $e = f(1)$.

The natural logarithm of a number is its logarithm to the base of the mathematical constant e . The natural logarithm of x is generally written as $\ln x$, $\log_e x$, or sometimes, if the base e is implicit, simply $\log x$. The natural logarithm, or logarithm to base e , is the inverse function to the natural exponential function.

Example 5

Classify the following functions

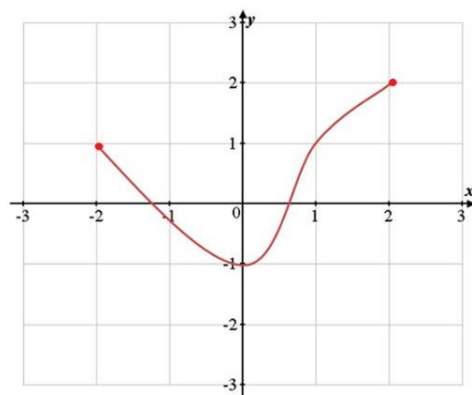
- a) $f(x) = 5^x$
- b) $g(x) = x^5$
- c) $h(x) = \frac{1+x}{1-\sqrt{x}}$
- d) $u(t) = 1 - t + 5t^4$

Answer:

- a) Exponential function
- b) Power function
- c) Algebraic function
- d) Polynomial

Example 6

Consider the following graph



Determine whether this graph represents a function. If so, find the domain and range.

Solution:

This graph is a function. Domain: $x \in [-2, 2]$. Range: $y \in [-1, 2]$.

Example 7

Determine the domain and range of the functions that follow.

- a) $y = x$
- b) $y = x^2$
- c) $y = \frac{1}{x-1}$
- d) $y = \sqrt{x}$
- e) $y = \frac{1}{\sqrt{x}}$

Solution:

- a) Domain: $x \in \mathbf{R}$. Range: $y \in \mathbf{R}$.
- b)

x	$y = x^2$
-2.5	6.25
-2	4
-1.5	2.25
-1	1
-0.5	0.25
0	0
0.5	0.25
1	1
1.5	2.25
2	4

2.5	6.25
-----	------

Domain: $x \in \mathbf{R}$.

Range: $y \in [0, \infty)$

c)

x	$y = 1 / (x - 1)$
-2.5	-0.2857143
-2	-0.3333333
-1.5	-0.4
-1	-0.5
-0.5	-0.666667
0	-1
0.5	-2
1	Undefined
1.5	2
2	1
2.5	0.6666667

Domain: $x \in \mathbf{R}$ except $x = 1$.

Range: $y \in \mathbf{R}$ except $y = 0$.

d)

x	$y = \sqrt{x}$
-2.5	Not a real number
-2	Not a real number
-1.5	Not a real number
-1	Not a real number
-0.5	Not a real number
0	0
0.5	0.7071068
1	1
1.5	1.224745
2	1.414214
2.5	1.581139

Domain: $x \in [0, \infty)$

Range: $y \in [0, \infty)$

e)

x	$y = 1/\sqrt{x}$
-2.5	Not a real number
-2	Not a real number
-1.5	Not a real number
-1	Not a real number
-0.5	Not a real number
0	Undefined
0.5	1.414214
1	1
1.5	0.8164966
2	0.7071068
2.5	0.6324555

Domain: $x \in (0, \infty)$.Range: $y \in (0, \infty)$.

Example 8

Determine the domain and range of the functions that follow.

f) $y = 4 - x^2$

g) $y = \sqrt{4 - x^2}$

h) $y = \sqrt{x^2 - 4}$

i) $y = \frac{1}{x^2 - 5x + 6}$

j) $y = x + \frac{x}{|x|}$

Solution:

f) Exercise: Fill in the following table.

x	$y = 4 - x^2$
-2.5	
-2	
-1.5	
-1	
-0.5	
0	
0.5	

1	
1.5	
2	
2.5	

Domain: $x \in \mathbf{R}$.

Range: $y \in [-\infty, 4]$

g) $y = \sqrt{4 - x^2}$

$$x^2 \leq 4 \rightarrow x^2 - 4 \leq 0$$

Let $x^2 = 4$. So

$$|x| = 2$$

$$x = \pm 2.$$

	-2	2	
y	$x < -2$	$-2 < x < 2$	$x > 2$
$y = \sqrt{4 - x^2}$	Undefined	$[0, 2]$	Undefined

Domain: $x \in [-2, 2]$

Range: $y \in [0, 2]$

Exercise: Fill in the following table

x	$y = \sqrt{4 - x^2}$
-2	0
-1.5	1.322876
-1	
-0.5	
0	
0.5	
1	
1.5	
2	

h) $y = \sqrt{x^2 - 4}$

$$x^2 \geq 4 \rightarrow x^2 - 4 \geq 0$$

Let $x^2 = 4$. So

$$|x| = 2$$

$$x = \pm 2.$$

	-2	2	
y	$x < -2$	$-2 < x < 2$	$x > 2$
$y = \sqrt{x^2 - 4}$	$[0, \infty)$	Undefined	$[0, \infty)$

Domain: $x \in (-\infty, -2] \cup [2, \infty)$

Range: $y \in [0, \infty)$

i) $y = \frac{1}{x^2 - 5x + 6}$
 $x^2 - 5x + 6 = (x - 3)(x - 2)$

	2	3	
y	$x < 2$	$2 < x < 3$	$x > 3$
$y = \frac{1}{x^2 - 5x + 6}$	$(0, \infty)$	$(-\infty, -4]$	$(0, \infty)$

Domain: $x \in (-\infty, 2) \cup (2, 3) \cup (3, \infty)$

Range: $y \in (-\infty, -4] \cup (0, \infty)$

$$x^2 - 5x + 6$$

$$x^2 - 5x + \left(-\frac{5}{2}\right)^2 - \left(-\frac{5}{2}\right)^2 + 6$$

$$\left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + 6$$

$$\left(x - \frac{5}{2}\right)^2 - \frac{25 + 24}{4}$$

x	$y = 1/(x^2 - 5x + 6)$
2	Undefined
2.2	-6.25
2.3	-4.76
2.4	-4.167
2.5	-4
2.6	-4.167
2.7	-4.76
2.8	-6.25
3	Undefined

$$\left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$$

$$y = \frac{1}{\left(x - \frac{5}{2}\right)^2 - \frac{1}{4}}$$

$$x = \frac{5}{2} = 2.5$$

$$y = \frac{1}{0 - \frac{1}{4}}$$

$$y = -4$$

j) $y = x + \frac{x}{|x|}$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

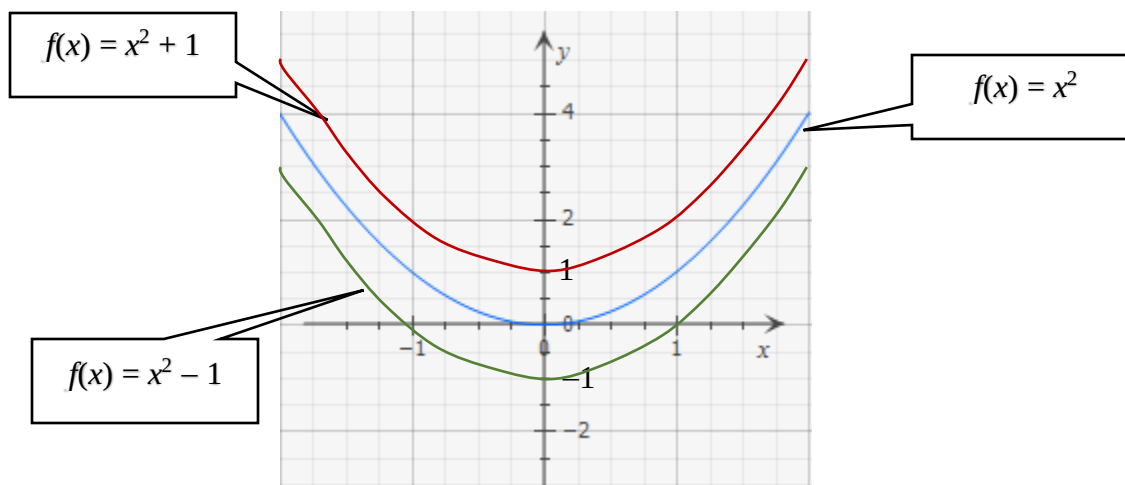
	0	
y	$x < 0$	$x > 0$
$y = x + \frac{x}{ x }$	$y = x + \frac{x}{-x} = x - 1$ $(-\infty, -1)$	$y = x + \frac{x}{x} = x + 1$ $(1, \infty)$

Domain: $x \in \mathbf{R}$ except $x = 0$

Range: $y \in (-\infty, -1) \cup (1, \infty)$

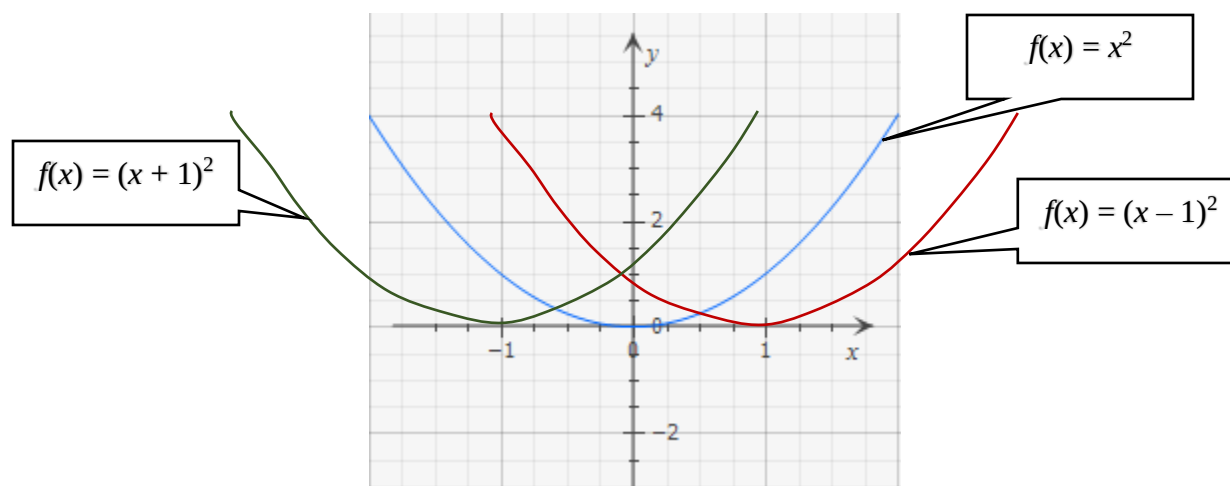
Transformations of functions

Let us start with a function $f(x) = x^2$. We can move it up or down by adding a constant to the y-value.



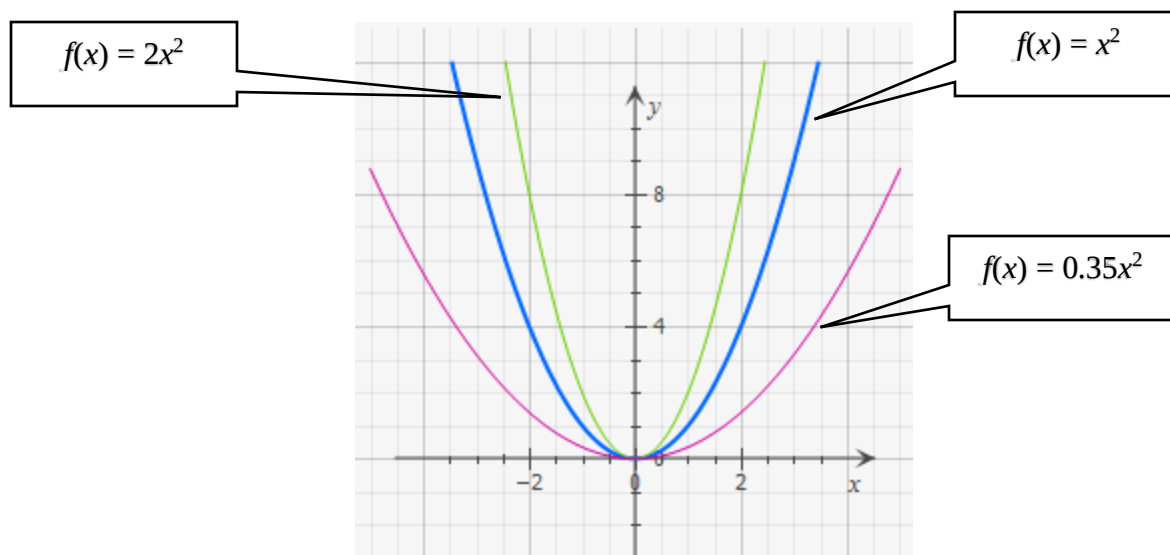
$f(x) = x^2 + c$ is just the graph of $f(x) = x^2$ shifted upward (or downward) a distance of c units.

We can move the graph of $f(x) = x^2$ to the left or right by adding a constant to the x-value.



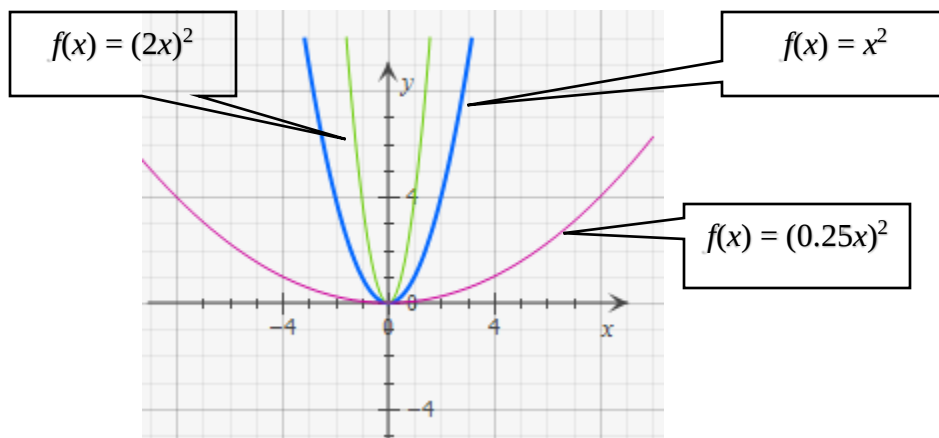
$f(x) = (x - c)^2$ is just the graph of $f(x) = x^2$ shifted to the left (or right) a distance of c units.

We can stretch or compress the graph of $f(x) = x^2$ vertically by multiplying the whole function by a constant c .



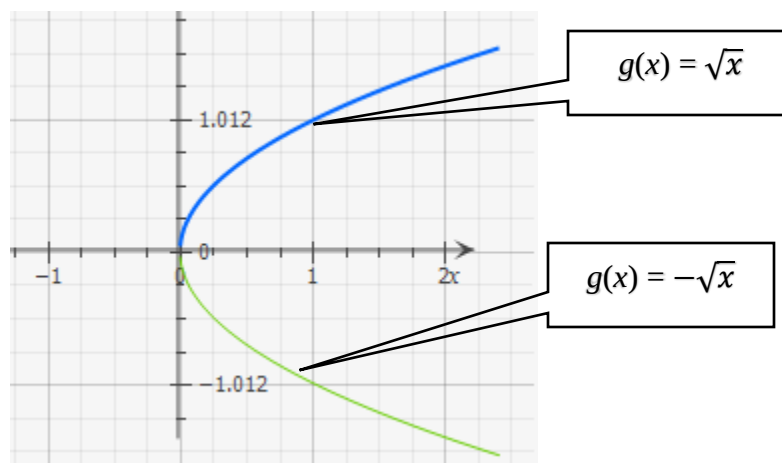
$c > 1$ stretches the graph vertically while $0 < c < 1$ compresses the graph vertically.

We can stretch or compress the graph of $f(x) = x^2$ horizontally by multiplying x by a constant c .



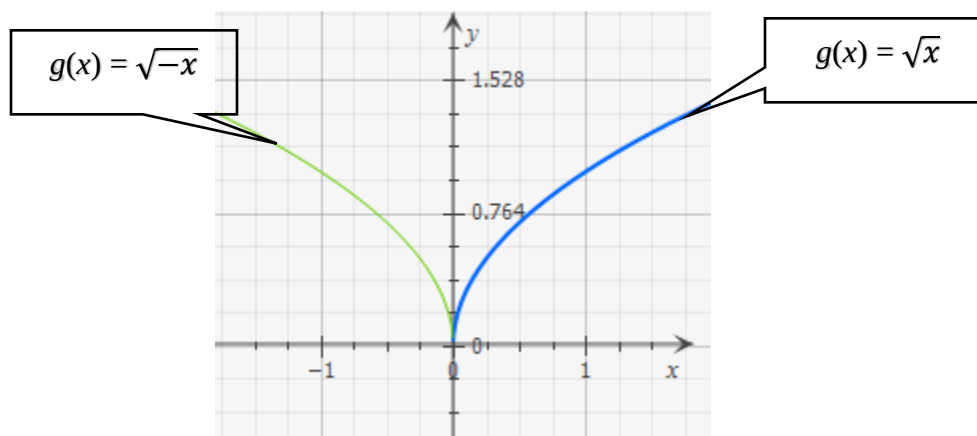
$c > 1$ compresses the graph horizontally while $0 < c < 1$ stretches the graph horizontally.

Consider the graph of $g(x) = \sqrt{x}$. We can flip it upside down by multiplying the whole function by -1 .



This is called the reflection about the x-axis.

We can flip the graph of $g(x) = \sqrt{x}$ left-right by multiplying the x-value by -1 .



This is called the reflection about the y-axis.

Function transformation rules:

$y = f(x) + c$	$c > 0$ moves it up $c < 0$ moves it down
$y = f(x + c)$	$c > 0$ moves it left $c < 0$ moves it right
$y = cf(x)$	$c > 1$ stretches it in the y-direction (vertically) $0 < c < 1$ compresses it
$y = f(cx)$	$c > 1$ compresses it in the x-direction (horizontally) $0 < c < 1$ stretches it
$y = -f(x)$	Reflects it about x-axis
$y = f(-x)$	Reflects it about y-axis

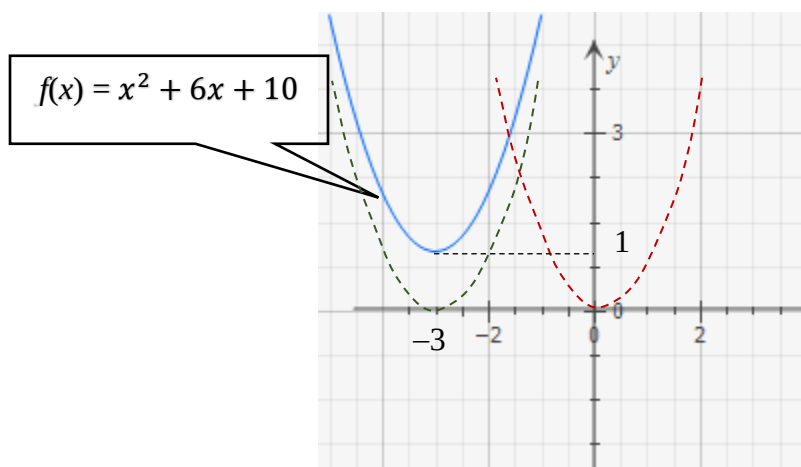
Example 9

Sketch the graph of the function $f(x) = x^2 + 6x + 10$.

Solution:

$$\begin{aligned}
 f(x) &= x^2 + 6x + 10 \\
 &= x^2 + 6x + \left(\frac{6}{2}\right)^2 - \left(\frac{6}{2}\right)^2 + 10 \\
 &= (x+3)^2 - 9 + 10 \\
 &= (x+3)^2 + 1
 \end{aligned}$$

Start with the graph of $f(x) = x^2$. Move the graph to the left a distance of 3 units. Move the graph upward a distance of 1 unit.



One to one functions

A function f is said to be one to one or injective if and only if $f(a) = f(b)$ implies $a = b$ for every a and b in the domain of f . A one-to-one function passes the horizontal line test; Every horizontal line intersects the graph at most once. If the horizontal line intersects the graph more than once, then the function is not one-to-one.

Example 10

Prove that $f(x) = \sqrt{x}$ from the set of real numbers to the set of real numbers, is one to one.

Solution:

Step 1: Assume that $f(a) = f(b)$.

$$f(a) = \sqrt{a}$$

$$f(b) = \sqrt{b}$$

$$f(a) = f(b)$$

$$\sqrt{a} = \sqrt{b}$$

Step 2: Prove that $a = b$.

$$\sqrt{a} = \sqrt{b}$$

$$a = b.$$

Example 11

Determine whether $f(x) = x^2$ from the set of real numbers to the set of real numbers, is one to one.

Solution:

Assume that $f(a) = f(b)$.

$$f(a) = a^2$$

$$f(b) = b^2$$

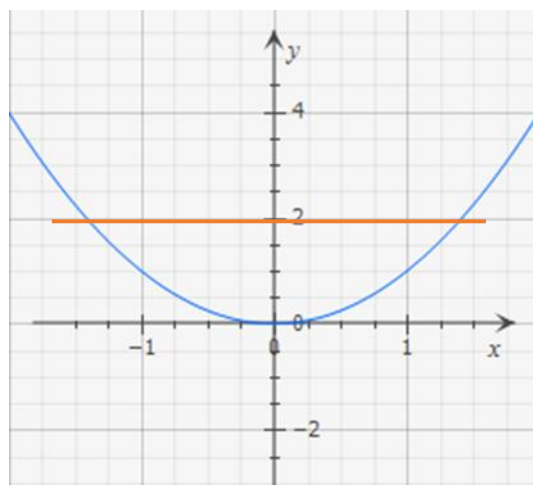
$$f(a) = f(b)$$

$$a^2 = b^2$$

$$a^2 - b^2 = 0$$

$$(a - b)(a + b) = 0$$

$$a = b \text{ or } a = -b$$



This shows that $f(a) = f(b)$ implies $a = b$ or $a = -b$. Thus, this function is not one-to-one.

Counter-example: $f(-1) = 1^2 = 1$

$$f(1) = 1^2 = 1$$

This function is not one-to-one because -1 and 1 are mapped to 1 .

Remark: A counterexample to a mathematical statement is an example that satisfies the statement's condition(s) but does not lead to the statement's conclusion. Identifying counterexamples is a way to disprove a mathematical statement or in other words, to show that a mathematical statement is false. In Example 11, the counter-example is given to disprove the following statement:

“ $f(x) = x^2$ is a one-to-one function for all real numbers x ”.

TUTORIAL 2

- Solve the following inequalities. Give your answer in the interval notation.
 - $5x - 2 \geq 4 + 3x$
 - $2x(x + 6) > 5 - x(x + 2)$
 - $|2x + 3| \geq |x - 4|$
 - $-\frac{3}{x-3} \leq |3 + x|$
- Determine whether a function f defined by $f(x) = \sin(x) - \cos^2(x)$ is even, odd or neither. Use “Desmos” online software (link: <https://www.desmos.com/calculator>) to draw the graph of f , then use the horizontal line test to determine whether this function is one-to-one.
- Determine the domain and range of
 - $g(x) = -2\sqrt{x+3} + 1$.
 - $g(x) = \frac{1}{2x^2 + x - 1}$
- Let $f(x) = e^x$. On the same axes, sketch the graph of $f(x)$, $f(x - 2)$ and $f(x) + 2$.
- Use transformations of functions to sketch the graph of h defined by $h(x) = 2x^2 + x - 1$.
- Use long division to factorize the cubic $p(x) = -x^3 - 2x^2 + x + 2$. Hence, find the values of x where $p(x) \geq 0$ and $p(x) < 0$. Give a counter-example to show that the statement

“ $p(x) = -x^3 - 2x^2 + x + 2$ is a one-to-one function for all real numbers x ”

 is false.
- Sketch the graph of the following function

$$f(x) = \begin{cases} \frac{1}{4}x^2 - 3, & x < 0 \\ x + 1, & 0 < x < 5 \\ (x - 3)^3 & x > 5 \end{cases}$$

- Express $g(x) = |x + 4| - |x - 4|$ in a piecewise-defined function form. Hence, sketch the graph.

9. Prove that the function h defined by $h(x) = e^{x+2} + 1$ is a one-to-one function. Hence, find h^{-1} and its domain.
10. Find the values of h and k if $(x + 1)$ and $(x + 2)$ are the factors of $x^3 + hx^2 + kx + 4$.

TOPIC 3: LIMIT AND CONTINUITY

3.1 LIMIT OF A FUNCTION

Objectives:

Students should be able to

- identify one-sided limits and infinite limits
- identify the application of limit to tangent and velocity problems
- calculate the limit of a function using limit laws
- calculate limits at infinity

Tangent and Velocity Problem

In this subtopic, we see how limits arise when we attempt to find the tangent to a curve or the velocity of an object. Tangent comes from Latin word “tangens” meaning touching.

Example 1.

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1,1)$.

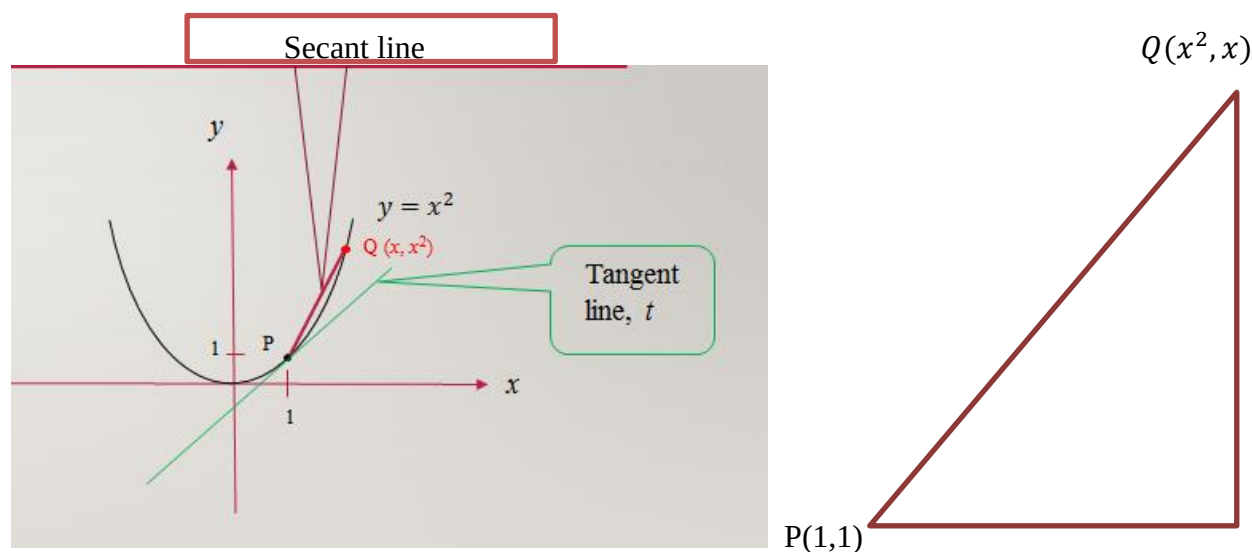
Solution.

Step 1: Find the slope of a secant line joining point $P(1,1)$ and point $Q(x, x^2)$

$$\text{Slope of the secant line, } m = \frac{x^2 - 1}{x - 1}.$$

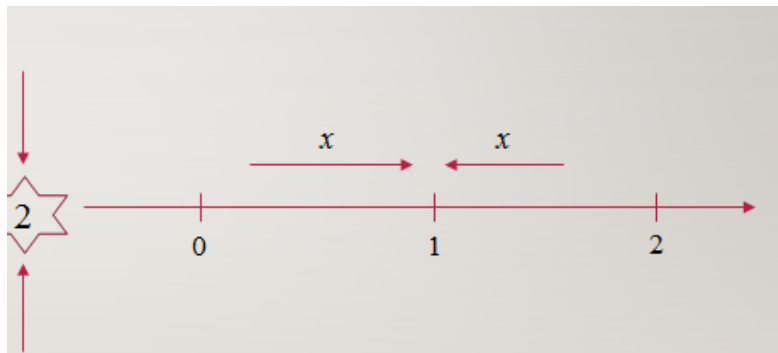
Step 2: Move point Q towards P

As Q approaches P along the parabola, the corresponding secant lines rotate about P and approach the tangent line t



Step 3: Find the limiting value of m

x	$\frac{x^2 - 1}{x - 1}$
0.9	1.9
0.99	1.99
1.01	2.01
1.001	2.001



Thus $m = 2$. Equation of the tangent line is $y = 2x - 1$.

Example 2.

Suppose that a ball is dropped from the upper observation deck of KLCC Tower in Kuala Lumpur, 450 m above the ground. Based on the Galileo's Law, the distance fallen after t seconds is given by

$$s(t) = 4.9t^2$$

Find the instantaneous velocity of the ball after 5 seconds.

Solution:

Instantaneous velocity when $t = 5$ is defined to be the limiting value of the average velocities over shorter and shorter time periods that start at $t = 5$.

Time interval	Average velocity (m/s)
$5 \leq t \leq 6$	53.9
$5 \leq t \leq 5.1$	49.49
$5 \leq t \leq 5.05$	49.245
$5 \leq t \leq 5.01$	49.049
$5 \leq t \leq 5.001$	49.0049

$$\begin{aligned}
 \text{Average velocity} &= \frac{\text{change in position}}{\text{time elapsed}} \\
 &= \frac{s(5.1) - s(5)}{0.1} \\
 &= \frac{4.9(5.1)^2 - 4.9(5)^2}{0.1} = 49.49 \text{ m/s}
 \end{aligned}$$

The instantaneous velocity $v = 49 \text{ m/s}$

CALCULATING LIMITS USING THE LIMIT LAWS

If L , M , c and k are real numbers and $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$ then

1. Sum Rule: $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$
2. Difference Rule: $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$
3. Product Rule: $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$
4. Quotient Rule: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}$, $M \neq 0$
5. Constant Multiple Rule: $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$

Power Rule: If r and s are integers with no common factor and $s \neq 0$, then $\lim_{x \rightarrow c} (f(x))^{r/s} = L^{r/s}$.

1. $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$ (Answer: 5)

$$\lim_{x \rightarrow 3} \frac{(x^2 - x - 6)}{x - 3} * \left(\frac{x + 3}{x + 3} \right) = \lim_{x \rightarrow 3} \frac{(x^2 - x - 6)(x + 3)}{(x - 3)(x + 3)} = \lim_{x \rightarrow 3} \frac{x^3 + 2x^2 - 9x - 18}{(x - 3)(x + 3)}$$

$$= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)(x + 2)}{(x - 3)(x + 3)} = \lim_{x \rightarrow 3} x + 2 = 3 + 2 = 5$$

2. $\lim_{h \rightarrow 0} \frac{\sqrt{3h+1}-1}{h}$ (Answer: 1.5)

$$\lim_{h \rightarrow 0} \frac{(\sqrt{3h+1}) - 1}{h} * \left(\frac{(\sqrt{3h+1}) + 1}{(\sqrt{3h+1}) + 1} \right) = \lim_{h \rightarrow 0} \frac{(3h+1) - 1}{h(\sqrt{3h+1} + 1)}$$

$$\lim_{h \rightarrow 0} \frac{3h}{h(\sqrt{3h+1} + 1)} = \lim_{h \rightarrow 0} \frac{3}{(\sqrt{3h+1} + 1)} = \frac{3}{\sqrt{0+1} + 1} = \frac{3}{2}$$

3. $\lim_{x \rightarrow -2} \sqrt{5x^2 - 4}$ (Answer: 4)

$$= \sqrt{5(-2)^2 - 4} = \sqrt{16} = 4$$

4. $\lim_{x \rightarrow 0} x \sin(x)$ (Answer: 0)

$$= 0 * \sin 0 = 0$$

$$5. \lim_{x \rightarrow 5} (2x^2 - 3x + 4) = \lim_{x \rightarrow 5} (2x^2) - \lim_{x \rightarrow 5} (3x) + \lim_{x \rightarrow 5} 4 \text{ (Answer: 39)} \\ = 2(5)^2 - 3(5) + 4 = 39$$

$$6. \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \frac{\lim_{x \rightarrow -2} (x^3 + 2x^2 - 1)}{\lim_{x \rightarrow -2} (5 - 3x)} = \frac{\lim_{x \rightarrow -2} x^3 + 2 \lim_{x \rightarrow -2} x^2 - \lim_{x \rightarrow -2} 1}{\lim_{x \rightarrow -2} 5 - 3 \lim_{x \rightarrow -2} x} \text{ (Answer: } -\frac{1}{11} \text{)} \\ = \frac{-2^3 + 2(-2)^2 - 1}{5 - 3(-2)} = -\frac{1}{11}$$

DEFINITION OF A LIMIT

Suppose $f(x)$ is defined when x is near the number a (This means that f is defined on some open interval that contains a , except possibly at a itself). Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

and say

“the limit of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily close to L (as close as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a .

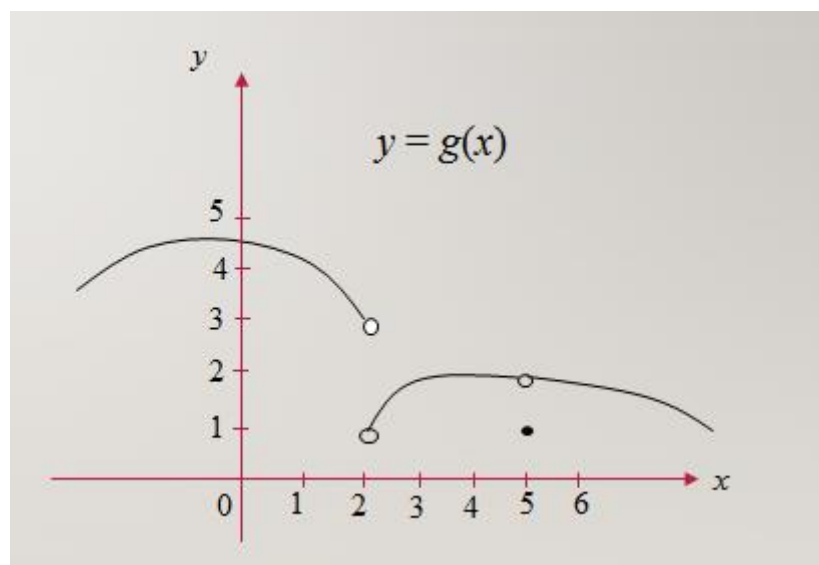
Definition of one-sided limits

- We write $\lim_{x \rightarrow a^-} f(x) = L$ and say the left-hand limit of $f(x)$ as x approaches a (or the limit of $f(x)$ as x approaches a from the left) is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x less than a .
- We write $\lim_{x \rightarrow a^+} f(x) = L$ and say the right-hand limit of $f(x)$ as x approaches a (or the limit of $f(x)$ as x approaches a from the right) is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x greater than a .

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Example 8

The graph of a function g is shown in Figure 1. Use it to state the values (if they exist) of the following $g(x)$

**Figure 1**

- (a) $\lim_{x \rightarrow 2^-} g(x)$ (b) $\lim_{x \rightarrow 2^+} g(x)$ (c) $\lim_{x \rightarrow 2} g(x)$ (d) $\lim_{x \rightarrow 5^-} g(x)$
 (e) $\lim_{x \rightarrow 5^+} g(x)$ (f) $\lim_{x \rightarrow 5} g(x)$

Answer: (c) limit does not exist

(f) $\lim_{x \rightarrow 5} g(x) = 1$

Example 9

Determine whether or not $\lim_{x \rightarrow 0} \frac{|x|}{x}$ exist.

Solution:

When $x \geq 0$, $|x| = x$

$x < 0$, $|x| = -x$

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \frac{x}{x} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \frac{-x}{x} = -1$$

Since $\lim_{x \rightarrow 0^+} \frac{|x|}{x} \neq \lim_{x \rightarrow 0^-} \frac{|x|}{x}$, thus limit does not exist.

DEFINITION OF AN INFINITE LIMIT

Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .

Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a , but not equal to a .

DEFINITION OF VERTICAL ASYMPTOTE

The vertical line $x = a$ is called a vertical asymptote of the curve $y = f(x)$ if at least one of the following statements is true.

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

Example 11

Find the vertical asymptotes of $f(x) = \tan x$

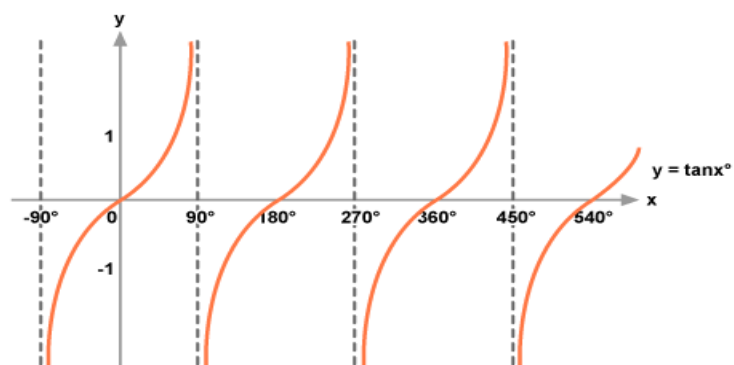
Solution:

$\tan x = \frac{\sin x}{\cos x}$ approaches to $\pm\infty$ when $\cos x$ approaches 0.

$\cos x = 0$ implies $x = 90, 270, 450, 540, \dots$

In general, x can be expressed as $x = \frac{\pi}{2} + n\pi$ (n is an integer).

This shows that lines $x = \frac{\pi}{2} + n\pi$ (n is an integer) are all vertical asymptotes of $f(x) = \tan x$



Example 13

Find the vertical asymptote of $f(x) = \frac{2x}{x-3}$.

Solution:

$$f(x) = \frac{2x}{x-3} \quad x \neq 3$$

$$\text{If } x - 3 = 0, \quad x = 3$$

$$\lim_{x \rightarrow 3} \frac{2x}{x-3} = \infty$$

Therefore, the function has vertical asymptote at $x = 3$.

COURSE MODULE: LOGICAL REASONING.

DEFINITION OF A LIMIT AT INFINITY (Positive Infinity)

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

x	$f(x) = \frac{x^2 - 1}{x^2 + 1}$
0	-1
1	0
2	0.6
3	0.8
4	0.882353
5	0.923077
10	0.980198
50	0.999200
100	0.999800
1000	0.999998

DEFINITION OF A LIMIT AT INFINITY (Negative Infinity)

Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.

x	$f(x) = \frac{x^2 - 1}{x^2 + 1}$
0	-1
-1	0
-2	0.6
-3	0.8
-4	0.882353
-5	0.923077
-10	0.980198
-50	0.999200
-100	0.999800
-1000	0.999998

Theorem

If $r > 0$ is a rational number, then $\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$

If $r > 0$ is a rational number such that x^r is defined for all x , then $\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$

Example

1. $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$ (Answer: $\frac{5}{3}$)

$$\lim_{x \rightarrow \infty} \frac{x^2 \left(5 + \frac{8}{x} - 3/x^2 \right)}{x^2 \left(3 + \frac{2}{x^2} \right)}$$

$$\lim_{x \rightarrow \infty} \frac{5 + \frac{8}{x} - 3/x^2}{3 + \frac{2}{x^2}} = \frac{5 + 0 - 0}{3 + 0} = 5/3$$

2. $\lim_{x \rightarrow \infty} \frac{x^2 - 3}{2x - 4}$ (Answer: ∞)

$$\lim_{x \rightarrow \infty} \frac{x^2 \left(1 - \frac{3}{x^2}\right)}{x \left(2 - \frac{4}{x}\right)}$$

$$\lim_{x \rightarrow \infty} x \left(\frac{1 - 0}{2 - 0}\right) = \lim_{x \rightarrow \infty} x \left(\frac{1}{2}\right) = \infty$$

3. $\lim_{x \rightarrow -\infty} \frac{6x + 1}{\sqrt{x^2 - 9}}$ (Answer: -6)

$$\lim_{x \rightarrow -\infty} \frac{\frac{6 + 1/x}{\sqrt{x^2}}}{\frac{\sqrt{x^2 - 9}}{\sqrt{x^2}}}$$

$$\lim_{x \rightarrow -\infty} \frac{\frac{6 + 1/x}{-x}}{\frac{\sqrt{x^2 - 9}}{-x}} = \lim_{x \rightarrow -\infty} \frac{x(6 + \frac{1}{x})}{\sqrt{-x^2(1 - \frac{9}{x^2})}} = \lim_{x \rightarrow -\infty} \frac{x(6 + \frac{1}{x})}{-x(1 - \frac{9}{x^2})} = \frac{(6 + 0)}{-(1 - 0)} = -6$$

4. $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16})$ (Answer: 0)

$$\lim_{x \rightarrow \infty} \frac{(x - \sqrt{x^2 + 16})(x + \sqrt{x^2 + 16})}{(x + \sqrt{x^2 + 16})}$$

$$\lim_{x \rightarrow \infty} \frac{x^2 - (x^2 + 16)}{(x + \sqrt{x^2 + 16})}$$

$$\lim_{x \rightarrow \infty} \frac{-16}{(x + \sqrt{x^2 + 16})}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{-16}{\sqrt{x^2}}}{\frac{x}{\sqrt{x^2}} + \sqrt{\frac{x^2 + 16}{x^2}}}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{-16}{x}}{1 + \sqrt{1 + \frac{16}{x^2}}} = 0$$

DEFINITION OF HORIZONTAL ASYMPTOTE

The line $y = L$ is called a horizontal asymptote of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \text{ or } \lim_{x \rightarrow -\infty} f(x) = L$$

Example

Determine the horizontal asymptote of the curve $f(x) = \frac{x^2-1}{x^2+1}$

Solution

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2-1}{x^2+1} = 1. \text{ Thus, the horizontal asymptote is the line } y = 1.$$

3.2 THE PRECISE DEFINITION OF A LIMIT

Recall definition of a limit

Suppose $f(x)$ is defined when x is near the number a . Then we write and say

“the limit of $f(x)$, as x approaches a , equate L ”

if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a but not equal to a .

Recall the concept of distance

Distance: Numerical description of how far apart objects are.

Distance between two points on a real line:

What is the distance between point -2 and 3?

Distance: $|-2 - 3| = 5$. Answer: the distance is 5.

Formula: Given two points x and y on a real line. The distance between x and y is $|x - y|$.

Precise definition of a limit

Assume that $f(x)$ exists for all x in some open interval containing a , except possibly at a . We say the limit of $f(x)$ as x approaches a is L , written

$$\lim_{x \rightarrow a} f(x) = L$$

if for any number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta$$

Example 1

Figure 1 shows the graph of a linear function f with $\lim_{x \rightarrow 3} f(x) = 5$

For each value of $\varepsilon > 0$, determine a value of $\delta > 0$ satisfying the statement

$$|f(x) - 5| < \varepsilon \text{ whenever } 0 < |x - 3| < \delta$$

a) $\varepsilon = 1$ b) $\varepsilon = \frac{1}{2}$

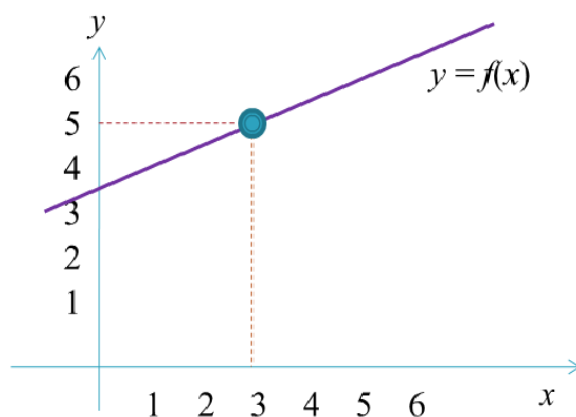


Figure 1

Solution

$$\varepsilon = 1$$

$$|f(x) - 5| < 1 \text{ whenever } 0 < |x - 3| < \delta$$

$$y = mx + c \text{ at } (0, 3), (3, 5)$$

$$\frac{5 - 3}{3 - 0} = \frac{2}{3}$$

$$y = \frac{2}{3}x + c$$

$$5 = \frac{2}{3}(3) + c,$$

$$c = 3$$

$$|f(x) - 5| < 1 \text{ whenever } 0 < |x - 3| < \delta$$

$$\left| \frac{2}{3}x + 3 - 5 \right| < 1 \text{ whenever } 0 < |x - 3| < \delta$$

$$\left| \frac{2}{3}x - 2 \right| < 1 \text{ whenever } 0 < |x - 3| < \delta$$

$$|2x - 6| = 2|x - 3| < 1 \Rightarrow |x - 3| < 1/2$$

which leads us to choose $\delta = 1/2$

$$\varepsilon = 1/2$$

$$|f(x) - 5| < \varepsilon \text{ whenever } 0 < |x - 3| < \delta$$

$$\left| \frac{2}{3}x + 3 - 5 \right| < 1/2 \text{ whenever } 0 < |x - 3| < \delta$$

$$\left| \frac{2}{3}x - 2 \right| < 1/2 \text{ whenever } 0 < |x - 3| < \delta$$

$$|2x - 6| = \frac{2}{3}|x - 3| < 1/2 \Rightarrow |x - 3| < 3/4$$

which leads us to choose $\delta = 3/4$

Example 2

By using definition of limit, prove that $\lim_{x \rightarrow 1} (5x - 3) = 2$

Step 1: Write in terms of $\varepsilon - \delta$

$$|5x - 3 - 2| < \varepsilon \text{ whenever } 0 < |x - 1| < \delta$$

$$|5x - 5| < \varepsilon \text{ whenever } 0 < |x - 1| < \delta$$

Step 2: Find δ

Let ε be a given positive number.

$$|5x - 5| = 5|x - 1| < \varepsilon \rightarrow |x - 1| < \frac{\varepsilon}{5}$$

which leads us to choose $\delta = \frac{\varepsilon}{5}$

Step 3: Write a proof

Let's now verify this guess. Let $\varepsilon > 0$ be any number and then choose $\delta = \frac{\varepsilon}{5}$. Next, assume that $0 < |x - 1| < \delta = \frac{\varepsilon}{5}$. We get the following

$$\begin{aligned} |5x - 5| &= 5|x - 1| \\ &< 5\left(\frac{\varepsilon}{5}\right) \\ &= \varepsilon \end{aligned}$$

Hence, we've shown that

$$|(5x - 5) - 2| < \varepsilon \text{ whenever } 0 < |x - 1| < \frac{\varepsilon}{5}$$

and by definition, we have

$$\lim_{x \rightarrow 2} 5x - 3 = 2$$

Example 3

By using definition of limit, prove that $\lim_{x \rightarrow 2} x^2 = 4$.

Solution

Step 1: Write in terms of $\varepsilon - \delta$

$$|x^2 - 4| < \varepsilon \text{ whenever } 0 < |x - 2| < \delta$$

Step 2: Find δ

Let ε be a given positive number.

$$|x^2 - 4| < \varepsilon$$

$$|(x - 2)(x + 2)| < \varepsilon$$

$$|x - 2||x + 2| < \varepsilon$$

Let $|x + 2| < C$.

We have $|x - 2||x + 2| < C|x - 2|$

Let $C|x + 2| < \varepsilon$. We have

$$\begin{aligned} |x - 2||x + 2| &< C|x - 2| < \varepsilon \\ |x - 2| &< \frac{\varepsilon}{C} \end{aligned}$$

Now, what is the value of C ?

We are interested in values of x that is closed to 2.

Assume that x is within a distance 1 from 2.

That is $|x - 2| < 1$

$$\begin{aligned} |x - 2| &< 1 \\ -1 &< x - 2 < 1 \\ 1 &< x < 3 \end{aligned}$$

This implies that

$$|x + 2| < 3 + 2 \Rightarrow |x + 2| < 5$$

Hence, $C = 5$.

Now, there are two restrictions on $|x - 2|$, namely

$$|x - 2| < 1 \quad \text{and} \quad |x - 2| < \frac{\varepsilon}{C} = \frac{\varepsilon}{5}$$

To make sure that both of these inequalities are satisfied, we take δ to be smaller of two numbers 1 and $\frac{\varepsilon}{5}$. The notation for this $\delta = \min\{1, \frac{\varepsilon}{5}\}$

Step 3: Write a proof

Let $\delta = \min\{1, \frac{\varepsilon}{5}\}$

If $0 < |x - 2| < \delta$, then

Case 1 ($\delta = 1$)

$$\begin{aligned} |x - 2| &< 1 \\ -1 &< x - 2 < 1 \\ 1 &< x < 3 \Rightarrow |x + 2| < 5 \end{aligned}$$

Case 2 ($\delta = \frac{\varepsilon}{5}$)

$$\begin{aligned} |x - 2| &< \frac{\varepsilon}{5} \\ |x - 2||x + 2| &< \frac{\varepsilon}{5} * 5 \\ |(x - 2)(x + 2)| &< \varepsilon \\ |x^2 - 4| &< \varepsilon \end{aligned}$$

Thus, if $0 < |x - 2| < \delta$, then $|x^2 - 4| < \varepsilon$.

COURSE MODULE: LOGICAL REASONING.

3.3 CONTINUITY

Objectives:

Students should be able to

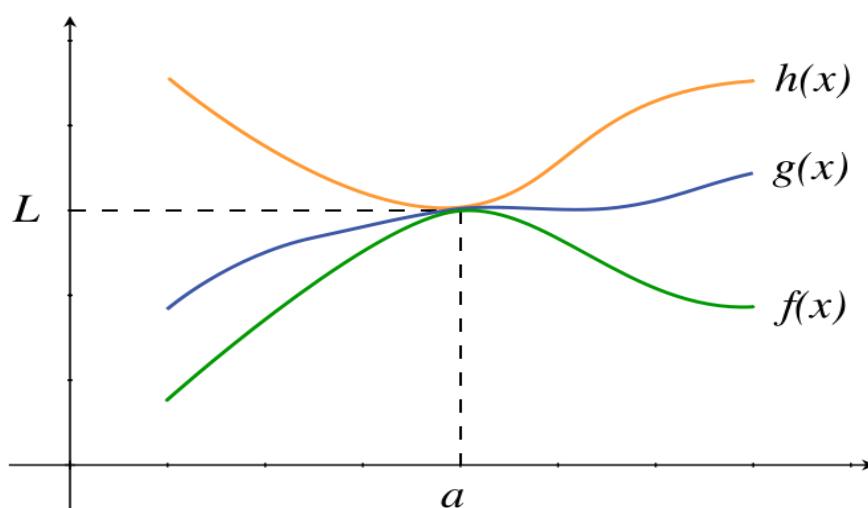
- Solve problem related to continuity.

THE SQUEEZE THEOREM

The Squeeze Theorem is also known as The Sandwich Theorem or The Pinching Theorem.

Assume the functions f , g and h satisfy $f(x) \leq g(x) \leq h(x)$ for all values of x near a (except possibly at a).

$$\text{If } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L \text{ then } \lim_{x \rightarrow a} g(x) = L.$$



Example 1

Show that $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

Solution

Since $\left|\sin\left(\frac{1}{x}\right)\right| \leq 1$ for all $x \neq 0$, we have that

$$|x| \left|\sin\left(\frac{1}{x}\right)\right| \leq |x|$$

$$\left|x \sin\left(\frac{1}{x}\right)\right| \leq |x|$$

This implies $-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$

$$-|x| \rightarrow 0 \text{ as } x \rightarrow 0$$

$$|x| \rightarrow 0 \text{ as } x \rightarrow 0$$

By the Squeeze Theorem, $x \sin\left(\frac{1}{x}\right) \rightarrow 0$ as $x \rightarrow 0$. Then $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

Example 2

Show that $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Solution

The limit $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$ cannot be determined through the limit law.

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

Because $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

However, by the definition of the sine function,

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

it follows that

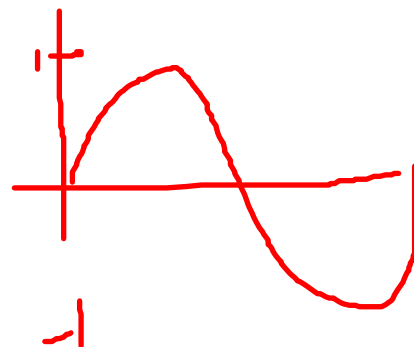
$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

Since

$$\lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$$

By the Squeeze Theorem,

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$



Continuous Function

- A function f is continuous at a number a if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

- f is continuous at a if $f(x)$ approaches $f(a)$ as x approaches a .
- It requires three things if f is continuous at a :
 - $f(a)$ is defined. (that is, a is in the domain of f)
 - $\lim_{x \rightarrow a} f(x)$ exists.
 - $\lim_{x \rightarrow a} f(x) = f(a)$

Example 3.

Figure 1 shows the graph of a function f .

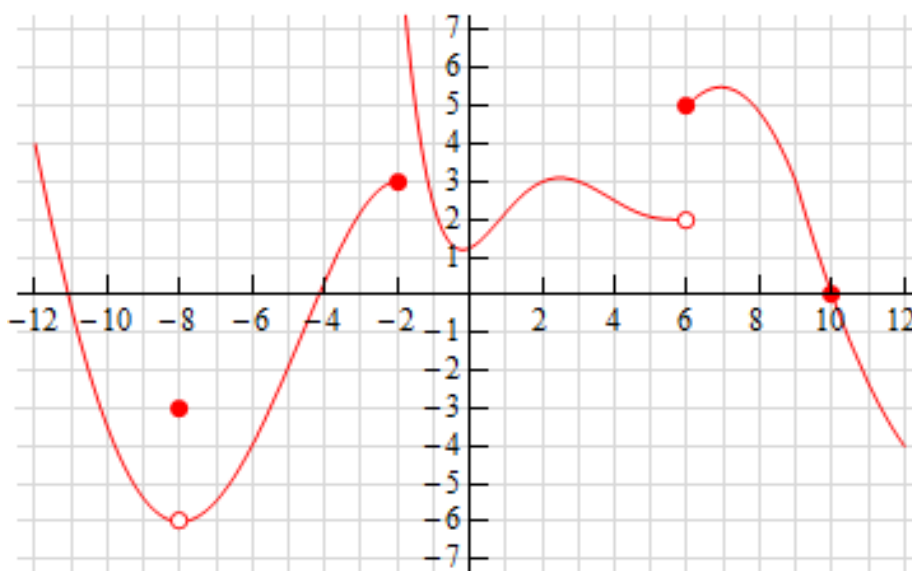


Figure 1

At which numbers is f discontinuous? Why?

Solution:

- i. The function f is discontinuous at $x = -8$ because

$$f(-8) = -3 \quad \lim_{x \rightarrow -8} f(x) = -6 \quad \lim_{x \rightarrow -8} f(x) \neq f(-8)$$

ii. The function f discontinuous at $x = -2$ because

$$f(-2) = 3 \quad \lim_{x \rightarrow -2} f(x) \text{ doesn't exist.}$$

The function value and the limit aren't the same and so the function is not continuous at this point.

iii. The function f discontinuous at $x = 6$ because

$$f(6) = 5 \quad \lim_{x \rightarrow 6} f(x) = 2 \quad \lim_{x \rightarrow 6} f(x) \neq f(6)$$

Example 4.

Find constants a and b so that the given function will be continuous for all x

$$f(x) = \begin{cases} ax + 3, & \text{if } x > 5 \\ 8, & \text{if } x = 5 \\ x^2 + bx + 1, & \text{if } x < 5 \end{cases}$$

Solution:

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} ax + 3 = 5a + 3 = 8$$

$$5a = 5$$

$$a = 1$$

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} x^2 + bx + 1 = 5^2 + b(5) + 1 = 8$$

$$26 + 5b = 8$$

$$5b = -18$$

$$b = -\frac{18}{5}$$

Example 5.

Determine whether the following functions are continuous at a . Justify your answer.

a. $f(x) = \frac{3x^2 + 2x + 1}{x - 1}; a = 1$

b. $g(x) = \frac{3x^2 + 2x + 1}{x - 1}; a = 2$

c. $h(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}; a = 0$

Solution

a. $f(1) = \frac{3(1)^2+2(1)+1}{1-1} = \frac{6}{0}$ is undefined.

Therefore, $f(x)$ is not continuous at $a = 1$.

b. $g(2) = \frac{3(2)^2+2(2)+1}{2-1} = 17$

$$\lim_{x \rightarrow 2} \frac{3x^2+2x+1}{x-1} = 17 = g(2)$$

Therefore, $g(x)$ is ~~not~~ continuous at $a = 2$.

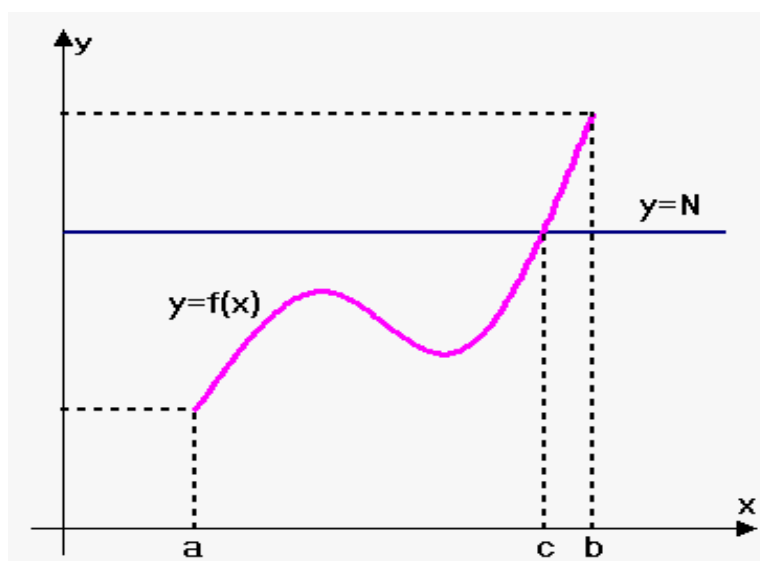
c. $h(0) = 0$

$$\lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right) = 0 = h(0)$$

By using squeeze Theorem to show that $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$. Therefore, $\lim_{x \rightarrow 0} h(x) = h(0)$, which implies that h is continuous at 0.

The Intermediate Value Theorem

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.



Example 5

The Intermediate Value Theorem says there is a number c between 1 and 2 such that $f(c) = N = 0$. Such c is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ in the interval $(1, 2)$. Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between $x = 1$ and $x = 2$.

Solution:

$$\text{Let } f(x) = 4x^3 - 6x^2 + 3x - 2$$

$$f(1) = 4 - 6 + 3 - 2 = -1 < 0$$

$$f(2) = 32 - 24 + 6 - 2 = 12 > 0$$

So, we have $f(1) < 0 < f(2)$; that is $N = 0$ is a number between $f(1)$ and $f(2)$.

TUTORIAL 3

LIMIT AND CONTINUITY

1. By using the definition of limit, prove that

a) $\lim_{x \rightarrow 3} (4x - 5) = 7$

b) $\lim_{x \rightarrow 3} x^2 = 9$

c) $\lim_{x \rightarrow 2} (x^2 + 2x - 7) = 1$

2. Evaluate the following limits.

a) $\lim_{x \rightarrow 5} (2x^2 - 3x + 4)$

b) $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$

c) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

d) $\lim_{x \rightarrow -3} \frac{x^2 + 3x}{x^2 - x - 12}$

e) $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x^2 + 2x - 3}$

3. Find the following limit, if it exist

a) $\lim_{x \rightarrow \infty} \frac{x^2 - 4x + 14}{x^3 + 9x^2}$

b) $\lim_{x \rightarrow -\infty} \frac{-8x^2 + 9x + 5}{-15x^2 + 7x + 14}$

c) $\lim_{x \rightarrow \infty} \frac{7x^3 - 3x^2 + 3x}{-x^3 - 2x + 7}$

4. Find the vertical asymptote for the following function

a) $f(x) = \frac{x-1}{x+5}$

b) $f(x) = \frac{2x}{x-3}$

c) $f(x) = \frac{3}{(4-x)^3}$

5. Determine the horizontal asymptote of the following curve

a) $f(x) = \frac{x^2-1}{x^2+1}$

b) $f(x) = \frac{3x^4+5x}{2x^4-3x^3+10}$

c) $f(x) = \frac{x+7}{3x+5}$

d) $f(x) = \frac{x^2-3x+7}{x^3+10x-4}$

6. Determine whether the following functions are continuous at a . Use the continuity checklist to justify your answer.

a) $f(x) = \frac{2x^2+3x+1}{x^2+5x}$, $a = 5$

b) $f(x) = \frac{2x^2+3x+1}{x^2+5x}$, $a = -5$

c) $f(x) = \sqrt{x-2}$, $a = 1$

d) $f(x) = \frac{1}{x-3}$, $a = 3$

$$\text{e) } f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & \text{if } x \neq 1 \\ 3 & \text{if } x = 1 \end{cases}; \quad a = 1$$

$$\text{f) } f(x) = \begin{cases} \frac{x^2 - 4x + 3}{x - 3} & \text{if } x \neq 3 \\ 2 & \text{if } x = 3 \end{cases}; \quad a = 3$$

$$\text{g) } f(x) = \frac{5x - 2}{x^2 - 9x + 20}, \quad a = 4$$

$$\text{h) } f(x) = \begin{cases} \frac{x^2 + x}{x + 1} & \text{if } x \neq -1 \\ 2 & \text{if } x = -1 \end{cases}; \quad a = -1$$

$$\text{i) } f(x) = \begin{cases} \frac{x^2 - 16}{x - 4} & \text{if } x \neq 4 \\ 9 & \text{if } x = 4 \end{cases}; \quad a = 4$$

$$\text{j) } f(x) = \sqrt{x^2 - 9}, \quad a = 9$$

$$\text{h) } f(x) = \begin{cases} \frac{x^2 - 16}{x - 4} & \text{if } x \neq 4 \\ 8 & \text{if } x = 4 \end{cases}; \quad a = 4$$

TOPIC 4: DIFFERENTIATION

4.1 Derivatives

Objectives:

Students should be able to

- recognize the secant line that approaches the tangent line of a curve
- use the definition of derivative to differentiate a given function
- sketch the graph of f' given the graph of f
- locate points on a graph of a function at which the function is not differentiable there

Tangents

Definition: The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

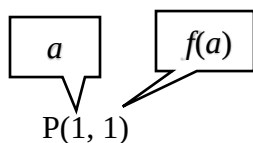
provided that this limit exists.

Remark: $\frac{f(x) - f(a)}{x - a}$ is the slope of the secant line connecting point $P(a, f(a))$ and point $Q(x, f(x))$. As x approaches a , the point Q approaches P and the secant line PQ approaches the tangent line of the curve at P .

Example 1

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Solution:



Slope of the tangent line is: $m = \lim_{x \rightarrow 1} \frac{f(x) - 1}{x - 1}$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1}$$

$$= 2$$

$$y = mx + c$$

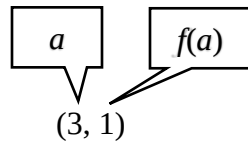
$$1 = 2(1) + c \rightarrow c = -1.$$

Equation of the tangent line is $y = 2x - 1$.

Example 2

Find an equation of the tangent line to the hyperbola $y = \frac{3}{x}$ at the point $(3, 1)$.

Solution:



Slope of the tangent line is: $m = \lim_{x \rightarrow 3} \frac{f(x) - 1}{x - 3}$

$$= \lim_{x \rightarrow 3} \frac{\frac{3}{x} - 1}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{\frac{3-x}{x}}{x-3}$$

$$= -\frac{1}{3}$$

$$y = mx + c$$

$$1 = -\frac{1}{3}(3) + c \rightarrow c = 2.$$

Equation of the tangent line is $y = -\frac{1}{3}x + 2$.

If $h = x - a$, then $x = a + h$ and so the slope of the tangent line is $m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

Definition: The derivative of a function f at a number a , denoted by $f'(a)$ is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Definition (derivative)

The derivative of a function f at x is $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ if this limit exists. For all x for which this limit exists, f' is a function at x .

Example 3

- a) By using definition of derivative, differentiate $f(x) = x^2$
- b) By using definition of derivative, differentiate $f(x) = \sqrt{x}$

Solution:

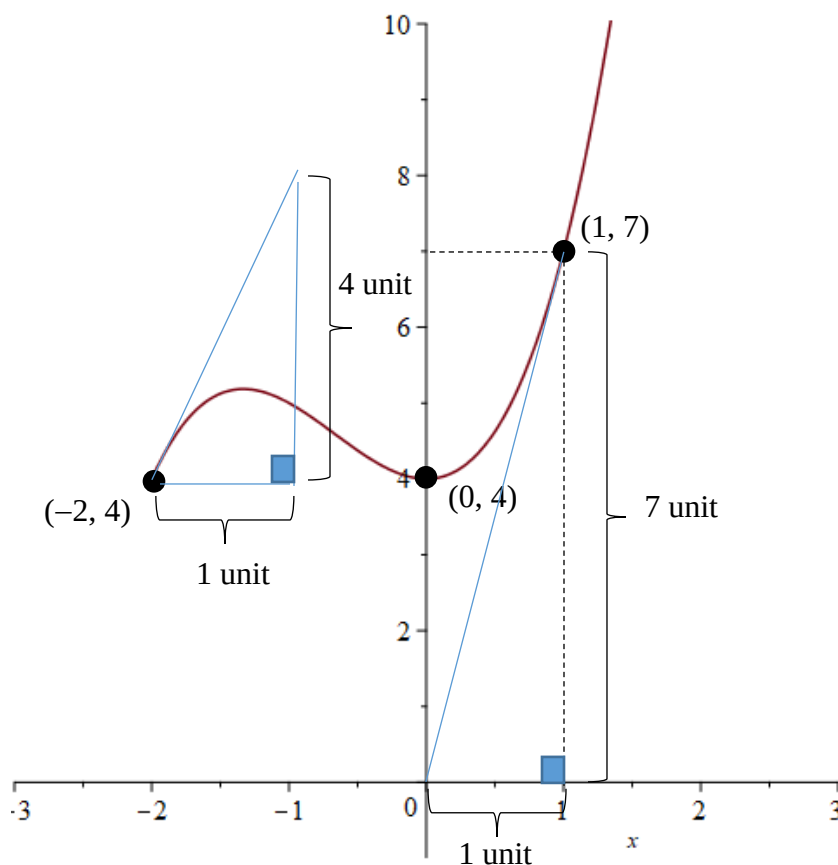
$$\begin{aligned} \text{a) } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x \end{aligned}$$

- b) Similar to a).

Example 4

The graph of a function $f(x) = x^3 + 2x^2 + 4$ is given. Sketch the graph of f' .

$$f(x) = x^3 + 2x^2 + 4$$



Solution:

Step 1: Choose several points on the graph of f (Refer the above graph).

Step 2: Determine the slope at each point in Step 1.

Slope at point $(0, 4)$ is 0

Slope at point $(-2, 4)$ is 4

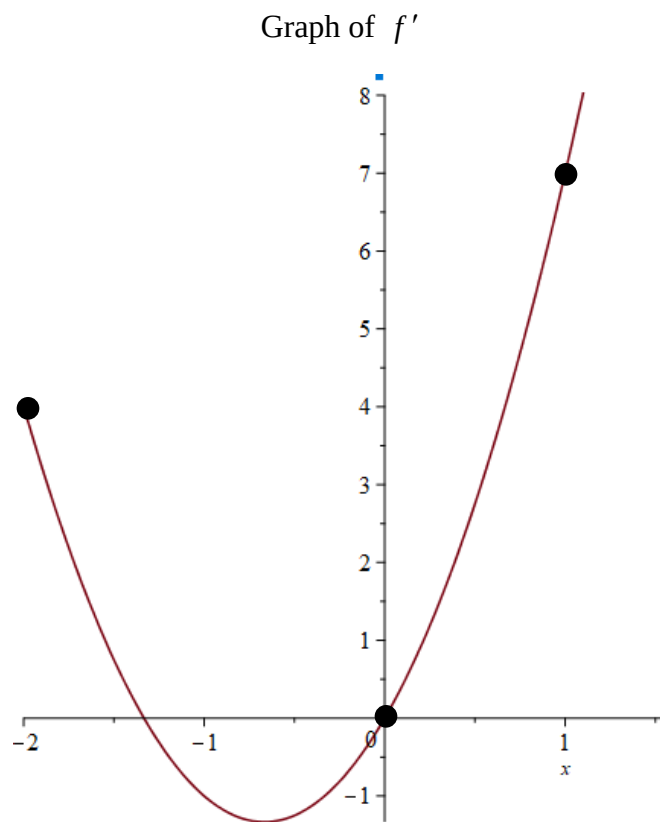
Slope at point $(1, 7)$ is 7

Step 3: Sketch the graph of f' .

x coordinate in the graph of $f \rightarrow x$ coordinate in the graph of f'

slope in the graph of $f \rightarrow y$ coordinate in the graph of f'

x	0	-2	1
y	0	4	7



Differentiability

Definition: A function f is differentiable at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Example 5

Where is the function $f(x) = |x|$ differentiable?

Solution:

Case 1 ($x > 0$)

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1 \end{aligned}$$

Since $f'(x)$ exists, therefore f is differentiable for any $x > 0$.

Case 2 ($x < 0$)

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1 \end{aligned}$$

Since $f'(x)$ exists, therefore f is differentiable for any $x < 0$.

Case 3 ($x = 0$)

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} \\ &= \lim_{h \rightarrow 0} \frac{|h|}{h} \text{ does not exist because the left hand limit is not equal to the right hand limit.} \end{aligned}$$

$$\text{Left hand limit} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

$$\text{Right hand limit} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

Therefore, f is not differentiable at $x = 0$.

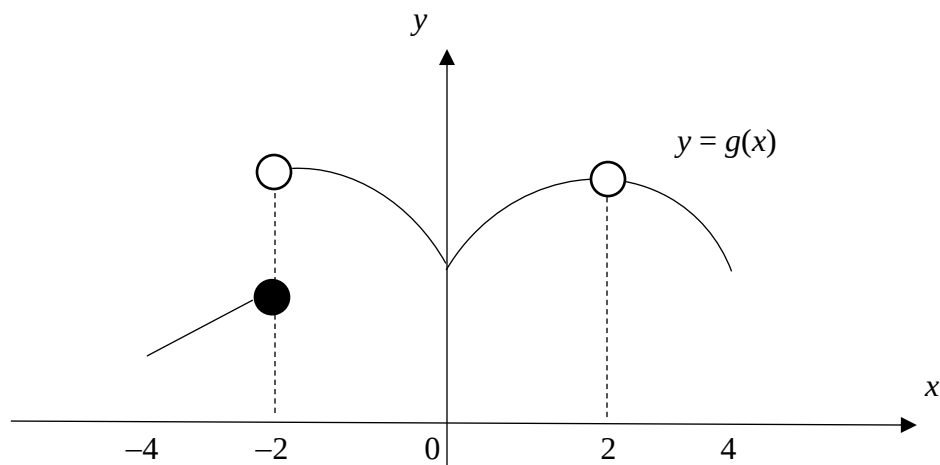
Thus, the function $f(x) = |x|$ is differentiable at all real x except 0.

A function f fails to be differentiable at a if

- 1) the function's graph has a corner or "kink" or "cusp" at $x = a$
- 2) the function's graph is not continuous at $x = a$
- 3) the function's graph has a vertical tangent at $x = a$

Example 6

Consider the graph of g :



- Find the values of x in the interval $(-4, 4)$ at which g is not continuous
- Find the values of x in the interval $(-4, 4)$ at which g is not differentiable

Solution:

- g is not continuous at $x = -2$ and $x = 2$.
- g is not differentiable at $x = -2$, $x = 2$ and $x = 0$

4.2 Rules of differentiation

Objectives:

Students should be able to

- calculate derivatives of functions using differentiation rules
- determine the implicit differentiation of a given equation

Notation

The process of calculating a derivative is called differentiation.

There are many ways to denote the derivative of a function $y = f(x)$ where the independent variable is x and the dependent variable is y . Some common notations for the derivative are

$$\underbrace{f'(x) = y'}_{\text{Newton's symbol for derivatives}} = \underbrace{\frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x)}_{\text{Leibniz's symbol for derivatives}}$$

The symbols $\frac{d}{dx}$ and D indicate the operation of differentiation and are called differentiation operators.

Rules of differentiation

1. Constant rule

$$\frac{d}{dx}(c) = 0. \text{ For example: } f(x) = 5. \quad f'(x) = 0.$$

2. Power rule

$$\frac{d}{dx}(x^n) = nx^{n-1}. \text{ For example: } f(x) = 5x^9. \quad f'(x) = 45x^8.$$

3. Product rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

$$\text{For example: } y = 5(x^3 - 8x)(x - 2)$$

$$f(x) = 5(x^3 - 8x), \quad g(x) = x - 2$$

$$f'(x) = 5(3x^2 - 8), \quad g'(x) = 1$$

$$\frac{dy}{dx} = 5(x^3 - 8x) + (x - 2) \cdot 5(3x^2 - 8)$$

$$= 20x^3 - 30x^2 - 80x + 80$$

4. Quotient rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}, \quad g(x) \neq 0$$

$$\text{For example: } y = \frac{x^2 + 1}{x + 1}$$

$$f(x) = x^2 + 1, \quad g(x) = x + 1$$

$$f'(x) = 2x, \quad g'(x) = 1$$

$$\frac{dy}{dx} = \frac{(x+1) \cdot 2x - (x^2 + 1)}{(x+1)^2}$$

$$= \frac{x^2 + 2x - 1}{(x+1)^2}$$

Derivatives of trigonometric functions

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

$$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

$$\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$$

$$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$$

The chain rule

If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f[g(x)]$ is differentiable at x , and

$$(f \circ g)'(x) = f'[g(x)] \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

Example 7

Find the derivatives of the following functions

a) $y = \sin(3x)$

b) $y = x\sin(x)$

c) $y = \frac{x}{\cos(x)}$

d) $y = \sin(x^2 + x)$

Solution:

a) $u = 3x, \quad y = \sin(u)$

$$\frac{du}{dx} = 3, \quad \frac{dy}{du} = \cos(u)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos(u) \cdot 3$$

Answer: $3\cos(3x)$.

- b) $f(x) = x, \quad g(x) = \sin(x)$
 $f'(x) = 1, \quad g'(x) = \cos(x)$
 Refer Product Rule's example

Answer: $\sin(x) + x\cos(x)$

- c) $f(x) = x, \quad g(x) = \cos(x)$
 $f'(x) = 1, \quad g'(x) = -\sin(x)$
 Refer Quotient Rule's example

$$\text{Answer: } \frac{1}{\cos(x)} + \frac{x \sin(x)}{(\cos(x))^2}$$

- d) $u = x^2 + x, \quad y = \sin(u)$
 $\frac{du}{dx} = 2x + 1, \quad \frac{dy}{du} = \cos(u)$
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (\cos(u)) \cdot (2x + 1)$

Answer: $(2x + 1)\cos(x^2 + x)$

Implicit differentiation

Given an equation in terms of x and y . Implicit differentiation is useful when $y'(x)$ is needed, but it is inconvenient to solve the equation for y in terms of x .

Technique:

- Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x
- Collect the terms with $\frac{dy}{dx}$ on one side of the equation
- Solve for $\frac{dy}{dx}$

Example 8

- a) Find $\frac{dy}{dx}$ if $y^2 = x$
- b) Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin(xy)$
- c) Find the slope of circle $x^2 + y^2 = 25$ at the point $(3, -4)$.
- d) Find $\frac{d^2y}{dx^2}$ if $2x^3 - 3y^2 = 8$.

Solution:

$$y^2 = x$$

$$a) \quad \frac{d}{dx}(y^2) = \frac{d}{dx}(x)$$

$$\text{Left side: } \frac{d}{dx}(y^2)$$

$$\text{Let } v = y^2.$$

Using chain rule, assume that $u = y, v = u^2$ so that $\frac{dv}{du} = 1, \frac{du}{dx} = 2u$.

$$\frac{dv}{dy} = \frac{dv}{du} \cdot \frac{du}{dy} = 1 \cdot (2u) = 2y$$

$$\frac{d}{dx}(y^2) = \frac{dv}{dx} = \frac{dv}{dy} \cdot \frac{dy}{dx} = 2y \cdot \frac{dy}{dx}$$

$$\text{Right side: } \frac{d}{dx}(x) = 1$$

Therefore, we have

$$2y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2y}$$

$$y^2 = x^2 + \sin(xy)$$

$$b) \quad \frac{d}{dx}(y^2) = \frac{d}{dx}(x^2) + \frac{d}{dx}(\sin(xy))$$

$$\text{Left side: } \frac{d}{dx}(y^2) = 2y \frac{dy}{dx} \text{ Refer a).}$$

Right side: $\frac{d}{dx}(x^2) + \frac{d}{dx}(\sin(xy)) = 2x + \frac{d}{dx}(\sin(xy))$

Let $v = \sin(xy)$. If $u = xy$ then $v = \sin(u)$. Therefore, $\frac{dv}{du} = \cos(u)$,

$$\frac{du}{dx} = x \frac{dy}{dx} + y \quad (\text{using Product Rule}).$$

$$\begin{aligned} \text{We have } \frac{d}{dx}(\sin(xy)) &= \frac{dv}{dx} = \frac{dv}{du} \cdot \frac{du}{dx} \\ &= \cos(u) \left[x \frac{dy}{dx} + y \right] \\ &= x \cos(xy) \frac{dy}{dx} + y \cos(xy) \end{aligned}$$

Hence, $2y \frac{dy}{dx} = 2x + x \cos(xy) \frac{dy}{dx} + y \cos(xy) \quad xy$

$$(2y - x \cos(xy)) \frac{dy}{dx} = 2x + y \cos(xy) \quad xy$$

$$\frac{dy}{dx} = \frac{2x + y \cos(xy)}{2y - x \cos(xy)} \quad xy$$

c) $\frac{dy}{dx} = -\frac{x}{y}$

Slope at $(3, -4)$ is $\frac{3}{4}$.

d) $2x^3 - 3y^2 = 8$

$$\frac{d}{dx}(2x^3) - \frac{d}{dx}(3y^2) = \frac{d}{dx}(8)$$

$$6x^2 - 6y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x^2}{y}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{x^2}{y} \right)$$

$$= \frac{y \cdot 2x - x^2 \frac{dy}{dx}}{y^2} \quad (\text{using Quotient Rule})$$

$$\begin{aligned}
 &= \frac{2xy - x^2 \left(\frac{x^2}{y} \right)}{y^2} \\
 &= \frac{2x}{y} - \frac{x^4}{y^3}.
 \end{aligned}$$

4.3 L' hospital's rule

L' hospital's rule is named after the 17th-century French mathematician Guillaume de l'Hôpital. L'hospital's rule is applied for indeterminate forms. What are indeterminate forms? An indeterminate form is an expression involving two functions whose limit cannot be determined solely from the limits of the individual functions.

Indeterminate forms of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$

Theorem

Suppose f and g are differentiable and $g'(x) \neq 0$ near a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \text{ and } \lim_{x \rightarrow a} g(x) = \pm\infty.$$

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ if the limit on the right side exists (or is ∞ or $-\infty$).

Moreover, this statement is also true in the case of a limit as $x \rightarrow a^-$, $x \rightarrow a^+$, $x \rightarrow -\infty$ or as $x \rightarrow +\infty$.

Remark: If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is called an indeterminate form of

type $\frac{0}{0}$. If $\lim_{x \rightarrow a} f(x) = \pm\infty$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is called an indeterminate form of

type $\frac{\infty}{\infty}$.

Example 9

Find $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$.

Solution:

Since $\lim_{x \rightarrow 1} \ln x = \ln 1 = 0$ and $\lim_{x \rightarrow 1} (x-1) = 1-1 = 0$, therefore, we can apply the L'Hospital's rule:

$$\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)} = \lim_{x \rightarrow 1} \frac{1/x}{1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1.$$

Example 10

Calculate $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

Solution:

We have $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow \infty} x^2 = \infty$ so L'Hospital's rule gives

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(x^2)} = \lim_{x \rightarrow \infty} \frac{e^x}{2x}.$$

Consider $\lim_{x \rightarrow \infty} \frac{e^x}{2x}$. We have $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow \infty} 2x = \infty$, therefore $\lim_{x \rightarrow \infty} \frac{e^x}{2x}$ is indeterminate. A

second application of L'Hospital's rule gives $\lim_{x \rightarrow \infty} \frac{e^x}{2x} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(2x)} = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$. Hence,

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \infty.$$

Example 11

Calculate $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}}$.

Solution:

Since $\lim_{x \rightarrow \infty} \ln x = \infty$ and $\lim_{x \rightarrow \infty} \sqrt[3]{x} = \infty$, applying L'Hospital's rule implies that:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{3}x^{-2/3}}$$

Notice that the limit on the right side is indeterminate of type $\frac{0}{0}$. But instead of applying L'Hospital's rule a second time as we did in Example 10, we simplify the expression:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{3}x^{-2/3}} = \lim_{x \rightarrow \infty} \frac{3}{\sqrt[3]{x}} = 0.$$

Indeterminate products

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ (or $-\infty$), then what is the value of $\lim_{x \rightarrow a} f(x)g(x)$? This kind of limit is called an indeterminate form of type $0 \cdot \infty$. We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \text{ or } fg = \frac{g}{1/f}$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ so that we can use L'Hospital's rule.

Example 12

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Solution:

Step 1: Let $f(x) = x$ and $g(x) = \ln x$.

Notice that $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$ and $\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} \ln x = -\infty$. Therefore

$\lim_{x \rightarrow 0^+} f(x)g(x) = \lim_{x \rightarrow 0^+} x \ln x$ is an indeterminate product $0 \cdot \infty$.

Step 2: Write

$$fg = \frac{f}{1/g} \text{ or } fg = \frac{g}{1/f}.$$

So, we have Option 1: $x \ln x = \frac{x}{1/\ln x}$ or Option 2: $x \ln x = \frac{\ln x}{1/x}$. Choose Option 2.

Step 3: Evaluate $\lim_{x \rightarrow 0^+} \ln x$ and $\lim_{x \rightarrow 0^+} \frac{1}{x}$.

We have $\lim_{x \rightarrow 0^+} \ln x = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$ so that $\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$ is the indeterminate form of type $\frac{\infty}{\infty}$.

Step 4: Apply the L'Hospital's rule

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(1/x)} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

Why use Option 2 ?

Because using Option 1 for solving Example 12 gives more complicated expression when we apply L'Hospital's rule. In general, when we rewrite an indeterminate product, we try to choose the option that leads to the simpler limit. In this case, Option 2 leads to the simpler limit.

Indeterminate differences

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, then the limit $\lim_{x \rightarrow a} [f(x) - g(x)] = \infty - \infty$ is called an indeterminate form of type $\infty - \infty$. We try to convert the difference into a quotient (for instance, by using a common denominator, or rationalization, or factoring out a common factor) so that we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Example 13

Compute $\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x)$.

Solution:

We have $\lim_{x \rightarrow (\pi/2)^-} \sec x = \infty$ and $\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty$.

Step 1: Convert into a quotient. Here we use a common denominator.

$$\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x) = \lim_{x \rightarrow (\pi/2)^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) = \lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x}.$$

Step 2: Evaluate $\lim_{x \rightarrow (\pi/2)^-} (1 - \sin x)$ and $\lim_{x \rightarrow (\pi/2)^-} \cos x$.

We have $\lim_{x \rightarrow (\pi/2)^-} (1 - \sin x) = 0$ and $\lim_{x \rightarrow (\pi/2)^-} \cos x = 0$. So, $\lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x}$ is the indeterminate form of type $\frac{0}{0}$.

Step 3: Apply the L'Hospital's rule.

$$\lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x} = \lim_{x \rightarrow (\pi/2)^-} \frac{\frac{d}{dx}(1 - \sin x)}{\frac{d}{dx}(\cos x)} = \lim_{x \rightarrow (\pi/2)^-} \frac{-\cos x}{-\sin x} = 0.$$

Exercises

1. Find the limit for the following functions

a) $\lim_{x \rightarrow -3} \frac{x^2 + 6x + 9}{x + 3}$ Answer: 0

b) $\lim_{x \rightarrow \infty} \frac{3x - 5}{6x + 8}$ Answer: $\frac{3}{6}$

c) $\lim_{x \rightarrow 4} \frac{8 - 2x}{x^2 - x - 12}$ Answer: $-\frac{2}{7}$

d) $\lim_{x \rightarrow -\infty} \frac{x - 4x^2}{5 - 6x^3}$ Answer: 0

2. Evaluate the following limits

a) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$ Answer: 0

b) $\lim_{x \rightarrow 0} \left(\frac{1}{\ln(x+1)} - \frac{1}{x} \right)$ Answer: $\frac{1}{2}$

c) $\lim_{x \rightarrow \infty} \left(\frac{1}{x} \ln(x) \right)$ Answer: 0

TUTORIAL 4

1. Find the derivative of

a) $y = x^3 e^x$

b) $y = e^{2x} \cos(4x)$

c) $y = 2x e^{4x+1}$

d) $y = \frac{e^x}{x^2}$

e) $y = \frac{4x - 2}{x^2 + 1}$

f) $y = \sin\left(\frac{2x}{x+1}\right)$

g) $y = \sin(\tan x^2)$

2. Differentiate both sides of the identity $\cos(2t) = \cos^2(t) - \sin^2(t)$ to prove that $\sin(2t) = 2\sin(t)\cos(t)$.
3. By using the definition of derivative, find the derivative of the function $f(x) = x^2 - 8x + 9$ at a number a .
4. Determine equations of the lines tangent to the graph $y = x\sqrt{5-x^2}$ at the points $(1, 2)$ and $(-2, -2)$.

5. Given that $xe^y - y\ln(x) = 5$. Find $\frac{dy}{dx}$.

6. Given that $y = a^x$ where a is a real number. Find $\frac{dy}{dx}$.

7. Given that $y = \frac{\cos(x)}{e^{2x}}$. Show that $5y + 4\frac{dy}{dx} + \frac{d^2y}{dx^2} = 0$.

8. Given that $v = \sqrt{\sin(u)}$. Show that $4v^3 \frac{d^2v}{du^2} + v^4 + 1 = 0$.

9. Find the slope of the curve $2x^3 + x^2y + y^2 = 4$ at the point $(-1, 2)$.

10. Evaluate the following limits

a) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$

b) $\lim_{x \rightarrow \infty} \frac{3x^4 - x^2}{6x^4 + 12}$

c) $\lim_{x \rightarrow (\pi/2)^-} \left(\frac{\pi}{2} - x \right) \sec x$

d) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 1} \right)$

TOPIC 5: APPLICATIONS OF DIFFERENTIATION

5.1 Graph sketching

Objectives:

Students should be able to

- locate the extreme values of a function f
- locate the critical number of a function f
- determine the concavity of a graph of f
- locate the inflection point of a graph of f
- sketch the graph of f

Extreme values

Definition: Let D be the domain of f . A function f has an *absolute maximum* at $x = c$ if $f(c) \geq f(x)$ for all x in D . Similarly, f has an *absolute minimum* at $x = c$ if $f(c) \leq f(x)$ for all x in D .

Definition: A function f has a *local maximum* at $x = c$ if $f(c) \geq f(x)$ when x is near c . Similarly, f has a *local minimum* at c if $f(c) \leq f(x)$ when x is near c .

Remark:

The word “absolute” is sometimes replaced by “global”.

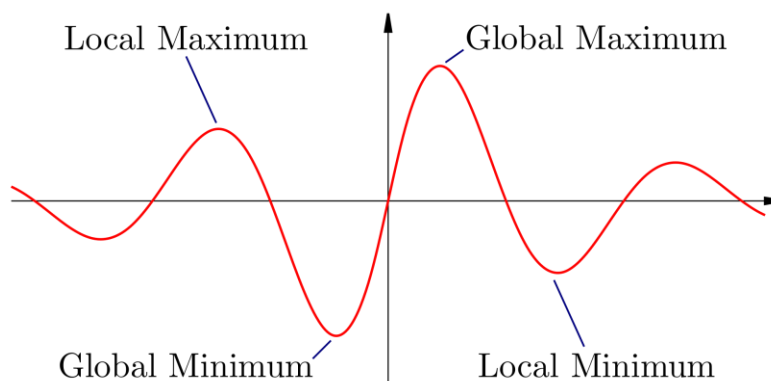
By “near”, we mean in an open interval containing c .

Absolute (local) maximum value is $f(c)$ where $f(c) \geq f(x)$ for all x in D .

Absolute (local) minimum value is $f(c)$ where $f(c) \leq f(x)$ for all x in D .

Absolute (local maximum point is $(c, f(c))$ where $f(c) \geq f(x)$ for all x in D .

Absolute (local) minimum point is $(c, f(c))$ where $f(c) \leq f(x)$ for all x in D



Critical number

A critical number of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Example 1

Find the critical numbers of $f(x) = x^{3/5}(4-x)$.

Solution:

Step 1: Find $f'(x)$

$$\begin{aligned} f'(x) &= x^{\frac{3}{5}}(-1) + \frac{3}{5}x^{\frac{3}{5}-1}(4-x) \\ &= -x^{\frac{3}{5}} + \frac{3}{5}x^{-\frac{2}{5}}(4-x) \end{aligned}$$

Step 2: Find values for which $f'(x) = 0$ or does not exist.

$$\begin{aligned} f'(x) &= x^{\frac{3}{5}} \left[-1 + \frac{3}{5}x^{-1}(4-x) \right] = 0 \\ x^{\frac{3}{5}} &= 0 \quad \text{or} \quad -1 + \frac{3(4-x)}{5x} = 0 \\ x &= 0 \qquad \qquad \qquad x = \frac{3}{2} \end{aligned}$$

The critical numbers are: $x = 0$, and $x = \frac{3}{2}$.

Example 2

Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 1$,
 $-\frac{1}{2} \leq x \leq 4$.

Solution:

Step 1: Find the critical numbers

$$f'(x) = 3x^2 - 6x$$

$$= 3x(x - 2) = 0$$

$$\begin{array}{lcl} 3x = 0 & \text{or} & x - 2 = 0 \\ x = 0 & & x = 2 \end{array}$$

The critical numbers are: $x = 0$ and $x = 2$.

At $x = 0$,

$$f(0) = 0^3 - 3(0)^2 + 1 = 1.$$

After students have learned the first derivative test and second derivative test, students must determine local maximum or local minimum values for questions like this

At $x = 2$,

$$f(2) = 2^3 - 3(2^2) + 1 = -3.$$

Step 2: Find the values of f at the endpoints of the interval $\left[-\frac{1}{2}, 4\right]$.

At $x = -\frac{1}{2}$,

$$f\left(-\frac{1}{2}\right) = \left(-\frac{1}{2}\right)^3 - 3\left(-\frac{1}{2}\right)^2 + 1 = \frac{1}{8}.$$

At $x = 4$,

$$f(4) = 4^3 - 3(4^2) + 1 = 17.$$

Step 3: Compare the values from Steps 1 and 2. The largest is the absolute maximum value. The smallest is the absolute minimum value.

The absolute maximum value is $f(4) = 17$. The absolute minimum value is $f(2) = -3$.

Increasing/Decreasing function

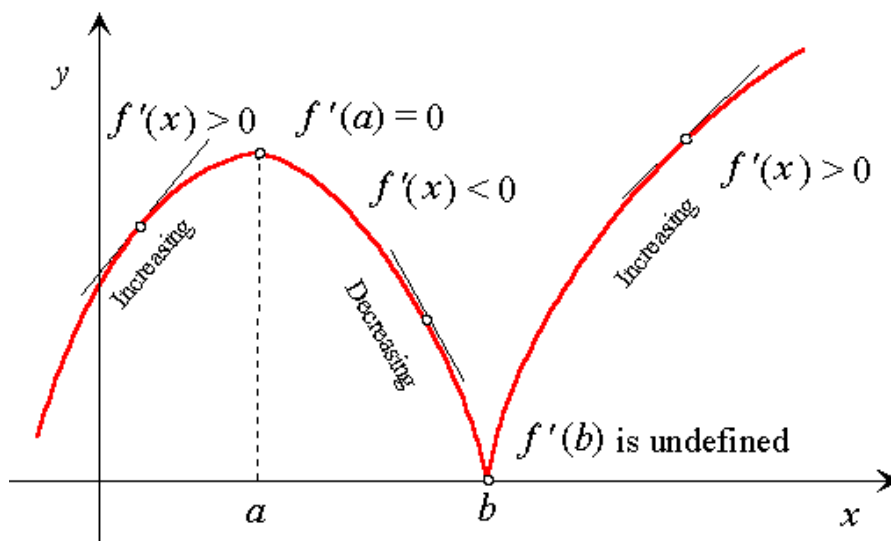
Definition: Let f be defined on an interval, and let x_1 and x_2 be numbers in that interval.

- (a) f is increasing on the interval if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (b) f is decreasing on the interval if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (c) f is constant on the interval if $f(x_1) = f(x_2)$ for all x_1 and x_2 .

Theorem

Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

- (a) If $f'(x) > 0$ for every number x in (a, b) , then f is increasing on $[a, b]$.
- (b) If $f'(x) < 0$ for every number x in (a, b) , then f is decreasing on $[a, b]$.
- (c) If $f'(x) = 0$ for every number x in (a, b) , then f is constant on $[a, b]$.



The first derivative test

Theorem (First derivative test)

Suppose that c is a critical number of a continuous function f .

- a) If f' changes from positive to negative at c , then f has a local maximum at c .
- b) If f' changes from negative to positive at c , then f has a local minimum at c .
- c) If f' does not change sign at c , then f has no local maximum or minimum at c .

Note: Number c has left and right sides.

Example 3

Find the local maximum and minimum values of the function using the first derivative test.

$$g(x) = x + 2\sin(x), \quad 0 \leq x \leq 2\pi$$

Solution:

Step 1: Find the critical numbers

$$g'(x) = 1 + 2 \cos(x)$$

$$g'(x) = 0 \rightarrow 1 + 2 \cos(x) = 0$$

$$\cos(x) = -0.5$$

$$x = 120^\circ, 240^\circ$$

Step 2: Determine the sign of $g'(x)$ at the left and right sides of each critical number.

$g'(x)$ sign				
	0°	+	–	+
				360°
		Increasing	Decreasing	Increasing
		Local maximum	Local minimum	

Step 3: Find the local maximum and the local minimum values.

$$\text{Local maximum value} = g(120^\circ) \approx 3.83$$

$$\text{Local minimum value} = g(240^\circ) \approx 2.46$$

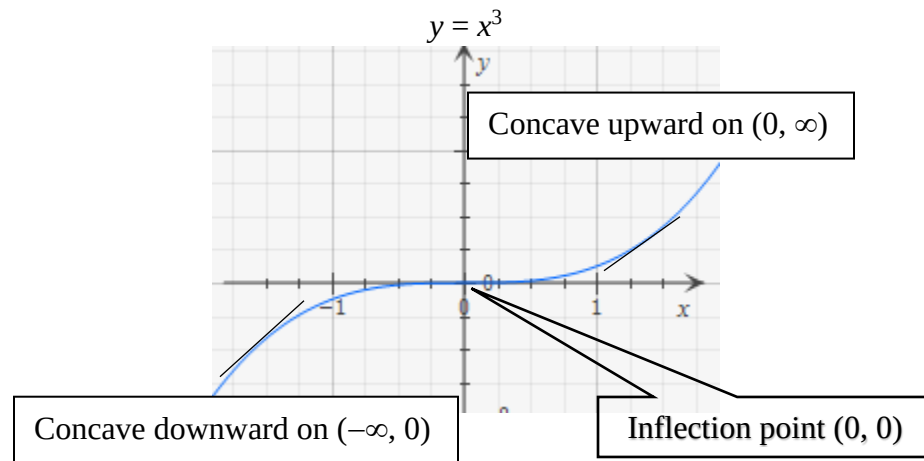
Concavity

Definition: If the graph of f lies above all of its tangents on an open interval I , then it is called *concave upward* on I . If the graph of f lies below all of its tangents on the open interval I , it is called *concave downward* on I .

Theorem

Let f be twice differentiable on an open interval I .

- If $f''(x) > 0$ for every number x in I , then f is concave upward on I .
- If $f''(x) < 0$ for every number x in I , then f is concave downward on I .



A point $P(c, f(c))$ on a curve $y = f(x)$ is called an *inflection point* if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

Example 4

Find the intervals of concavity and the inflection points of the function

$$f(x) = 2x^3 - 3x^2 - 12x$$

Solution:

Step 1: Find $f''(x)$.

$$f'(x) = 6x^2 - 6x - 12$$

$$f''(x) = 12x - 6$$

Step 2: Find interval on which the graph is concave upward.

The graph is concave upward if $f''(x) > 0$. Assume that $f''(x) > 0$.

Therefore, we have $12x - 6 > 0$. Solving this inequality, we get

$$x > \frac{1}{2}.$$

Thus, the interval on which the graph is concave upward is $\left(\frac{1}{2}, \infty\right)$.

Step 3: Find interval on which the graph is concave downward.

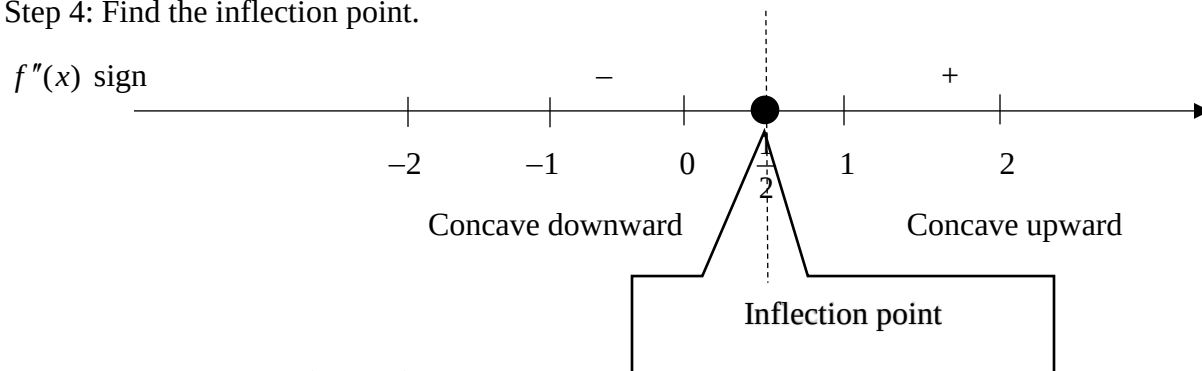
The graph is concave upward if $f''(x) < 0$. Assume that $f''(x) < 0$.

Therefore, we have $12x - 6 < 0$. Solving this inequality, we get

$$x < \frac{1}{2}.$$

Thus, the interval on which the graph is concave downward is $\left(-\infty, \frac{1}{2}\right)$.

Step 4: Find the inflection point.



The inflection point is $\left(\frac{1}{2}, -\frac{13}{2}\right)$.

The second derivative test

Theorem

Suppose that f is twice differentiable at $x = c$.

- a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .
- c) If $f'(c) = 0$ and $f''(c) = 0$, then the test is inconclusive; that is, f may have a local minimum, local maximum, or neither at c .

Example 5

Find the local maximum and minimum values of the function using the second derivative test.

$$g(x) = x + 2\sin(x), \quad 0 \leq x \leq 2\pi$$

Step 1: Find the critical numbers

$$g'(x) = 1 + 2\cos(x)$$

$$g'(x) = 0 \rightarrow 1 + 2\cos(x) = 0$$

$$\begin{aligned}\cos(x) &= -0.5 \\ x &= 120^\circ, 240^\circ\end{aligned}$$

Step 2: Find the second derivative of g at the critical numbers of g

$$g''(x) = -2 \sin(x)$$

$$g''(120^\circ) = -2 \sin(120^\circ) = -\sqrt{3} < 0 \rightarrow g \text{ has a local maximum at } x = 120^\circ.$$

$$g''(240^\circ) = -2 \sin(240^\circ) = \sqrt{3} > 0 \rightarrow g \text{ has a local minimum at } x = 240^\circ.$$

Step 3: Find the local maximum and the local minimum values.

$$\text{Local maximum value} = g(120^\circ) \approx 3.83$$

$$\text{Local minimum value} = g(240^\circ) \approx 2.46$$

Example 6

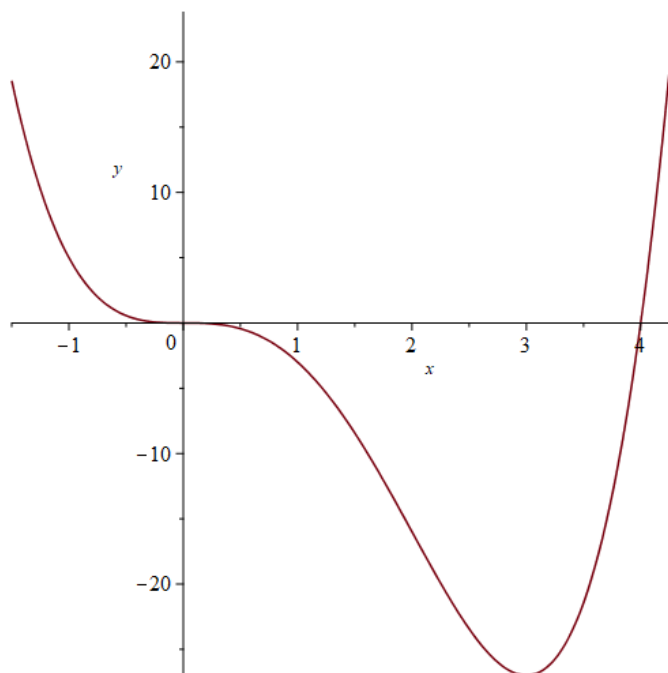
A function f is defined by $f(x) = x^4 - 4x^3$. Find

- Local minimum point
- Interval of concavity
- Inflection point(s)

Hence, sketch the graph of f .

Solution:

- Refer steps in Example 3 and 5..
Local minimum point is $(3, -27)$.
- Refer steps in Example 4.
The graph is concave upward when $x \in (-\infty, 0)$ and $(2, \infty)$.
The graph is concave downward when $x \in (0, 2)$.
- Refer steps in Example 4.
The inflection points are $(0, 0)$ and $(2, -16)$.



Exercises

- Find the local maximum and minimum values of f using both the First and Second Derivative Tests. Which method do you prefer? Explain.

$$f(x) = \frac{x}{x^2 + 4}$$

- Find the intervals of concavity and the inflection points of the function:

$$f(x) = x^3 - 12x + 1$$

- A function h is defined by $h(x) = x\sqrt{x+3}$. Find

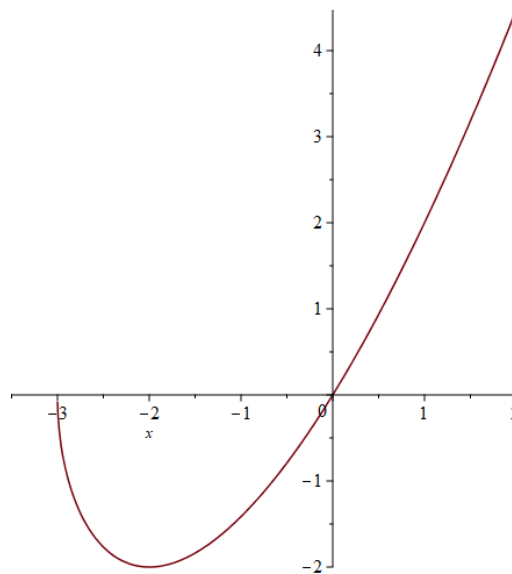
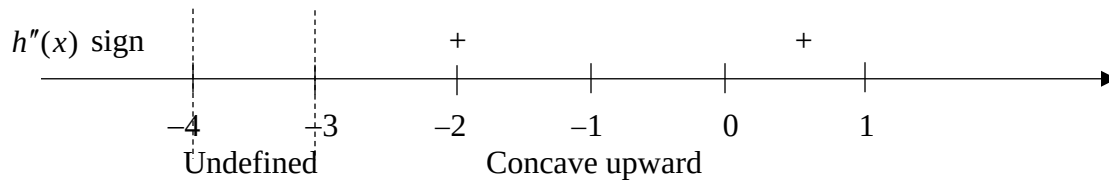
- $h'(x)$ and $h''(x)$
- Absolute minimum point
- Interval of concavity

Hence, sketch the graph of h .

Answers:

- Local minimum value : $f(-2) = -\frac{1}{4}$. Local maximum value: $f(2) = \frac{1}{4}$. Write explanation using your own words.

- 2) The graph is concave upward when $x \in (0, \infty)$. The graph is concave downward when $x \in (-\infty, 0)$. The inflection point is $(0, 1)$.
- 3) $h'(x) = \frac{3x+6}{2\sqrt{x+3}}$. $h''(x) = \frac{3x+12}{4(x+3)\sqrt{x+3}}$. Absolute minimum point: $(-2, -2)$. The graph is concave upward when $x \in (-3, 0) \cup \infty$

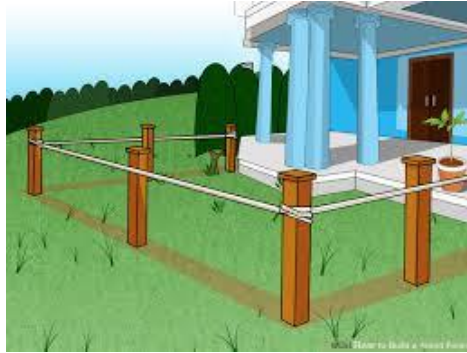


5.2 Optimization problem

In optimization problems we are looking for the largest value or the smallest value that a function can take.

Example 7

We need to enclose a field with a rectangular fence. We have 1200m of fencing material and a building is on one side of the field and so won't need any fencing. Determine the dimensions of the field that will enclose the largest area.



Solution:

Step 1: Understand the problem

Step 2: Formulate the equation. Let x and y be the width and length of the fence respectively.

$$\text{Area } A = xy \text{ ---(1)}$$

$$\text{Perimeter } 2x + y = 1200 \text{ ---(2)}$$

From (1) and (2),

$$A(x) = 1200x - 2x^2$$

Step 3: Find the domain of $A(x)$.

Since $A(x)$ cannot be a negative number, we find x values such that $A(x) \geq 0$.

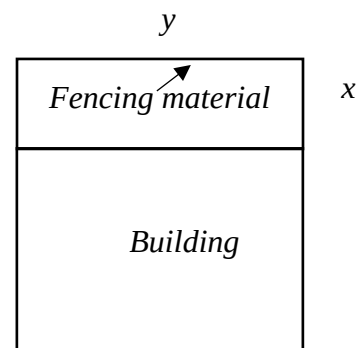
$1200x - 2x^2 \geq 0$ implies that $x \geq 0$ and $x \leq 600$. Thus, the domain is $0 \leq x \leq 600$.

Step 4: Maximize $A(x) = 1200x - 2x^2$ subject to $0 \leq x \leq 600$.

$$\frac{dA}{dx} = 1200 - 4x = 0$$

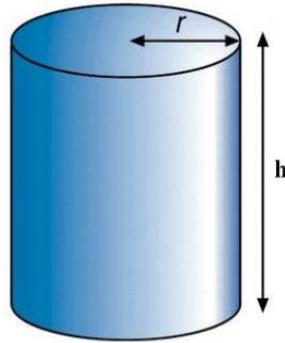
$$x = 300 \text{ (this value satisfies } 0 \leq x \leq 600 \text{)}$$

Answer: $x = 300m$, $y = 1200 - 2x = 600m$. Maximum area = $A(300) = 180,000m^2$.



Example 2

A cylindrical can is to be made to hold 1 liter of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.



Step 1: Understand the problem.

Step 2: Formulate the equation. Let r and h be the radius and height of the cylinder respectively.

Total surface area is $A = 2\pi r^2 + 2\pi rh$ (1)

Volume is given as 1 liter which is equivalent to 1000cm^3 . Thus,

$$\pi r^2 h = 1000 \quad (2)$$

From (1) and (2), we have $A(r) = 2\pi r^2 + \frac{2000}{r}$.

Step 3: Find the domain of $A(r)$.

Domain: $r > 0$.

Step 4: Minimize $A(r) = 2\pi r^2 + \frac{2000}{r}$ subject to $r > 0$.

$$\frac{dA}{dr} = 4\pi r - \frac{2000}{r^2} = 0$$

$$4\pi r^3 = 2000$$

$$r^3 = \frac{2000}{4\pi}$$

$$r = \sqrt[3]{\frac{500}{\pi}}$$

Substitute $r = \sqrt[3]{\frac{500}{\pi}}$ into (2) to get

$$\pi \left(\sqrt[3]{\frac{500}{\pi}} \right)^2 h = 1000$$

$$\pi \cdot \pi^{-2/3} \cdot 500^{2/3} h = 1000$$

$$\pi^{1/3} \cdot h = \frac{1000}{500^{\frac{2}{3}}}$$

$$\pi^{1/3} \cdot h = \left(\frac{500 \cdot 2}{500 \cdot 500^{\frac{1}{3}}} \right)$$

$$\pi^{1/3} \cdot h = \left(\frac{2}{500^{\frac{1}{3}}} \right)$$

$$h = 2 \cdot \frac{500^{\frac{1}{3}}}{\pi^{1/3}} = 2r$$

Answer: $r = \sqrt[3]{\frac{500}{\pi}}$, $h = 2r$.

Derivatives of inverse trigonometric functions

$$1. \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$4. \quad \frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

$$2. \quad \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$5. \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$3. \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$6. \quad \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

Proof for derivative of inverse sine.

Let $\sin^{-1} x = y$ where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. We have

$$\sin y = x$$

Differentiate both sides: $\frac{d}{dx}(\sin y) = \frac{d}{dx}(x)$

$$(\cos y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

From trigonometric identity $\sin^2 y + \cos^2 y = 1$, we have $\cos y = \pm\sqrt{1 - \sin^2 y} = \pm\sqrt{1 - x^2}$. Since $\cos y$ is positive for all $0 \leq y \leq \pi$, therefore $\cos y = \sqrt{1 - x^2}$. Hence,

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$$

Proof for derivative of inverse cosine.

Let $\cos^{-1} x = y$ where $0 \leq y \leq \pi$. We have

$$\cos y = x$$

Differentiate both sides: $\frac{d}{dx}(\cos y) = \frac{d}{dx}(x)$

$$(-\sin y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\sin y}$$

From trigonometric identity $\sin^2 y + \cos^2 y = 1$, we have $\sin y = \pm\sqrt{1 - \cos^2 y} = \pm\sqrt{1 - x^2}$. Since $\sin y$ is positive for all $0 \leq y \leq \pi$, therefore $\sin y = \sqrt{1 - x^2}$. Hence,

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}}$$

Proof for derivative of inverse tangent.

Let $\tan^{-1} x = y$. We have

$$\tan y = x$$

Differentiate both sides: $\frac{d}{dx}(\tan y) = \frac{d}{dx}(x)$

$$(\sec^2 y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

From trigonometric identity $1 + \tan^2 y = \sec^2 y$, we have

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}.$$

Exercises

1. If a resistor of R ohms is connected across a battery of E volts with internal resistance r ohms, then the powers (in watts) in the external resistor is

$$P = \frac{E^2 R}{(R+r)^2}$$

If E and r are fixed but R varies, what is the maximum value of the power?

Solution:

Step 1: Find $\frac{dP}{dR}$.

Let $f(R) = E^2 R$ and $g(R) = (R+r)^2$. So $f'(R) = E^2$ and $g'(R) = 2(R+r)$.

$$\frac{dP}{dR} = \frac{g(R)f'(R) - f(R)g'(R)}{(g(R))^2} \quad (\text{Refer 'Quotient Rule' in topic 4})$$

$$= \frac{(R+r)^2 E^2 - E^2 R \cdot 2(R+r)}{[(R+r)^2]^2}$$

$$= \frac{E^2 r - E^2 R}{(R+r)^3}$$

$$= \frac{E^2 (r - R)}{(R+r)^3}$$

Step 2: Setting $\frac{dP}{dR} = 0$.

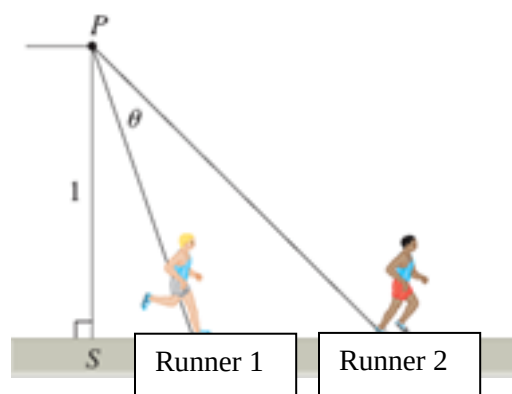
$$\frac{dP}{dR} = \frac{E^2 (r - R)}{(R+r)^3} = 0. \text{ This implies that } R = r.$$

Step 3: Substitute $R = r$ into P .

$$P = \frac{E^2 r}{(r+r)^2} = \frac{E^2 r}{(2r)^2} = \frac{E^2}{4r}.$$

Answer: The maximum value is $P = \frac{E^2}{4r}$.

2. An observer stands at a point P, one unit away from a track. Two runners start at the point S in the figure and run along the track. One runner runs three times as fast as the other. Find the maximum value of the observer's angle of sight θ between the runners.



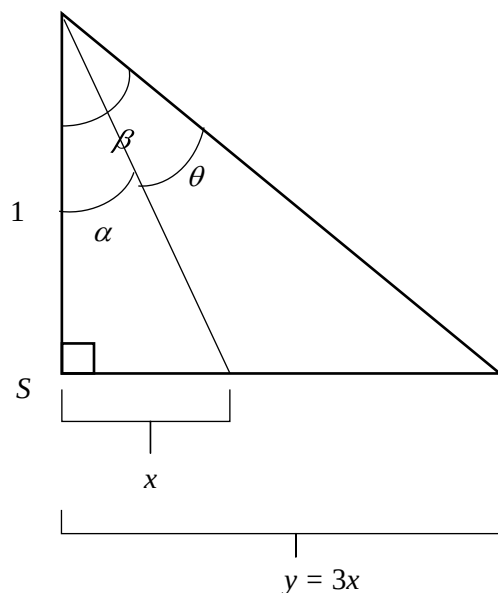
Solution:

Let x : The distance between Runner 1 and point S

y : The distance between Runner 2 and point S

It is obvious from the picture that Runner 2 runs three times as fast as Runner 1. Therefore, the distance between Runner 2 and point S must be three times longer than that for Runner 1.

$$y = 3x$$



$$\tan \alpha = \frac{x}{1} = x \Rightarrow \alpha = \tan^{-1} x$$

$$\tan \beta = \frac{y}{1} = y \Rightarrow \beta = \tan^{-1} y$$

$$\theta = \beta - \alpha$$

$$\theta = \tan^{-1} y - \tan^{-1} x$$

$$\theta = \tan^{-1}(3x) - \tan^{-1} x$$

$$\begin{aligned}\frac{d\theta}{dx} &= \frac{d}{dx}(\tan^{-1}(3x)) - \frac{d}{dx}(\tan^{-1} x) \\ &= \frac{3}{1+(3x)^2} - \frac{1}{1+x^2}\end{aligned}$$

$$\frac{d\theta}{dx} = 0 \Rightarrow \frac{3}{1+(3x)^2} - \frac{1}{1+x^2} = 0$$

$$x = \pm \frac{1}{\sqrt{3}}. \text{ Since } x \text{ represents 'distance', therefore } x = \frac{1}{\sqrt{3}}.$$

$$\text{The maximum value of } \theta \text{ is } \theta = \tan^{-1}\left(3 \cdot \frac{1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 60^\circ - 30^\circ = 30^\circ.$$

Answer: The maximum value of θ is $30^\circ (= \frac{\pi}{6})$.

5.3 Rates of change

Objectives:

Students should be able to

- compute the rate of change of one quantity in terms of the rate of change of another quantity

Let

$$\frac{\Delta y}{\Delta x} = \frac{f(x) - f(a)}{x - a}$$

We have

$$f'(x) = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$\frac{\Delta y}{\Delta x}$ is the average rate of change of y with respect to x .

Example 3

The position of a particle is given by the equation

$$s = f(t) = t^3 - 6t^2 + 9t$$

where t is measured in seconds and s in meters. Find the velocity at time t . When is the particle at rest?

Solution:

Velocity is the average rate of change of displacement (s) with respect to time (t). Therefore,

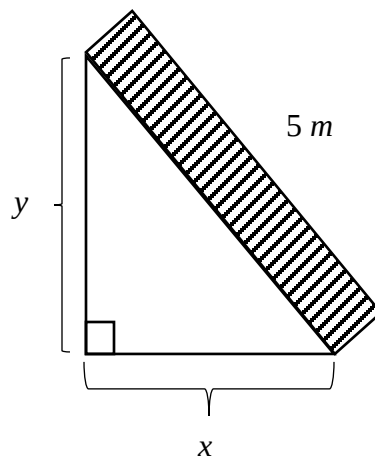
$v(t) = f'(t) = 3t^2 - 12t + 9$. The particle is at rest when $v(t) = 0$. Letting

$v(t) = f'(t) = 3t^2 - 12t + 9 = 0$ implies that $t = 1$ or $t = 3$. Hence, the particle is at rest after 1 second and after 3 seconds.

Example 4

A ladder 5 m long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 m/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 3 m from the wall?

Solution:



Given that $\frac{dx}{dt} = 1$. What is $\frac{dy}{dt}$ when $x = 3$?

From the Pythagoras' Theorem, we know that $x^2 + y^2 = 5^2 = 25$. Differentiating both sides of the equation:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

From the Chain Rule: $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

$$= -\frac{x}{y}(1) = -\frac{x}{y}.$$

Substitute $x = 3$ into $x^2 + y^2 = 25$ we get $y = 4$. Hence,

$$\frac{dy}{dt} = -\frac{3}{4}.$$

The top of the ladder is sliding down the wall at a rate of $\frac{3}{4}$ m/s.

5.4 Newton's method

Methods such factoring, completing the square and formulas are used to get the root(s) of functions. However, such methods and formulas do not give roots of all functions. For example, we cannot find the root(s) of the function f defined by $f(x) = \cos(x) - x$ by factoring, completing the square and by using existing formulas. In this case, we need another approach. That approach is to approximate roots using numerical methods, such as Newton's method.

To approximate a root of the equation $f(x) = 0$, start with an initial estimate x_0 and generate a sequence of approximation $x_1, x_2, \dots$ using the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad f'(x_n) \neq 0, n = 0, 1, 2, \dots$$

The value x_{n+1} will converge to a number c for some $n > N$. The root of the equation is the number c .

Example 5

Approximate the roots of $f(x) = x^3 - 5x + 1$ using Newton's method. Use $x_0 = -3$, $x_0 = 1$, and $x_0 = 4$ as initial approximations.

Solution:

Step 1: Find $f'(x_n)$.

$$f'(x_n) = 3x_n^2 - 5$$

Step 2: Use formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, $f'(x_n) \neq 0, n = 0, 1, 2, \dots$

$$\begin{aligned} x_{n+1} &= x_n - \frac{x_n^3 - 5x_n + 1}{3x_n^2 - 5} \\ &= \frac{x_n(3x_n^2 - 5) - (x_n^3 - 5x_n + 1)}{3x_n^2 - 5} \\ &= \frac{3x_n^3 - 5x_n - x_n^3 + 5x_n - 1}{3x_n^2 - 5} \\ &= \frac{2x_n^3 - 1}{3x_n^2 - 5} \end{aligned}$$

Step 3: Use the given initial approximations.

C3		X		f_x		=((2*B3^3)-1)/((3*B3^2)-5)	
	A	B	C	D	E	F	G
1	x0=-3						
2	n	xn	xn+1				
3	0	-3	-2.50000				
4	1	-2.5	-2.34545				
5	2	-2.34545	-2.33020				
6	3	-2.3302	-2.33006				
7	4	-2.33006	-2.33006				
8	5	-2.33006	-2.33006				
9	6	-2.33006	-2.33006				
10	7	-2.33006	-2.33006				
11	8	-2.33006	-2.33006				
12	9	-2.33006	-2.33006				
13	10	-2.33006	-2.33006				
14	11	-2.33006	-2.33006				

Using $x_0 = -3$, we get -2.33006 . Exercise: Study the above Microsoft Excel calculations for finding roots of f , then use it to find the other two approximations using $x_0 = 1$ and $x_0 = 4$.

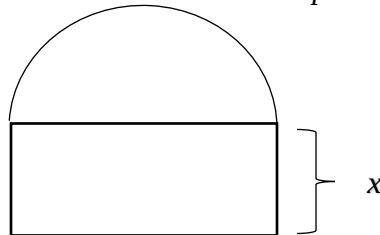
The other two approximations are 0.20164 and 2.12842.

TUTORIAL 5

- Find the critical number(s) of the following functions
 - $f(x) = 2x^3 - 3x^2 - 36x$
 - $f(x) = x^{\frac{4}{5}}(x-4)^2$
- Find the absolute maximum and absolute minimum values of g on the given interval
 - $g(x) = 2x^3 - 3x^2 - 12x + 1$ $[-2, 3]$
 - $g(x) = 2\cos(x) + \sin(2x)$ $\left[0, \frac{\pi}{2}\right]$
- Identify the interval of concavity and the inflection points of the following functions
 - $h(x) = 5x^4 - 20x^3 + 10$
 - $h(x) = (x-6)(x+6)^2$
- A function m is defined by $m(x) = \frac{3x}{x^2 - 1}$. Find
 - $m'(x)$ and $m''(x)$
 - the vertical asymptote of m
 - the interval of concavity

Hence, sketch the graph of m .

- Find the local maximum point, local minimum point and inflection point of $f(x) = \frac{1}{3}x^3 - 3x^2 + 8x + 1$.
- Three hundred twenty Ringgit Malaysia are available to fence a rectangular garden. The fencing for the side of the garden facing the road costs RM6 per foot and the fencing for the other sides costs RM2 per foot. Find the dimensions of the largest possible area of the garden enclosed by the fence.
- A church window consisting of a rectangle topped by a semicircle is to have a perimeter p . Formulate the area, A of the window in terms of x and p .



- A spherical balloon is inflated and its volume increases at a rate of $15 \text{ inch}^3/\text{min}$. What is the rate of change of its radius when the radius is 10 inches?

9. A water heater that has the shape of a right cylindrical tank with a radius of 1 ft and a height of 4 ft is being drained. How fast is water draining out of the tank (in ft^3/min) if the water level is dropping at 6 ft/min ?
10. Approximate the root of $f(x) = x - e^{-x}$ using Newton's method up to seven decimal places. Use $x_0 = 0.5$ as an initial approximation.

COURSE MODULE: LOGICAL REASONING.

TOPIC 6: CONIC SECTION

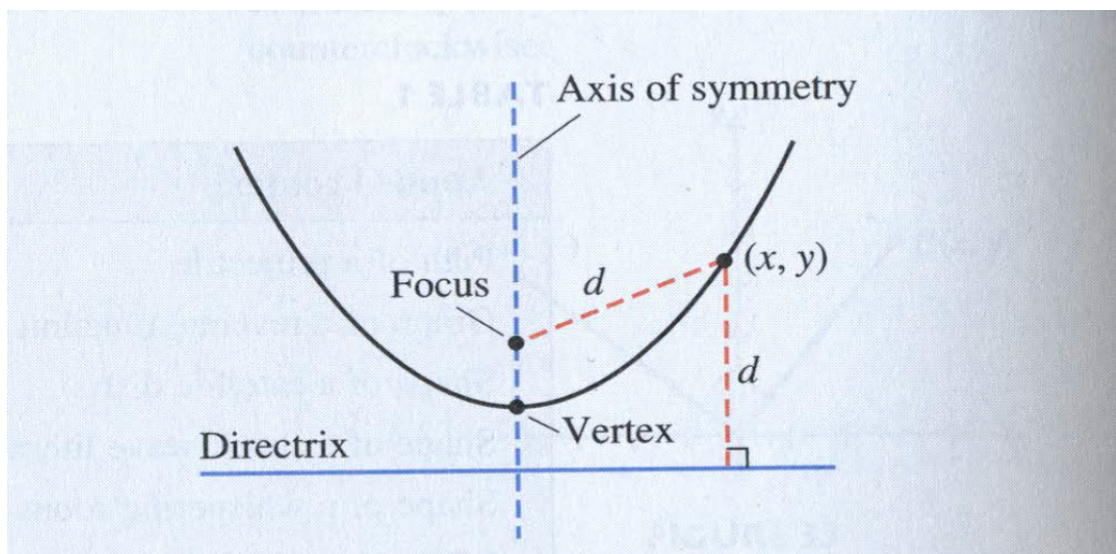
Objectives:

Students should be able to

- sketch a graph of the parabola, ellipse or hyperbola
- identify the formula for parabola, ellipse or hyperbola
- solve related problems

PARABOLA

- ⊙ A parabola is the set of all points (x, y) that are equidistant from a fixed line (called the **directrix**) and a fixed point (called the **focus**) not on the line. The point midway between the focus and directrix is called the **vertex**, and the line passing through the focus and vertex is called the **axis of symmetry** or the **axis of the parabola**.



Parabolas With Vertex At The Origin (0, 0)

Equation	Axis	Description
$y = ax^2$	y – axis	Opens upward if $a > 0$, downward if $a < 0$.
$x = ay^2$	x – axis	Opens to the right if $a > 0$, to the left if $a < 0$.

The focus is on the axis of the parabola and in the interior of the parabola, $|p|$ units from the vertex, where

$$p = \frac{1}{4a}$$

The directrix is perpendicular to the axis of the parabola and is $|p|$ units from the vertex on the side of the parabola opposite the focus.

Example 1

Sketch a graph of the parabola $y = \frac{1}{8}x^2$ and find its focus and directrix.

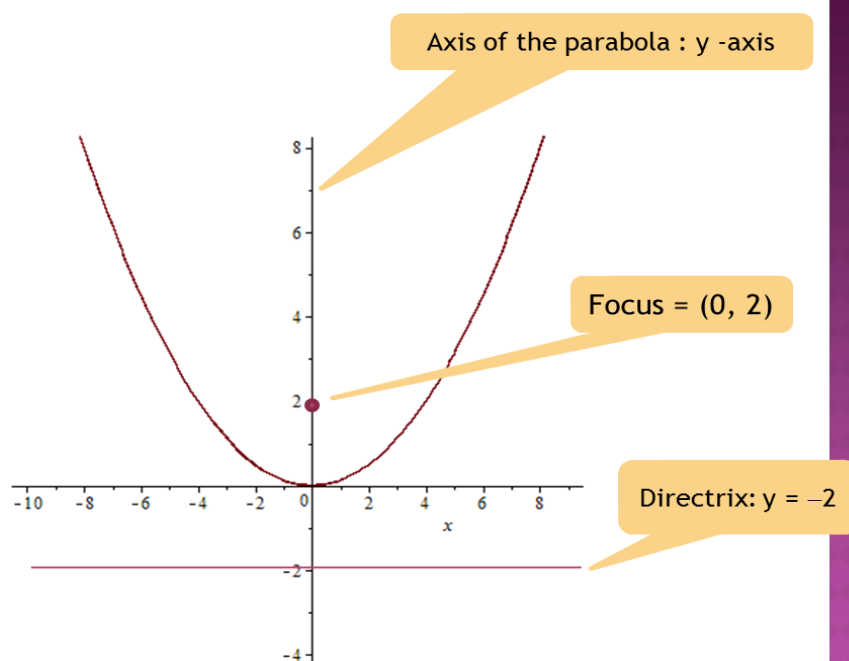
Solution:

$$y = ax^2 = \frac{1}{8}x^2 \Rightarrow a = \frac{1}{8}.$$

$a > 0$, so the parabola opens upward

$$p = \frac{1}{4a} = \frac{1}{4(1/8)} = 2$$

So, the focus and directrix are respectively, 2 units from the vertex.



Example 2

Sketch a graph of the parabola $x = \frac{1}{8}y^2$ and find its focus and directrix.

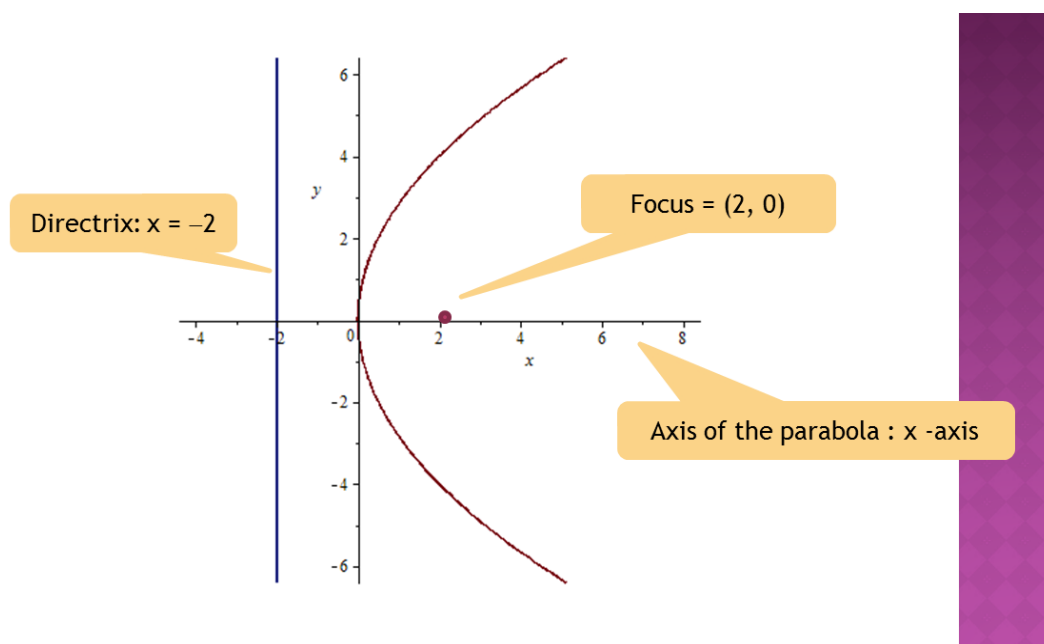
Solution:

$$x = ay^2 = \frac{1}{8}y^2 \Rightarrow a = \frac{1}{8}.$$

$a > 0$, so the parabola opens to the right

$$p = \frac{1}{4a} = \frac{1}{4(1/8)} = 2$$

So, the focus and directrix are respectively, 2 units from the vertex



Parabola With Vertex (h, k)

Equation	Axis	Description
$y - k = a(x - h)^2$	Vertical axis	Opens upward if $a > 0$, downward if $a < 0$.
$x - h = a(y - k)^2$	Horizontal axis	Opens to the right if $a > 0$, to the left if $a < 0$.

The focus and directrix can be found in a similar manner as with parabolas having vertex at the origin $(0, 0)$.

Example 3

Sketch a graph of the parabola $x = y^2 + 2y - 3$. Identify the vertex, focus and directrix.

Solution:

Step 1: Write the equation in standard form

$$x = y^2 + 2y - 3$$

$$x + 3 = y^2 + 2y$$

$$x + 3 + \left(\frac{2}{2}\right)^2 = y^2 + 2y + \left(\frac{2}{2}\right)^2$$

$$x + 3 + 1 = y^2 + 2y + 1$$

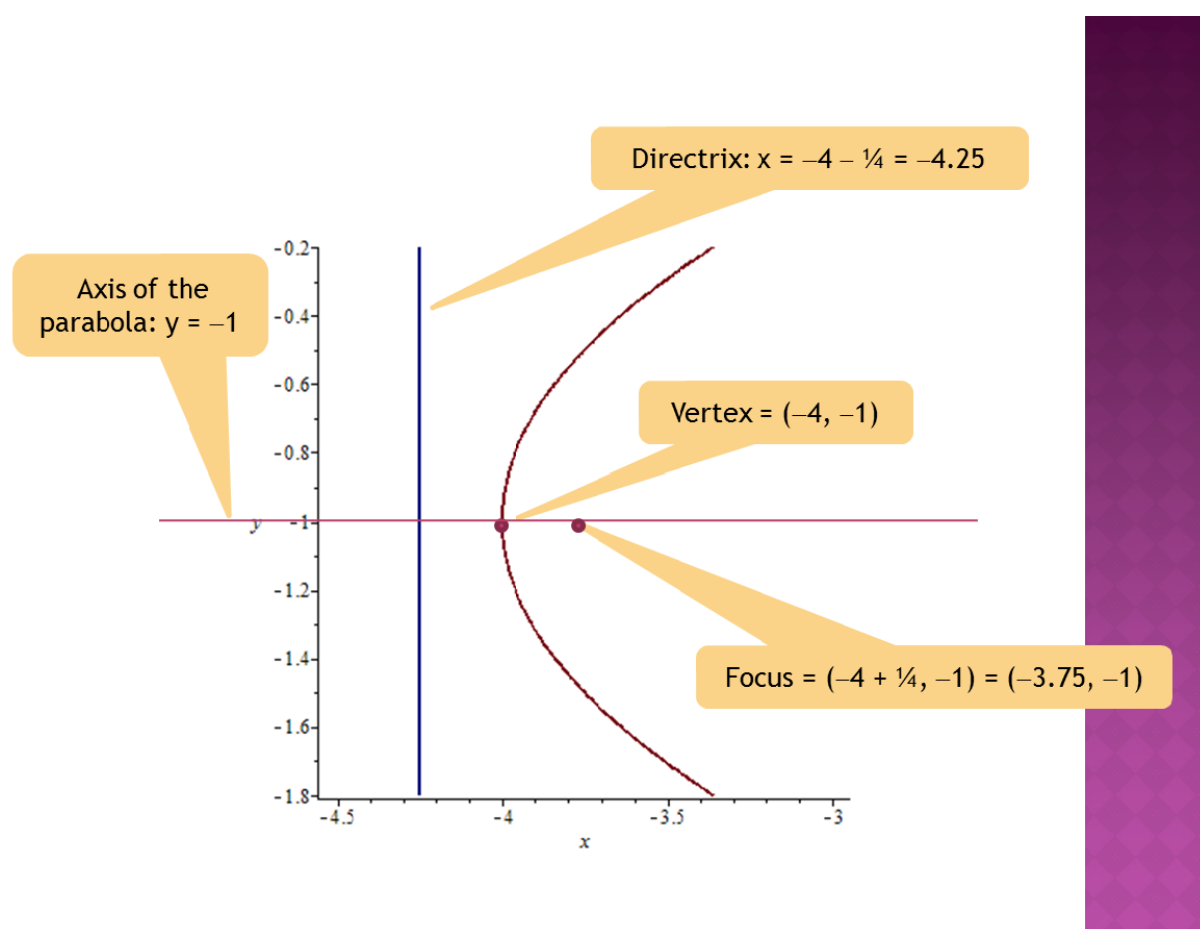
$$x + 4 = (y + 1)^2.$$

Step 2: Identify the vertex, focus and directrix

Comparing $x + 4 = (y + 1)^2$ with $x - h = a(y - k)^2$, we get $h = -4$, $k = -1$, $a = 1$. The vertex is $(-4, -1)$. The parabola opens to the right.

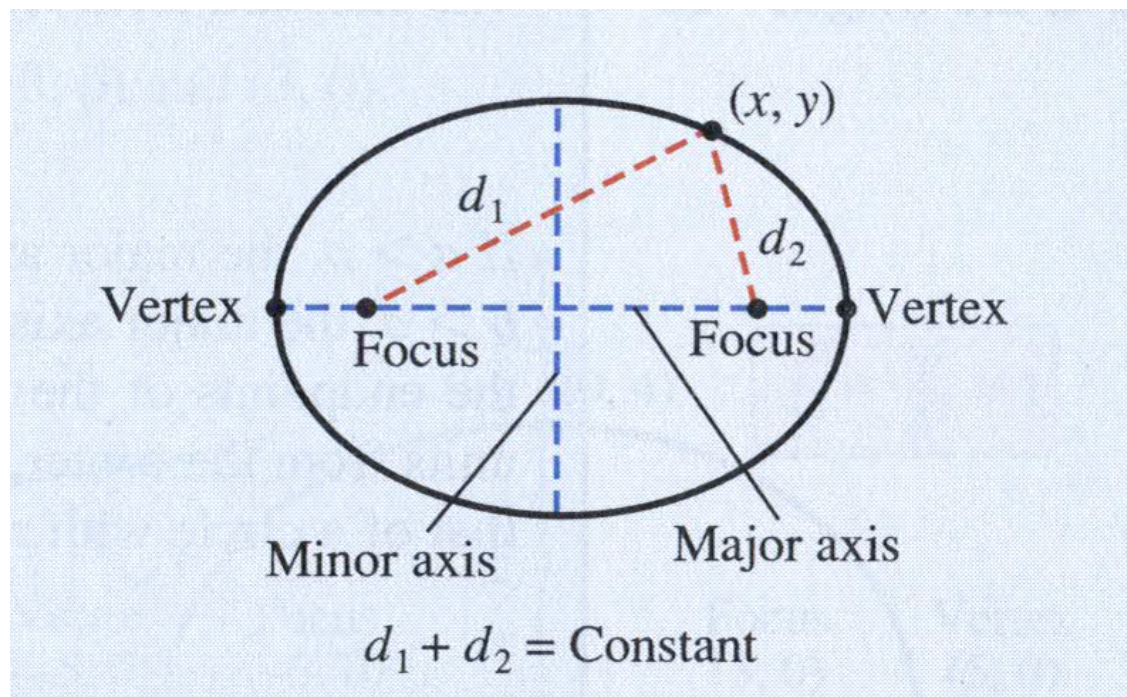
$$p = \frac{1}{4a} = \frac{1}{4(1)} = \frac{1}{4}$$

So, the focus and directrix are respectively, $1/4$ units from the vertex



ELLIPSE

An ellipse is the set of all points (x, y) such that the sum of the distances from fixed points (called the foci) is constant. The line through the foci intersects the ellipse at two points, called the vertices. The line segment connecting the two vertices is called the major axis, and the midpoint of the major axis is called the center. The line segment perpendicular to the major axis, passing through the center, and connecting two points on the ellipse is called the minor axis.



Ellipses centered at the origin $(0, 0)$

Equation	Major axis	Graph shape
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a \neq b$	If $a > b$, the major axis is horizontal with length $2a$	
	If $b > a$, the major axis is vertical with length $2b$	

The vertices are the endpoints of the major axis. The foci lie on the major axis a distance of c units from the center, where

$$c = \sqrt{a^2 - b^2}$$

Note that if $a = b$, the equation is that of a circle with radius a .

Example 4

Sketch the graph of the ellipse with equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Identify the vertices and foci.

Solution:

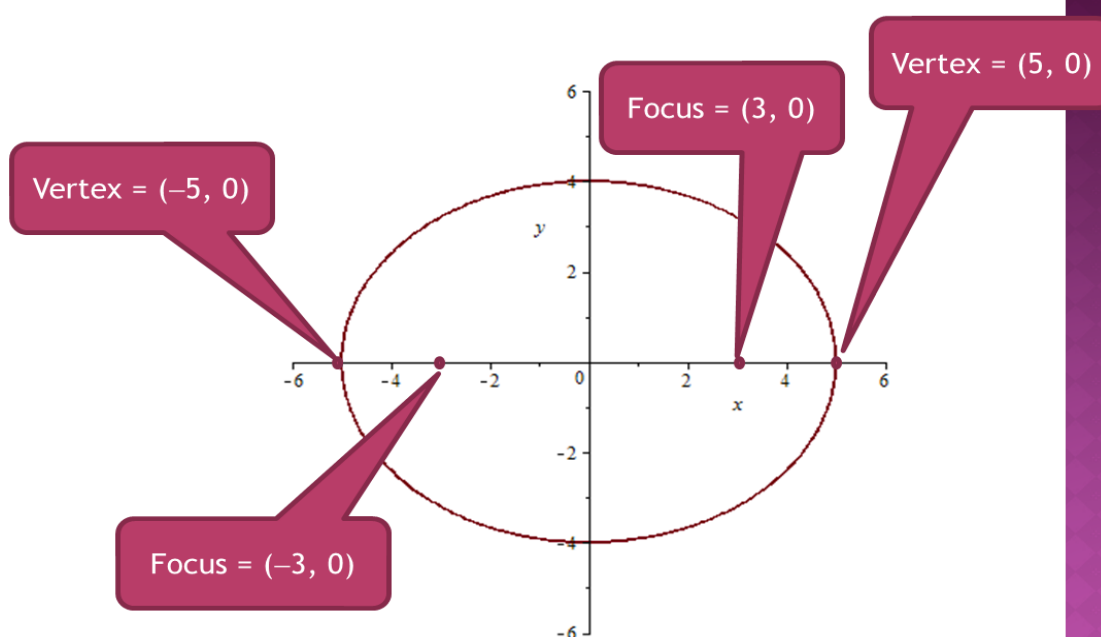
$$\frac{x^2}{25} + \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{5^2} + \frac{y^2}{4^2} = 1.$$

$a = 5$, $b = 4$. Here, $a > b$. The ellipse should look like this:



Length of the major axis is $2a = 2(5) = 10$. The vertices are the endpoints of the major axis. The foci lie c units from the center where

$$c = \sqrt{a^2 - b^2} = \sqrt{25 - 16} = 9 = 3$$



Example 5

Sketch the graph of the ellipse with equation $9x^2 + 4y^2 - 36 = 0$. Identify the vertices and foci.

Solution:

Write the equation in standard form:

$$9x^2 + 4y^2 - 36 = 0$$

$$9x^2 + 4y^2 = 36$$

$$\frac{9x^2}{36} + \frac{4y^2}{36} = 1$$

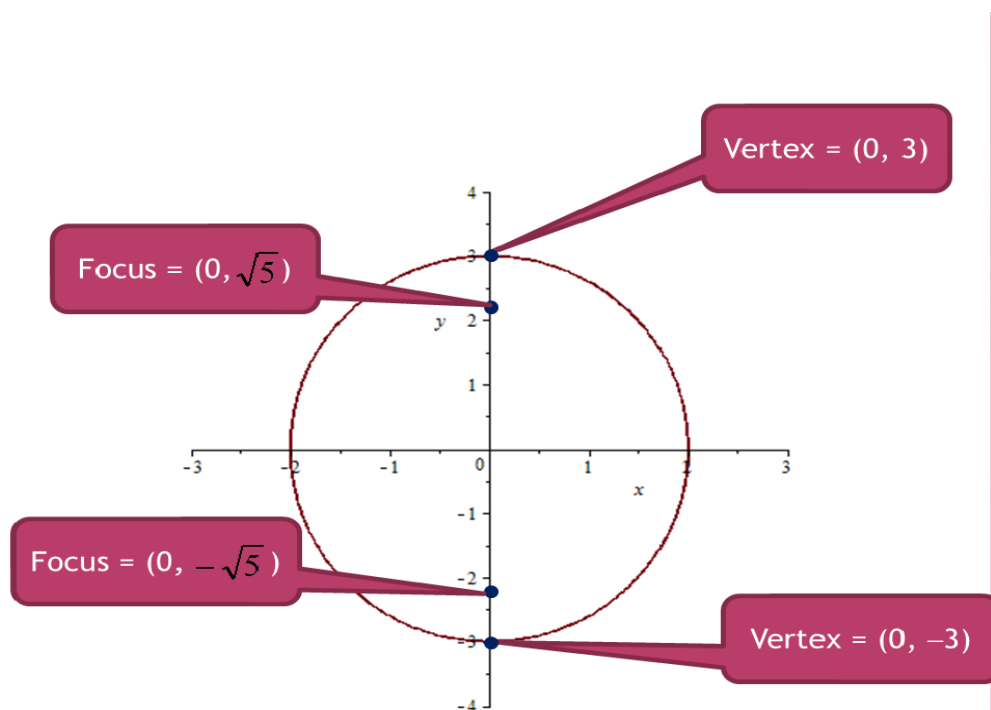
$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

$a = 2$, $b = 3$. Here, $b > a$. The ellipse should look like this:



Length of the major axis is $2b = 2(3) = 6$. The vertices are the endpoints of the major axis. The foci lie c units from the center where

$$c = \sqrt{|2^2 - 3^2|} = |4 - 9| = 5 = \sqrt{5}$$



Ellipses centered at (h, k)

Equation	Major axis	Graph shape
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, a \neq b$	If $a > b$, the major axis is horizontal with length $2a$	
	If $b > a$, the major axis is vertical with length $2b$	

The vertices and foci can be found in a similar manner as with ellipses centered at the origin $(0, 0)$.

Example 6

Sketch the graph of the ellipse with equation $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{25} = 1$. Identify the center, vertices and foci.

Solution:

Comparing

$$\frac{(x-2)^2}{9} + \frac{(y+3)^2}{25} = 1$$

with the standard equation

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, a \neq b$$

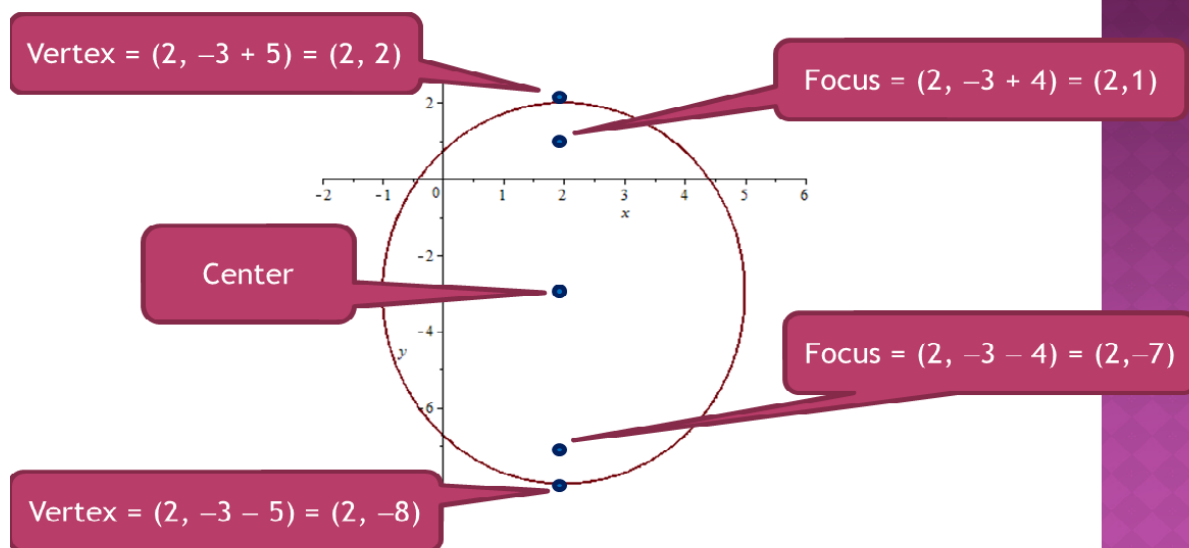
we get $h = 2, k = -3$. So, the center is $(2, -3)$.

From equation $a = 3, b = 5$. So, $b > a$. Therefore, the graph shape is .

Length of the major axis is $2b = 2(5) = 10$. The vertices are the endpoints of the major axis.

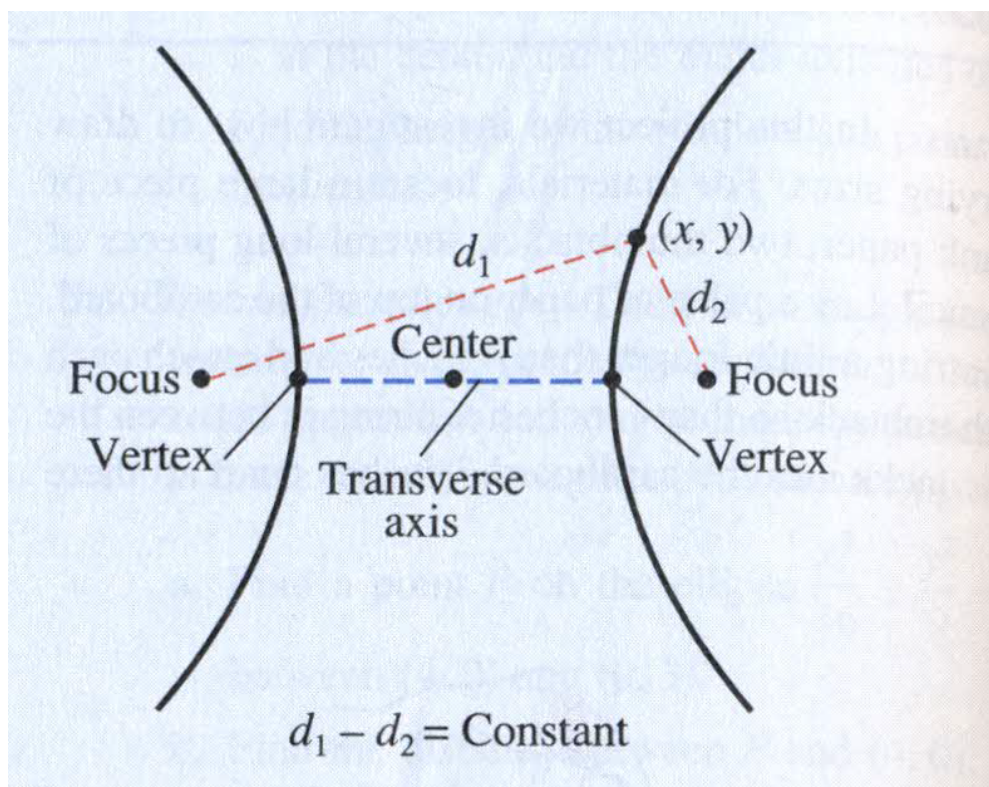
The foci lie c units from the center where

$$c = \sqrt{|3^2 - 5^2|} = \sqrt{|9 - 25|} = \sqrt{16} = 4$$

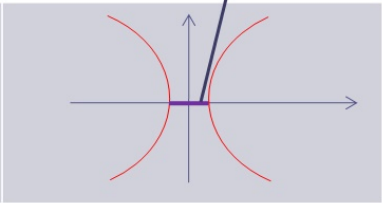
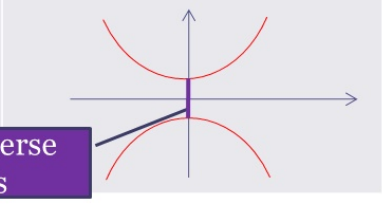


HYPERBOLAS

A hyperbola is the set of all points (x, y) such that the difference of the distances from two fixed points (called the foci) is constant. The line through the foci intersects the hyperbola at two points, called the vertices. The line segment connecting the two vertices is called the transverse axis, and the midpoint of the transverse axis is called the center.



Hyperbolas centered at the origin (0, 0)

Equation	Description	Graph shape
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	Horizontal transverse axis	
$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$	Vertical transverse axis	

The foci are a distance of $c = \sqrt{a^2 + b^2}$ units from the center on the same line as the vertices.

Asymptotes of a hyperbola centered at the origin

The asymptotes for a hyperbola centered at the origin are

$$y = \frac{b}{a}x \quad \text{and} \quad y = -\frac{b}{a}x$$

Example 7

Find the vertices, foci and asymptotes for the hyperbola with equation

$$\frac{y^2}{36} - \frac{x^2}{64} = 1$$

Solution

Step 1: Find a , b and c .

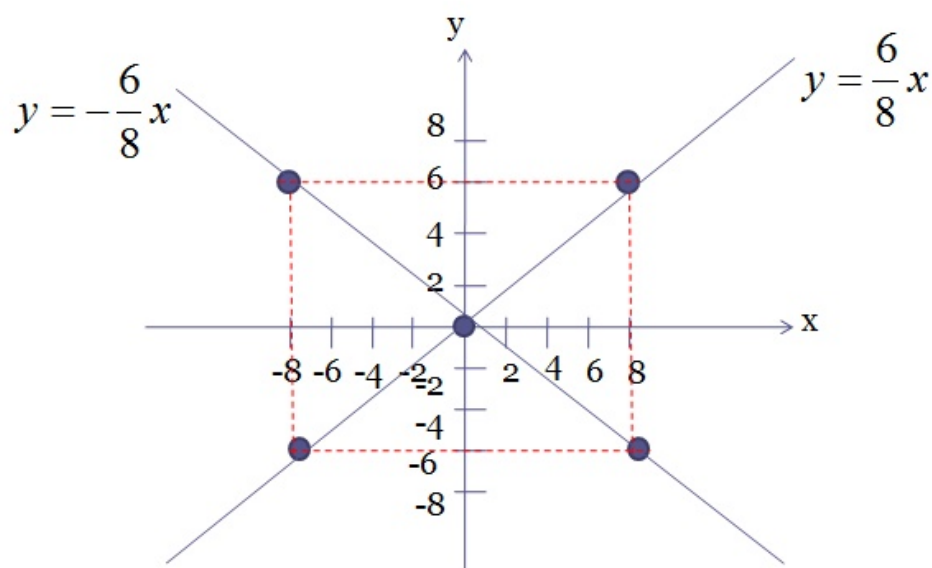
- Comparing $\frac{y^2}{36} - \frac{x^2}{64} = 1$ with $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ we get $b = 6, a = 8$. So, $c = \sqrt{36 + 64} = 10$.

Step 2: Determine the asymptotes.

- The asymptotes for the hyperbola:

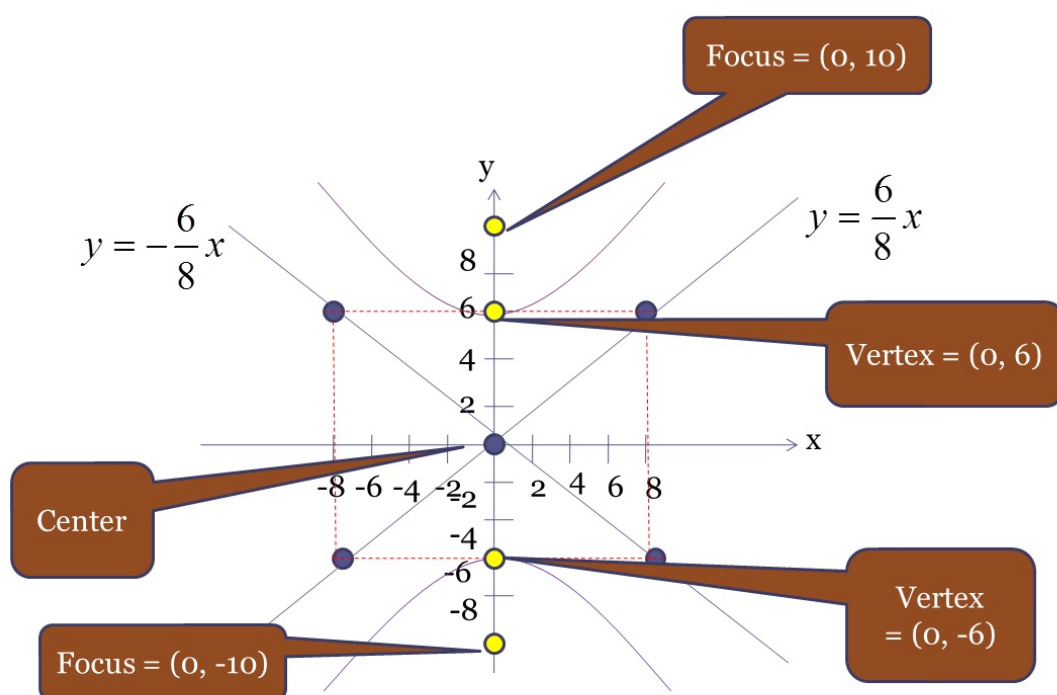
$$y = \frac{6}{8}x \quad \text{and} \quad y = -\frac{6}{8}x$$

Step 3: Sketch the asymptotes



Step 4: Sketch the hyperbola and locate the vertices and foci

- The foci are $c = 10$ units from the center on the same line as the vertices.



Hyperbolas centered at (h, k)

Equation	Description
$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	Horizontal transverse axis. The vertices are the endpoints of the transverse axis: a units from the center.
$\frac{(y-k)^2}{b^2} - \frac{(x-h)^2}{a^2} = 1$	Vertical transverse axis. The vertices are the endpoints of the transverse axis: b units from the center.

Example 8

Find the center, vertices, foci, and asymptotes for the hyperbola with equation

$$\frac{(x-2)^2}{4} - \frac{(y-1)^2}{16} = 1$$

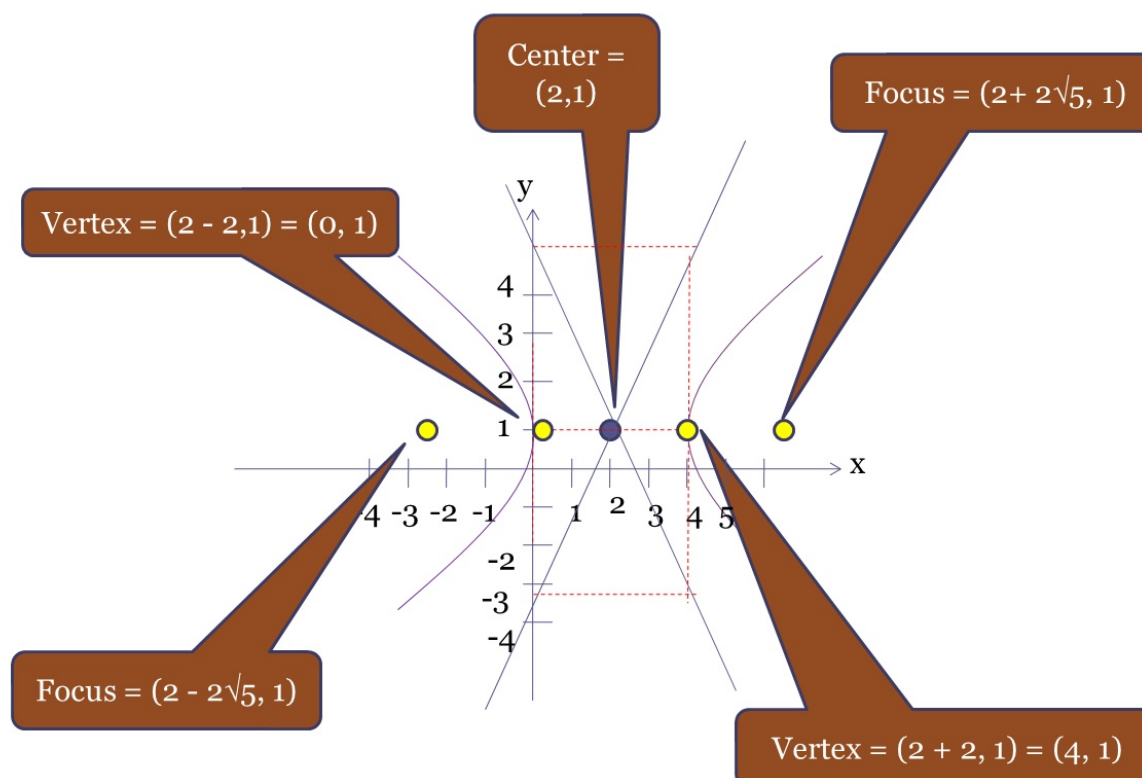
and sketch its graph.

Solution:

Comparing $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{16} = 1$ with $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ we get

$$h = 2, k = 1, a = 2, b = 4, \text{ and } c = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$

So, the center of the hyperbola is $(2, 1)$. The vertices are 2 units from the center. The foci are $2\sqrt{5}$ units from the center on the same line as the vertices.



The asymptotes have slope $m = \pm \frac{b}{a} = \pm 2$.

The asymptotes pass through the center (2, 1). Thus, the equations of the asymptotes are $y = 2x - 3$ and $y = -2x + 5$.

Example 9

Rewrite the following equation in standard form.

$$16x^2 - y^2 + 64x - 2y + 67 = 0$$

Solution

$$\begin{aligned} [16x^2 + 64x] - [y^2 + 2y] &= -67 \\ [16(x^2 + 4x)] - \left[y^2 + 2y + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 \right] &= -67 \\ \left[16 \left(x^2 + 4x + \left(\frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 \right) \right] - [(y+1)^2 - 1] &= -67 \\ [16((x+2)^2 - 4)] - (y+1)^2 + 1 &= -67 \\ 16(x+2)^2 - 64 - (y+1)^2 + 1 &= -67 \\ 16(x+2)^2 - (y+1)^2 &= -67 + 64 - 1 \\ 16(x+2)^2 - (y+1)^2 &= -4 \end{aligned}$$

Dividing both sides by -4

$$\frac{16(x+2)^2 - (y+1)^2}{-4} = \frac{-4}{-4}$$

$$-4(x+2)^2 + \frac{(y+1)^2}{4} = 1$$

Thus, we have

$$\frac{(y+1)^2}{4} - \frac{(x+2)^2}{1/4} = 1.$$

Example 10

Find the equation of the hyperbola with vertices $(\pm 1, 0)$ and foci $(\pm \sqrt{5}, 0)$.

Solution:

Step 1: Plot the given points

Step 2: Find a and b

Step 3: Substitute the values a and b into the standard equation of hyperbola.

CONIC SECTION

1. Sketch a graph of the following parabola. Identify the focus and directrix.

a. $x^2 + 12y$

b. $y^2 + 20x$

c. $x = \frac{-y^2}{16}$

d. $4x = -y^2$

e. $8y = -3x^2$

f. $12x = 5y^2$

2. Find an equation of the following ellipse assume the center is at the origin. Sketch the graph. Labeling the vertices and foci.

a. An ellipse with vertices $(\pm 6, 0)$ and foci $(\pm 4, 0)$

b. An ellipse with vertices $(\pm 5, 0)$ and foci $(\pm 2, 0)$

c. An ellipse with foci $(0, \pm 5)$ and vertices $(0, \pm 13)$

3. Sketch a graph of the following ellipses. Identify the vertices and foci.

a. $\frac{x^2}{4} + y^2 = 1$

b. $12x^2 + 5y^2 = 60$

4. Sketch a graph of the following hyperbolas. Find the vertices, foci and asymptotes.

a. $\frac{y^2}{16} - \frac{x^2}{9} = 1$

b. $4x^2 - y^2 = 16$

5. Find the equation of the following hyperbolas.

a. A hyperbola with vertices $(\pm 4, 0)$ and foci $(\pm 6, 0)$.

b. A hyperbola with vertices $(\pm 3, 0)$ and foci $(\pm 5, 0)$.

c. A hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 5)$.

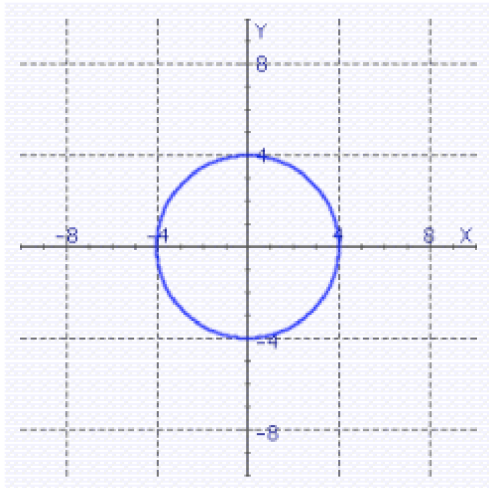
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Conic Sections Practice Test

1. Give the coordinates of the circle's center and its radius.

$$(x - 2)^2 + (y + 9)^2 = 1$$

- _____ 2. Find the equation of the circle graphed below.



A) $x^2 + y^2 = 4$

C) $x^2 + y^2 = 16$

E) $x^2 + y = 16$

B) $y^2 = x^2 + 16$

D) $x^2 + y^2 = 1$

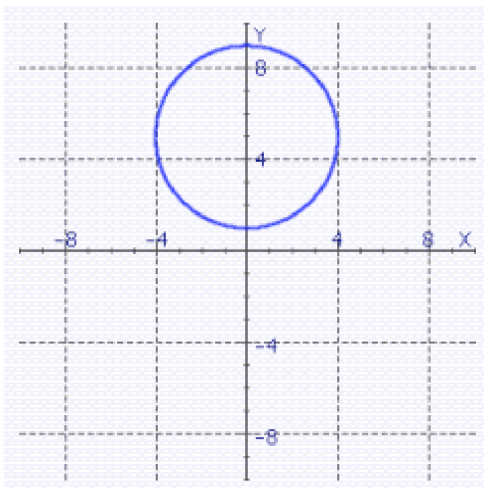
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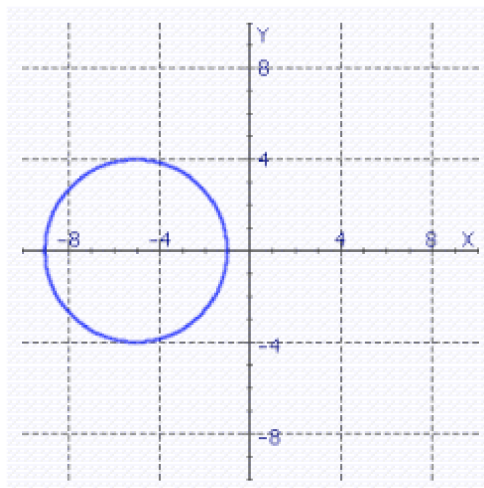
_____ 3. Graph the following equation.

$$x^2 - 10x + y^2 = -9$$

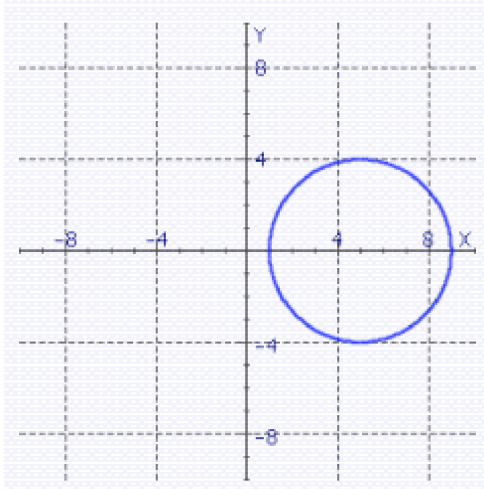
A)



C)



B)



_____ 4. Find the vertex and focus of the parabola.

$$(y - 2)^2 + 16(x - 3) = 0$$

- A) vertex: $(-3, -2)$ focus: $(-3, 14)$
B) vertex: $(-3, -2)$ focus: $(-3, -18)$
C) vertex: $(-3, -2)$ focus: $(-7, -2)$
D) vertex: $(3, 2)$ focus: $(-1, 2)$

- _____ 5. Find the standard form of the equation of the parabola with the given characteristic and vertex at the origin.
focus: $(0, 7)$
A) $x^2 = 28y$ C) $x^2 = -7y$ E) $y^2 = 7x$
B) $x^2 = 7y$ D) $y^2 = 28x$
- _____ 6. Find the standard form of the equation of the parabola with the given characteristic and vertex at the origin.
directrix: $x = 1$
A) $x^2 = -4y$ C) $x^2 = y$ E) $y^2 = -4x$
B) $x^2 = 4y$ D) $y^2 = x$
- _____ 7. Find the vertex and focus of the parabola.
 $y^2 = -\frac{9}{8}x$
A) vertex: $\left(0, -\frac{5}{4}\right)$ focus: $\left(-\frac{9}{8}, -\frac{9}{8}\right)$
B) vertex: $(0, 0)$ focus: $\left(0, -\frac{9}{8}\right)$
C) vertex: $(0, 0)$ focus: $\left(-\frac{9}{8}, 0\right)$
D) vertex: $(0, 0)$ focus: $\left(-\frac{9}{32}, 0\right)$
- _____ 8. Find the equation of the parabola with vertex at $(5, 4)$ and focus at $(-3, 4)$.
A) $(y - 4)^2 = -32(x - 5)$ D) $(y + 4)^2 = -32(x - 5)$
B) $(y - 4)^2 = 32(x - 5)$ E) $(y - 4)^2 = 8(x - 5)$
C) $(y + 4)^2 = 32(x + 5)$
9. Find the equation of the parabola with vertex at $(0, 0)$ and focus at $(0, 5)$. Express the equation in standard form.

____ 10. Find the center and vertices of the ellipse.

$$\frac{x^2}{49} + \frac{y^2}{4} = 1$$

- A) center: (7, 0) vertices: (0, -2), (0, 2)
B) center: (0, 0) vertices: (-2, 0), (2, 0)
C) center: (0, 0) vertices: (0, -7), (0, 7)
D) center: (0, 0) vertices: (-7, 0), (7, 0)

____ 11. Find the center and foci of the ellipse.

$$\frac{(x+5)^2}{5} + \frac{(y+9)^2}{9}$$

- A) center: (5, 9) foci: (5, 7), (5, 11)
B) center: (-5, -9) foci: (-5, -11), (-5, -7)
C) center: (-5, -9) foci: (-7, -9), (-3, -9)
D) center: (5, 9) foci: (3, -9), (7, -9)

____ 12. Find the center and vertices of the ellipse.

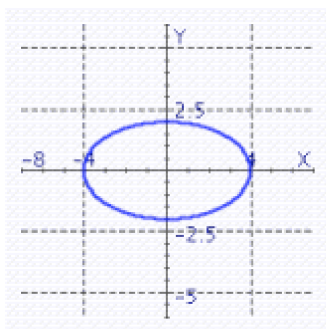
$$4x^2 + 9y^2 - 24x + 72y + 144 = 0$$

- A) center: (-4, 3) vertices: (-7, 3), (-1, 3)
B) center: (-3, 4) vertices: (-5, 4), (-1, 4)
C) center: (3, -4) vertices: (1, -4), (5, -4)
D) center: (3, -4) vertices: (0, -4), (6, -4)
E) center: (-3, 4) vertices: (-6, 4), (0, 4)

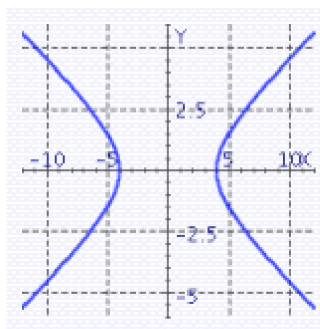
_____ 13. Identify the graph of the following ellipse.

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

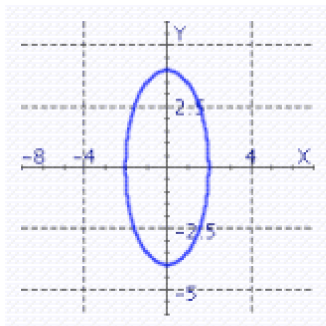
A)



C)



B)



_____ 14. Find the center and vertices of the hyperbola.

$$11x^2 - 25y^2 + 22x + 250y - 889 = 0$$

- A) center: (1, -5), vertices: (1, -10), (1, 0)
B) center: (-1, 5), vertices: (-1, 0), (-1, 10)
C) center: (-1, 5), vertices: (-6, 5), (4, 5)
D) center: (1, -5), vertices: (-4, -5), (6, -5)

Name: _____

ID: A

_____ 15. Find the vertices and asymptotes of the hyperbola.

$$9y^2 - 16x^2 = 144$$

- A) vertices: $(0, \pm 4)$ asymptote: $y = \pm \frac{4}{3}x$
B) vertices: $(0, \pm 4)$ asymptote: $y = \pm \frac{3}{4}x$
C) vertices: $(\pm 4, 0)$ asymptote: $y = \pm \frac{4}{3}x$
D) vertices: $(\pm 4, 0)$ asymptote: $y = \pm \frac{3}{4}x$

_____ 16. Find the standard form of the equation of the hyperbola with the given characteristics.

vertices: $(0, \pm 6)$ foci: $(0, \pm 7)$

A) $\frac{y^2}{36} - \frac{x^2}{49} = 1$

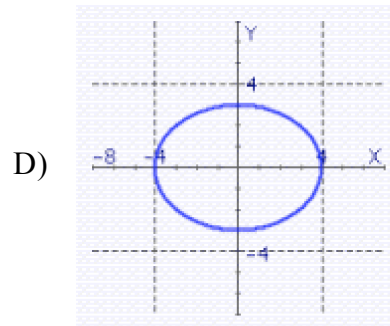
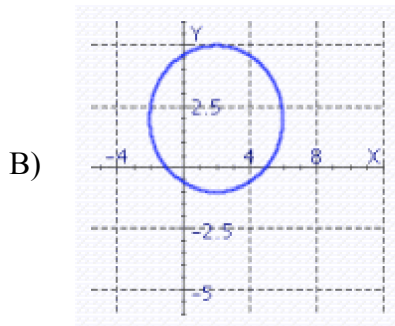
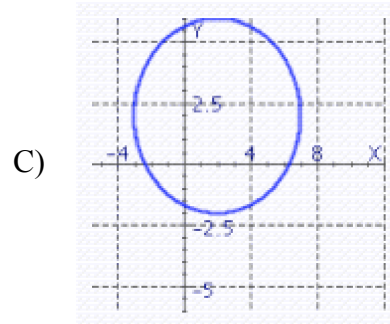
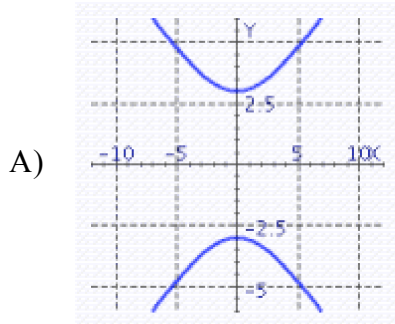
C) $\frac{x^2}{36} - \frac{y^2}{13} = 1$

B) $\frac{y^2}{36} - \frac{x^2}{13} = 1$

D) $\frac{x^2}{36} - \frac{y^2}{13} = 49$

_____ 17. Find the graph of the following ellipse.

$$9x^2 + 16y^2 - 36x - 64y + -44 = 0$$



_____ 18. Write the equation of the ellipse that has its center at the origin with focus at (0, 4) and vertex at (0, 7).

A) $\frac{x^2}{49} + \frac{y^2}{33} = 1$

C) $\frac{x^2}{33} + \frac{y^2}{49} = -1$

B) $\frac{x^2}{33} - \frac{y^2}{49} = 1$

D) $\frac{x^2}{33} + \frac{y^2}{49} = 1$

_____ 19. Find the center and vertices of the ellipse.

$$x^2 + 9y^2 + 16x - 54y + 136 = 0$$

- | | |
|--------------------|-----------------------------|
| A) center: (3, -8) | vertices: (0, -8), (6, -8) |
| B) center: (8, -3) | vertices: (7, -3), (9, -3) |
| C) center: (-8, 3) | vertices: (-9, 3), (-7, 3) |
| D) center: (-8, 3) | vertices: (-11, 3), (-5, 3) |
| E) center: (8, -3) | vertices: (5, -3), (11, -3) |

_____ 20. Find the standard form of the equation of the ellipse with the following characteristics.

foci: $(\pm 4, 0)$ major axis of length: 12

A) $\frac{x^2}{36} + \frac{y^2}{20} = 1$

D) $\frac{x^2}{144} + \frac{y^2}{16} = 1$

B) $\frac{x^2}{36} + \frac{y^2}{16} = 1$

E) $\frac{x^2}{144} + \frac{y^2}{128} = 1$

C) $\frac{x^2}{16} + \frac{y^2}{36} = 1$

_____ 21. Find the standard form of the equation of the hyperbola with the given characteristics.

vertices: $(-2, -4), (-2, 6)$ foci: $(-2, -5), (-2, 7)$

A) $\frac{(y-1)^2}{25} - \frac{(x+2)^2}{11} = 1$

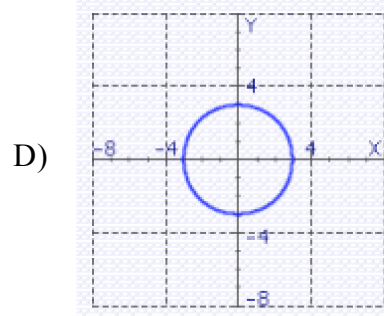
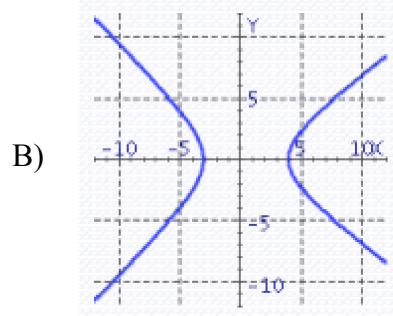
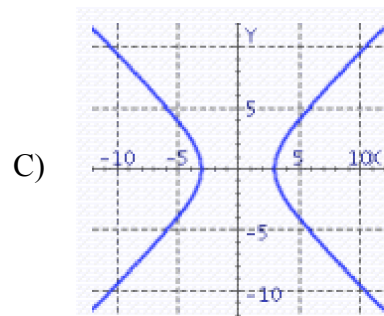
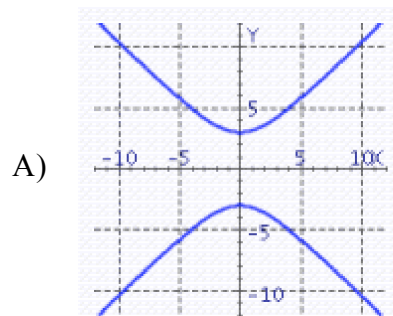
C) $\frac{(y-2)^2}{11} - \frac{(x+1)^2}{25} = 1$

B) $\frac{(y+1)^2}{25} - \frac{(x-2)^2}{11} = 1$

D) $\frac{(y-1)^2}{25} - \frac{(x+2)^2}{36} = 1$

_____ 22. Graph the hyperbola.

$$9x^2 - 9y^2 = 81$$



____ 23. Identify the conic by writing the equation in standard form.

$$10y^2 - 20x^2 + 60y + 160x - 255 = 0$$

A) $\frac{(y-3)^2}{\frac{5}{2}} - \frac{(x-4)^2}{\frac{5}{4}} = 1$; hyperbola

B) $\frac{(y+3)^2}{\frac{5}{2}} - \frac{(x-4)^2}{\frac{5}{4}} = 1$; hyperbola

C) $\frac{(y+3)^2}{\frac{97}{2}} - \frac{(x-4)^2}{\frac{97}{4}} = 1$; hyperbola

____ 24. Identify the conic by writing the equation in standard form.

$$4x^2 + 4y^2 + 40x + 16y + 40 = 0$$

A) $(x+5)^2 + (y+2)^2 = 19$; circle

B) $(x+5)^2 + (y+2)^2 = 39$; circle

C) $\frac{(x+5)^2}{\frac{11}{4}} + \frac{(y+2)^2}{\frac{11}{4}} = 1$; ellipse

**Conic Sections Practice Test
Answer Section**

1. $(2, -9), r = 1$
2. C
3. B
4. D
5. A
6. E
7. D
8. A
9. $x^2 = 20y$
10. D
11. B
12. D
13. A
14. C
15. A
16. B
17. B
18. D
19. D
20. A
21. A
22. C
23. B
24. A

TUTORIAL TOPIC 7 : FUNCTIONS OF TWO VARIABLES

1. In 2016, an investment of RM1000 was made in a bond earning 6% compounded annually. Assume that the buyer pays tax at rate R and the annual rate of inflation is I . In the year 2026, the value V of the investment in Ringgit Malaysia is

$$V(I, R) = 1000 \left[\frac{1+0.06(1-R)}{1+I} \right]^{10}.$$

Use this function of two variables to complete the table for V :

	Inflation Rate		
Tax Rate	0	0.03	0.05
0			
0.28			
0.35			

2. The average length of time that a customer waits in line for service is

$$W(x, y) = \frac{1}{x - y}, x > y$$

where y is the number of customers arriving per hour, and x is the number of services per hour. Evaluate

- (a) $W(15, 9)$.
(b) $W(15, 13)$
(c) $W(12, 7)$
(d) $W(5, 2)$
3. Sketch the graph of the surface given by $f(x, y) = \frac{x^2}{25} + \frac{y^2}{16} - 1$ where $0 \leq z \leq 17$
4. Find the limit

- (a) $\lim_{(x,y) \rightarrow (1,2)} \frac{5x^2y}{x^2+y^2}$
(b) $\lim_{(x,y) \rightarrow (1,1)} \frac{x}{\sqrt{x+y}}$
(c) $\lim_{(x,y) \rightarrow (1,1)} \frac{xy}{x^2+y^2}$
(d) $\lim_{(x,y) \rightarrow (1,2)} e^{xy}$
(e) $\lim_{(x,y) \rightarrow (1,0)} \frac{1}{x+y}$

5. Find the limit

- (a) $\lim_{(x,y) \rightarrow (1,-2)} \frac{y^2+2xy}{y+2x}$

$$(b) \lim_{(x,y) \rightarrow (-1,1)} \frac{2x^2 - xy - 3y^2}{x+y}$$

$$(c) \lim_{(x,y) \rightarrow (4,5)} \frac{\sqrt{x+y}-3}{x+y-9}$$

6. Show that the following limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{\sqrt{x}-\sqrt{y}}$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{x^2+y}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4}$$

7. A function f is defined by

$$f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^2 + y^2} & , (x, y) \neq (0, 0) \\ 1 & , (x, y) = (0, 0) \end{cases}$$

(a) Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the lines $y = 0$, $y = x$ and $x = 0$.

(b) Determine whether f is continuous at $(0, 0)$.

8. At what points of $\mathbf{R}^2$ are the following functions continuous?

$$(a) f(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$(b) f(x, y) = \begin{cases} \frac{1 - \cos(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

TUTORIAL TOPIC 7 : FUNCTIONS OF TWO VARIABLES

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where y is the number of customers arriving per hour, and x is the number of services per hour. Evaluate

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(c) $W(12, 7)$
(d) $W(5, 2)$
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4. Find the limit

- (a) $\lim_{(x,y) \rightarrow (1,2)} \frac{5x^2y}{x^2+y^2}$
(b) $\lim_{(x,y) \rightarrow (1,1)} \frac{x}{\sqrt{x+y}}$
(c) $\lim_{(x,y) \rightarrow (1,1)} \frac{xy}{x^2+y^2}$
(d) $\lim_{(x,y) \rightarrow (1,2)} e^{xy}$
(e) $\lim_{(x,y) \rightarrow (1,0)} \frac{1}{x+y}$

5. Find the limit

- (a) $\lim_{(x,y) \rightarrow (1,-2)} \frac{y^2+2xy}{y+2x}$

$$(b) \lim_{(x,y) \rightarrow (-1,1)} \frac{2x^2 - xy - 3y^2}{x+y}$$

$$(c) \lim_{(x,y) \rightarrow (4,5)} \frac{\sqrt{x+y}-3}{x+y-9}$$

6. Show that the following limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{\sqrt{x}-\sqrt{y}} \quad \leftarrow \text{This limit exists and equals zero}$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{x^2+y}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4}$$

7. A function f is defined by

$$f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^2 + y^2} & , (x, y) \neq (0, 0) \\ 1 & , (x, y) = (0, 0) \end{cases}$$

Make a conjecture about the value of

(a) Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the lines $y = 0$, $y = x$ and $x = 0$.

(b) Determine whether f is continuous at $(0, 0)$.

8. At what points of $\mathbf{R}^2$ are the following functions continuous?

$$(a) f(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$(b) f(x, y) = \begin{cases} \frac{1 - \cos(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

COURSE MODULE: LOGICAL REASONING.

TOPIC 8: SEQUENCES AND SERIES

Objectives:

Students should be able to

- recognize some basic sequences and series: arithmetic sequence and series, geometric sequence and series, Fibonacci sequence, p-series
- define convergent and divergent series using partial sums
- determine the convergence and divergence of a geometric series
- determine the convergence and divergence of p-series
- use the divergence test to show the divergence of a given series
- use the comparison test and the limit comparison test to determine the convergence and divergence of a given series
- represent a function as Maclaurin series

Definition and Notation

- Sequence is a list of numbers written in a definite order:

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots$$

- Series is a sum of sequence:

$$a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$$

- The number a_1 is called the first term, a_2 is the second term, and in general a_n is the n th term.
- Notation:

Sequence $\{a_1, a_2, a_3, a_4, \dots\}$ is denoted by $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$

Series $\{a_1 + a_2 + a_3 + a_4 + \dots\}$ is denoted by $\sum a_n$ or $\sum_{n=1}^{\infty} a_n$

Examples

$$\left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n}{n+1}, \dots \right\}.$$

$$\left\{ \frac{(-1)^n(n+1)}{3^n} \right\} = \left\{ -\frac{2}{3}, \frac{3}{9}, -\frac{4}{27}, \frac{5}{81}, \dots, \frac{(-1)^n(n+1)}{3^n}, \dots \right\}.$$

$$\sum_{n=1}^{\infty} \frac{n}{n+1} = \left\{ \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \dots + \frac{n}{n+1} + \dots \right\}.$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n(n+1)}{3^n} = \left\{ -\frac{2}{3} + \frac{3}{9} - \frac{4}{27} + \frac{5}{81} - \dots \right\}.$$

Basic Sequences and Series

- **Arithmetic sequence (progression):**

$$\{a + (n-1)d\}_{n=1}^{\infty} = \{a, a+d, a+2d, a+3d, \dots, a+(n-1)d, \dots\}$$

where a is the first term and d is the common difference

Arithmetic series:

$$\sum_{n=1}^{\infty} (a + (n-1)d) = \{a + (a+d) + (a+2d) + (a+3d) + \dots + (a+(n-1)d) + \dots\}$$

- **Geometric sequence (progression):**

$$\{ar^{n-1}\}_{n=1}^{\infty} = \{a, ar, ar^2, ar^3, \dots, ar^{n-1}, \dots\}$$

where a is the first term and r is the common ratio.

Geometric series:

$$\sum_{n=1}^{\infty} ar^{n-1} = \{a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots\}$$

Fibonacci Sequences and p -series

- Fibonacci sequence $\{f_n\}$ is defined recursively by the conditions: $f_1 = 1, f_2 = 1, f_n = f_{n-1} + f_{n-2}, n \geq 3$.

$$\{f_n\} = \{1, 1, 2, 3, 5, 8, 13, 21, \dots\}$$

- This sequence arose when the 13th-century Italian mathematician known as Fibonacci solved a problem concerning the breeding of rabbits.
- p -series is a series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where p is a positive real number.
- If $p = 1$, then it is called harmonic series.

Convergent and Divergent Series

Given that

$$\sum_{n=1}^{\infty} a_n = \{a_1 + a_2 + a_3 + \dots + a_n + \dots\}$$

Consider the partial sum

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$s_4 = a_1 + a_2 + a_3 + a_4$$

and in general, $s_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{i=1}^n a_i$

- If $\lim_{n \rightarrow \infty} s_n = s$ exists as a real number, then the series is called convergent and we write $a_1 + a_2 + a_3 + \dots + a_n + \dots = s$ or $\sum_{n=1}^{\infty} a_n = s$

The number s is called the sum of the series. Otherwise, the series $\sum_{n=1}^{\infty} a_n$ is called divergent.

Example 1

Show that if $-1 < r < 1$ then the sum of geometric series is $\frac{a}{1-r}$.

Solution:

Suppose $-1 < r < 1$.

$$s_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} \quad \dots \text{Eq 1}$$

$$rs_n = ar + ar^2 + ar^3 + ar^4 + \dots + ar^n \quad \dots \text{Eq 2}$$

Eq 1 – Eq 2 gives:

$$s_n - rs_n = a - ar^n$$

$$s_n = \frac{a - ar^n}{1 - r}$$

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \frac{a(1 - r^n)}{1 - r}$$

$$= \frac{a}{1 - r} \lim_{n \rightarrow \infty} (1 - r^n)$$

$$= \frac{a}{1 - r} (1) - \lim_{n \rightarrow \infty} r^n$$

$$= \frac{a}{1 - r} - 0$$

$$= \frac{a}{1 - r}$$

Why $\lim_{n \rightarrow \infty} r^n = 0$??

Let say $r = 0.1$

n	$r^n = 0.1^n$
1	0.1
2	0.01
3	0.001
4	0.0001
5	0.00001

r^n is approaching 0 as $n \rightarrow \infty$

Convergent and Divergent Series.

The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \dots$$

is convergent if $|r| < 1$ and its sum is $\frac{a}{1-r}$.

If $|r| \geq 1$, the geometric series is divergent.

Example 2

Is the geometric series $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$ convergent or divergent?

Solution:

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$$

$$a_1 = 2^{2(1)} 3^{1-1} = 4$$

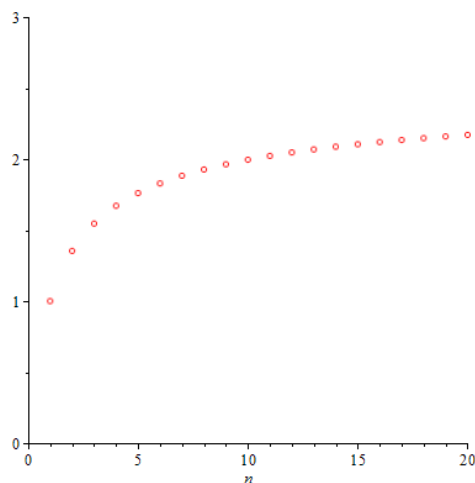
$$a_2 = 2^{2(2)} 3^{1-2} = 16/3$$

$$a = 4. \quad r = \frac{16/3}{4} = \frac{4}{3}$$

Hence, the geometric series diverges.

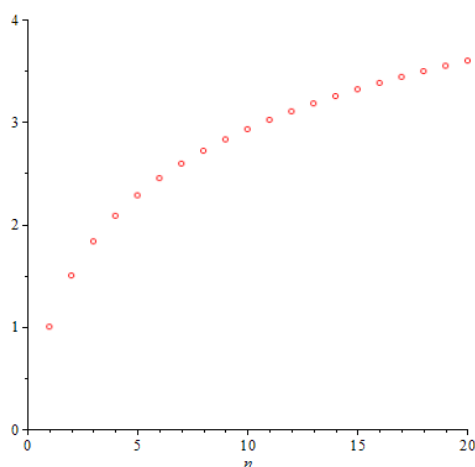
Example of convergent p-series.

- Graph of partial sums of series $\sum_{n=1}^{\infty} \frac{1}{n^{1.5}}$



Example of divergent p-series

- Graph of partial sums of series $\sum_{n=1}^{\infty} \frac{1}{n}$



The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots$$

1. Converges if $p > 1$, and
2. Diverges if $0 < p \leq 1$.

Test of Divergence

If $\lim_{n \rightarrow \infty} a_n$ does not exist or $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Example 3

Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$ diverges.

Solution:

$$\lim_{n \rightarrow \infty} \frac{n^2}{5n^2+4} = \frac{1}{5 + \frac{4}{n^2}} = \frac{1}{5+0} = \frac{1}{5} \neq 0$$

Since $\lim_{n \rightarrow \infty} \frac{n^2}{5n^2+4} \neq 0$, hence, the series is diverges.

The Comparison Test

- Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.
 - (i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent.
 - (ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent.

Example 4

- a. Determine whether the series $\sum_{n=1}^{\infty} \frac{n^3}{2n^4-1}$ converges or diverges.
- b. Determine whether the series $\sum \frac{5}{2n^2+4n+3}$ converges or diverges.

Solution:

- a. Step 1: Let $a_n = \frac{n^3}{2n^4-1}$

Step 2: Choose term which dominate the numerator n^3 . The term is n^3 .

Choose term which dominate denominator $2n^4 - 1$. The term is $2n^4$.

Step 3: Let $b_n = \frac{n^3}{2n^4} = \frac{1}{2n}$.

Step 4: Determine whether $\sum b_n$ converges or diverges.

Step 5: Conclude for $\sum a_n$ using the comparison test.

Since

$$a_n = \frac{n^3}{2n^4-1} \geq \frac{1}{2n}$$

Then, $\sum b_n$ is divergent.

Therefore, the series diverges.

b. Step 1: Let $a_n = \frac{5}{2n^2+4n+3}$

Step 2: Choose term which dominate the numerator 5. The term is 5.

Choose term which dominate denominator $2n^2 + 4n + 3$. The term is $2n^2$.

Step 3: Let $b_n = \frac{5}{2n^2}$

Step 4: Determine whether $\sum b_n$ converges or diverges. $\sum \frac{5}{2n^2}$ is p -series is convergent.

Step 5: Conclude for $\sum a_n$ using the comparison test.

Since

$$a_n = \frac{5}{2n^2 + 4n + 3} \leq \frac{5}{2n^2}$$

Then, $\sum a_n$ is convergent.

Therefore, the series converges.

The Limit Comparison Test

- Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and let $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$
 - (i) If c is a finite number and $c > 0$, then either both series converge or both diverges.
 - (ii) If $c = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
 - (iii) If $c = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Example 5

Determine whether $\sum_{n=1}^{\infty} \frac{n-2}{n^3-n^2+3}$ converges or diverges.

a. Step 1: Let $a_n = \frac{n-2}{n^3-n^2+3}$

Step 2: Choose term which dominate the numerator $n - 2$. The term is n .

Choose term which dominate denominator $n^3 - n^2 + 3$. The term is n^3 .

Step 3: Let $b_n = \frac{n}{n^3} = \frac{1}{n^2}$.

Step 4: Calculate $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$

Step 5: Use the condition in limit comparison test. If $\sum a_n$ and $\sum b_n$ are either both converge or both diverge, then use the comparison test to conclude for $\sum a_n$.

Since

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{n-2}{n^3 - n^2 + 3}}{\frac{1}{n^2}} = 1$$

We get $c = 1$ and $\sum b_n$ converges (because $\sum b_n$ is a convergent p-series).

Therefore, the series $\sum a_n$ converges.

Exercise

Determine whether the following series converges or diverges

a) $\sum_{n=3}^{\infty} \frac{1}{\ln(n)}$ ---- Answer: diverges

b) $\sum_{n=1}^{\infty} 12(0.73)^{n-1}$

c) $\sum_{n=1}^{\infty} \frac{1}{n^3+8}$

d) $\sum_{n=1}^{\infty} \frac{1}{2^n}$ ---- Answer: converges

e) $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ ---- Answer: converges

Power Series

A power series is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

where x is a variable and the c_n 's are constants called the coefficients of the series.

If we take $c_n = 1$ for all n , the power series becomes the geometric series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$$

More generally, a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

Representing Functions as Power Series

The sum of geometric series

$$1 + x + x^2 + \dots + x^n + \dots$$

(with $a = 1$ and $r = x$) is $\frac{1}{1-x}$.

Thus, we can write

$$\underbrace{\frac{1}{1-x}}_{\text{function}} = \underbrace{1 + x + x^2 + \dots + x^n + \dots}_{\text{power series}}$$

In general, a function $f(x)$ can be represented as power series:

$$f(x) = \sum_{n=0}^{\infty} c_n x^n$$

Expand $f(x) = \sum_{n=0}^{\infty} c_n x^n$:

$$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots \quad (1)$$

From (1), $f(0) = c_0$.

Differentiate (1) with respect to x :

$$f'(x) = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots \quad (2)$$

From (2), $f'(0) = c_1$

Differentiate (2) with respect to x :

$$f''(x) = 2c_2 + 3(2)c_3x + 4(3)c_4x^2 + \dots \quad (3)$$

From (3), $f''(0) = 2c_2$ which implies

$$c_2 = \frac{f''(0)}{2!}$$

In the same way,

Differentiate (3) with respect to x :

$$f'''(x) = 3(2)c_3 + 4(3)(2)c_4x + \dots \quad (4)$$

From (4), $f'''(0) = 3(2)c_3$ which implies

$$c_3 = \frac{f'''(0)}{3!}$$

Based on the above, after differentiating $f(x)$ for r times, we conclude that

$$c_n = \frac{f^{(n)}(0)}{n!}$$

Hence,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

The Maclaurin series

The Maclaurin series generated by a function f is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

Exercises

- a) Find the first four terms in Maclaurin series expansion of e^x .
- b) Find the first four terms in Maclaurin series expansion of $\ln(1+x)$.
- c) Find the first four terms in Maclaurin series expansion of $\frac{\cos(x)}{1-x}$.
 ([Hint: Find the expansion for $\cos(x)$ and expansion for $(1-x)^{-1}$. Then multiply expansion for $\cos(x)$ with expansion for $(1-x)^{-1}$].

TUTORIAL TEST FOR CONVERGENCE

1. Show that the following series diverges

$$\text{i) } \sum_{n=1}^{\infty} \frac{n^2 + 3n + 1}{2n^2 + 1}$$

$$\text{ii) } \sum_{n=1}^{\infty} \frac{n^2 - 1}{n^2 + n}$$

2. Is that the geometric series convergent or divergent?

$$\text{i) } \sum_{k=0}^{\infty} 1.1^k$$

$$\text{ii) } \sum_{n=1}^{\infty} 3(-0.75)^k$$

3. Determine whether the following series converges or diverges.

i) $\sum_{n=1}^{\infty} \frac{4}{\sqrt[3]{n}}$

ii) $\sum_{k=1}^{\infty} \frac{k+1}{k^2+2k+3}$

iii) $\sum_{k=1}^{\infty} \frac{2k}{k^3+1}$

iv) $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{3}{k}$

v) $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k+1}$

vi) $\sum_{k=1}^{\infty} (-1)^k \frac{k}{2^k}$

vii) $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k}{k^2+1}$

4. By using Limit Comparison Test, determine whether the following series

$$\sum_{k=1}^{\infty} \frac{k^2 - 2k + 7}{k^5 + 5k^4 - 3k^3 + 2k - 1}$$

converges or diverges.

Practice problems (Maclaurin series and conic sections)

1. Find the first four terms of Maclaurin series for the function $f(x) = \frac{e^x - 1 - x}{x^2}$
2. Use the Maclaurin series for $\cos(x)$ to find the first four nonzero terms of Maclaurin series for the function $f(x) = x \cos(2x)$
3. Find the first five terms of Maclaurin series for the function
 - a) $f(x) = \frac{1}{1+x}$
 - b) $f(x) = \frac{3}{1-2x}$
 - c) $f(x) = \frac{1}{2-x}$
 - d) $f(x) = e^{x^2}$
4. Use the Maclaurin series for e^x to find the first five terms for $g(x) = e^{3x} - e^{2x}$
5. Use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ to find the first four nonzero terms of Maclaurin series for the function $h(x) = \sin^2 x$
6. Find an equation of the conic that satisfies the given conditions
 - a) parabola, focus (2, -1), vertex (2, 3)
 - b) ellipse, center (-1, 4), vertex (-1, 0), focus (-1, 6)
 - c) hyperbola, vertices (-1, 2), (7, 2), foci (-2, 2), (8, 2)
7. Identify the type of conic section whose equation is given by
$$x^2 - 2x + 2y^2 - 8y + 7 = 0$$
8. Find the vertex, focus and directrix for the parabola $2x^2 - 16x - 3y + 38 = 0$ and sketch its graph
9. Find the vertices and foci of the ellipse $x^2 + 3y^2 + 2x - 12y + 10 = 0$ and sketch its graph
10. Find the vertices, foci and asymptotes of the hyperbola $9y^2 - 4x^2 - 36y - 8x = 4$ and sketch its graph